

SkyRoof v.1.34

software for Hams and satellite enthusiasts

User's Guide



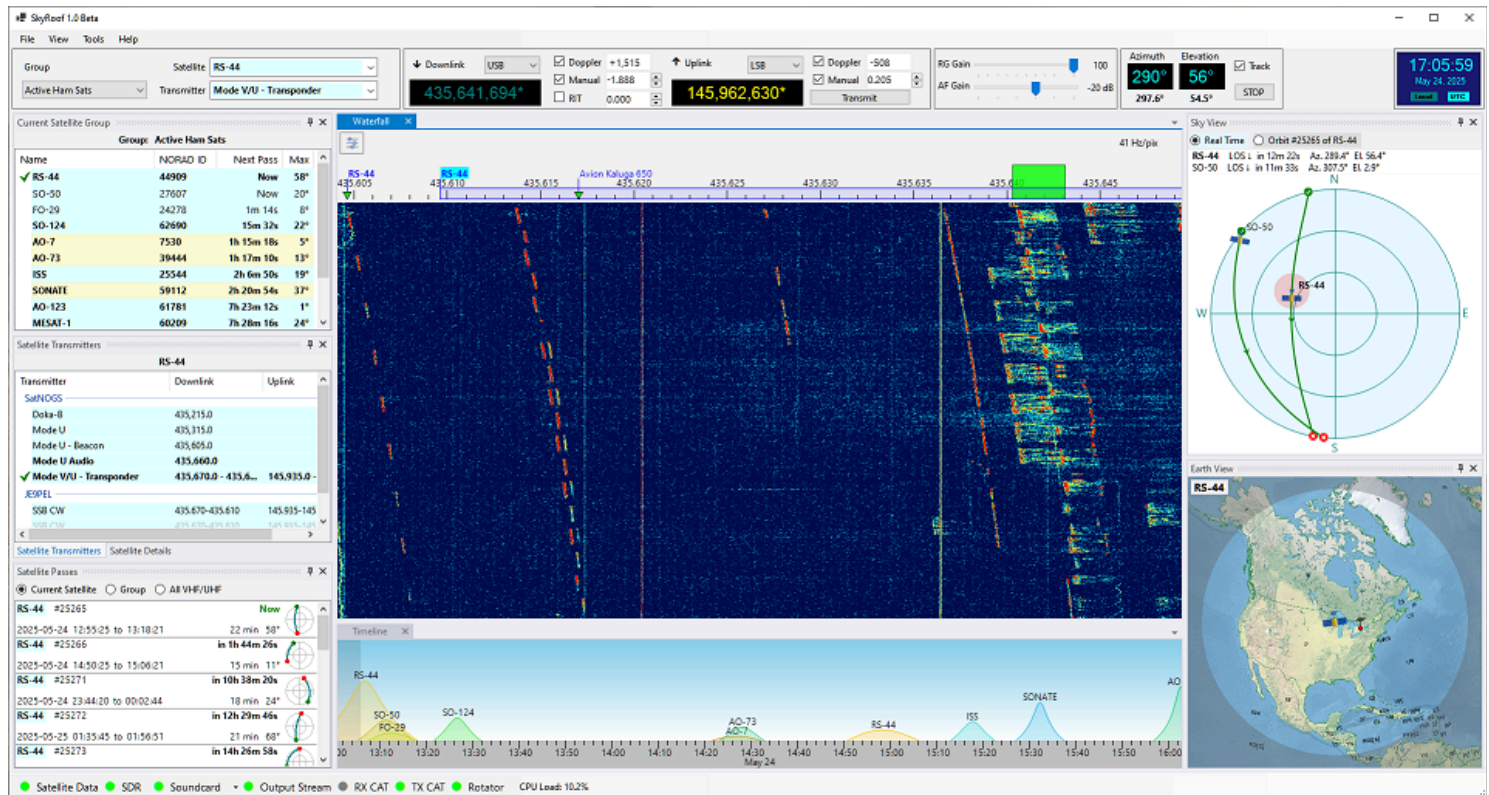
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Overview

SkyRoof is an open source, 64-bit Windows application for Hams and satellite enthusiasts, available on the terms of the GPL v.3 license. It combines satellite tracking and SDR functions in one program, which opens some interesting possibilities. For example, all satellite traces on the waterfall are labeled with satellite names, the boundaries of the transponder segments follow the Doppler shift, and all frequency tuning is done visually, with a mouse.



Features

The main features of SkyRoof are:

- detailed information about all satellites that transmit in the Ham bands;
- satellite tracking in real time;
- pass prediction for the selected satellites;
- visual representation of the current satellite position and future passes, using:
 - Sky View - the view of the sky from your location;
 - Earth View - the view of the Earth from the satellite;
 - Time Line - the satellite passes on the time scale;
 - Pass List - the details of the predicted passes;
- SDR-based waterfall display that covers the whole satellite segments on the VHF and UHF bands, with zoom and pan;
- SDR-based SSB/CW/FM receiver with RIT and Doppler tracking;

- audio and I/Q output to external programs via VAC or UDP;
- frequency scale with satellite names and transponder segments, Doppler-corrected;
- CAT control of an external transceiver;
- antenna rotator control.

The program can work without an SDR, or even without any radio at all, but many useful functions are not available in this mode.

System Requirements

Hardware

- **Computer:** 64-bit PC. 3-GHz Quad-core CPU is recommended;
- **Video Card:** OpenGL 3.3 or higher, 512 Mb of texture memory;
- **Monitor:** screen resolution 1900x1280 or higher, 4K recommended;
- **Internet:** required, to download satellite data;
- **SDR:** optional, but highly recommended. Currently plugins are available for:
 - Airspy;
 - AirspyHF+;
 - SDRplay;
 - RTL-SDR;
 - HackRF;
 - PlutoSDR;
- **Transceiver:** optional. Currently the command definition files are available for:
 - FT-817;
 - FT-818;
 - FT-847;
 - FT-991a;
 - IC-705;
 - IC-706MKIIG;
 - IC-915;
 - IC-910H;
 - IC-9700;
 - TS-2000;

Many other radios are supported via HamLib;

- **Antenna rotator** - optional, any rotator supported by HamLib.

Software

- **OS:** Windows 10 or Windows 11, 64-bit only;
- **SkyCAT:** optional, for CAT control;
- **HamLib:** optional, for CAT and rotator control.

Quick Start

Installation

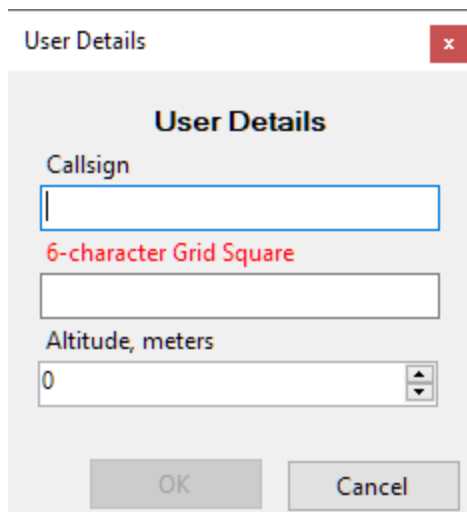
To install SkyRoof, download the installer from the [Download](#) page, run it, and follow the on-screen instructions.

First Run

When you run SkyRoof for the first time, the program performs several important, but somewhat lengthy steps. Fortunately, they need to be done only once.

User Information Input

You will be presented with the User Details dialog:

A screenshot of the 'User Details' dialog box. The dialog has a title bar with 'User Details' and a close button. Inside, the title 'User Details' is centered. Below it, there are three input fields: 'Callsign' (a text box), '6-character Grid Square' (a text box with a red label), and 'Altitude, meters' (a spinner box with '0' selected). At the bottom are 'OK' and 'Cancel' buttons.

User Details

User Details

Callsign

6-character Grid Square

Altitude, meters

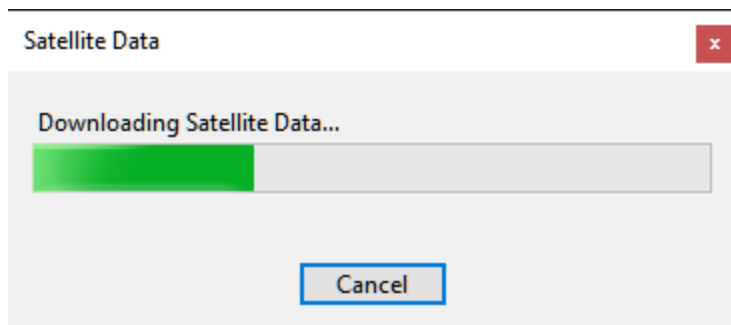
0

OK Cancel

Enter your callsign, 6-character grid square and your altitude above the sea level. The grid square is required, the program cannot proceed without that information. The other two values are optional.

Satellite Data Download

Then SkyRoof downloads the satellite data: make sure that your computer is connected to the Internet.

A screenshot of the 'Satellite Data' dialog box. The dialog has a title bar with 'Satellite Data' and a close button. Inside, the text 'Downloading Satellite Data...' is displayed above a progress bar. The progress bar is partially filled with green. At the bottom is a 'Cancel' button.

Satellite Data

Downloading Satellite Data...

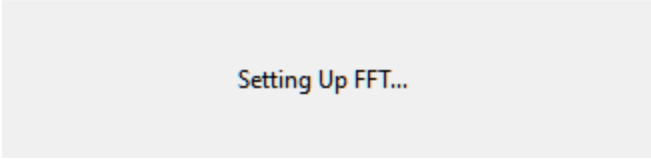
Cancel

Wait until the data are downloaded and imported. Again, SkyRoof cannot proceed without this data downloaded at least once, so if you click on Cancel, the program terminates.

FFT Setup

Wait for SkyRoof to try different ways of computing the FFT transform and to find the one that works best on your system. This may take quite some time!

Please Wait



Setting Up FFT...

That's all for the quick start! Now you can use the program for tracking the satellites in frequency and space, and for predicting the satellite passes over your location. To do more than that, you have to perform the rest of the setup steps described in the next sections.

Creating Satellite Groups

SkyRoof comes with two pre-defined groups of satellites created for your convenience, **Active Ham Sats** and **CW Beacons**. The first group lists the satellites carrying the linear transponders, FM repeaters or digital systems that were available to Hams at the time of this writing. The second group includes the satellites that send beacon signals or telemetry in Morse Code, or just transmit an unmodulated carrier (a.k.a. Continuous Wave, CW). Most likely, you will want to modify or delete these groups and add your own ones. Here is how to do this.

Click on **Tools / Satellites and Groups** in the main menu to open the **Satellites and Groups window**:

Satellites and Groups

Satellites

Status: ☒ Alive ☒ Future ☐ Re-Entered Bands: ☒ VHF ☒ UHF ☐ Other

Radio: ☒ Transponder ☒ Transceiver ☒ Transmitter Service: ☒ Ham ☐ Non-Ham

Search: Updated 2025-05-26

Name	NORAD ID	Launched	Service
Bufeng-1A	44312		
BUGSAT-1	40014	2014-06-19	Amateur
BUZZER-1	98693	2025-01-14	
CAKRA-1	98861	2024-08-16	
CALIPSO	29108	2006-04-28	Meteorological
CANX-4	40055	2014-06-30	Amateur
CANX-5	40056	2014-06-30	Amateur
CANYVAL-C 1U	99732	2021-03-22	
CANYVAL-C 2U	99731	2021-03-22	
CAPE-1	31130	2007-04-17	Amateur
CARBONITE-1	40710	2015-07-10	

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Satellite Groups

- Active Ham Sats**
 - AO-73
 - RS-44
 - MO-122
 - FO-29
 - AO-7
 - AO-91
 - ARISS
 - JO-97
 - SO-50
 - SO-124
 - SONATE
 - AO-123
- CW Beacons**
 - KKS-1
 - CUTE-1
 - KOSEN-1
 - Veronika
 - SEEDS II

OK Cancel

The left panel lists [all satellites](#) known to SkyRoof, the right panel shows the groups.

- to create a group, click on the **[+]** button, then enter the group name;
- to add a satellite to the group, drag it from the satellite list onto the group, or click on the **[>]** button;
- to delete a group, or a satellite from the group, select it in the right panel and press the Delete key, or click on the **[<]** button.
- click on OK to save the changes.

The **Satellites and Groups window** has many commands to filter and search satellites, to rename them, and to view detailed information about the satellites and their transmitters. These commands are described in the [Satellites and Groups Window](#) section of this document.

Configuring the Window Layout

The layout of SkyRoof's main window is under your full control. Any panel may be shown or hidden, docked anywhere in the window, or left floating.

Show and Hide

Show the panels using the menu commands in the **View** section, hide them using the same command again, or by clicking on the Close button on panel's caption bar.

Docking

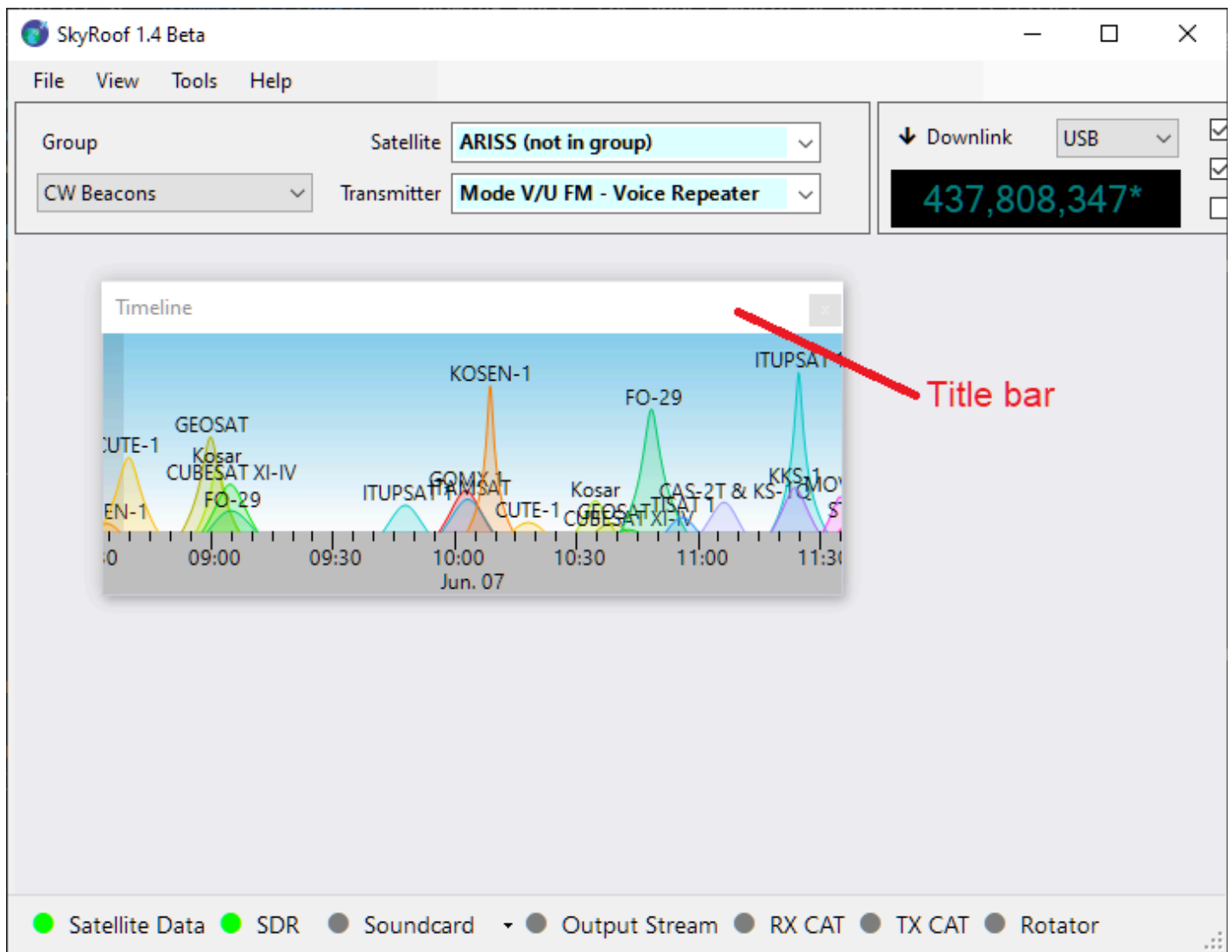
The panels you open are initially in the floating state. This is not very convenient as you cannot move or resize the main window without breaking your panel arrangement. You can dock the panels so that they move and resize when you move/resize the main window.

How to Dock a Floating Panel

The [Timeline](#) panel in the screenshot below is floating. Let us dock it to the bottom of the main window.

1. Find the Panel's Title Bar

Locate the top bar of the floating panel labeled "*Timeline*". This is known as the **title bar**.



2. Click and Hold

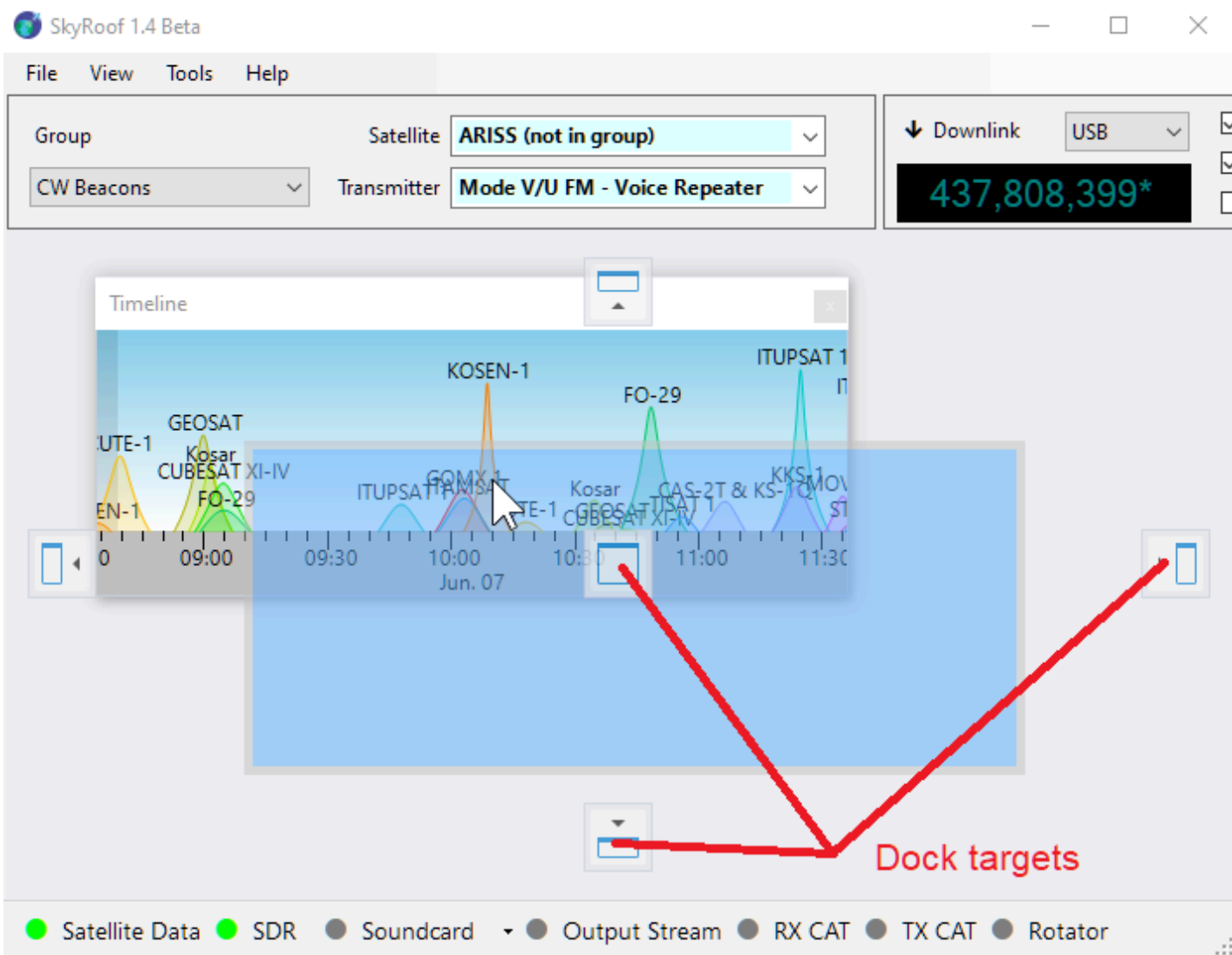
Move your mouse cursor to the title bar and press the **left mouse button**. Keep holding the button down.

3. Drag the Panel

While holding the mouse button, move the panel by dragging it with the mouse. As you begin to drag, you will see **dock target icons** appear in the main window—these icons represent the available docking positions.

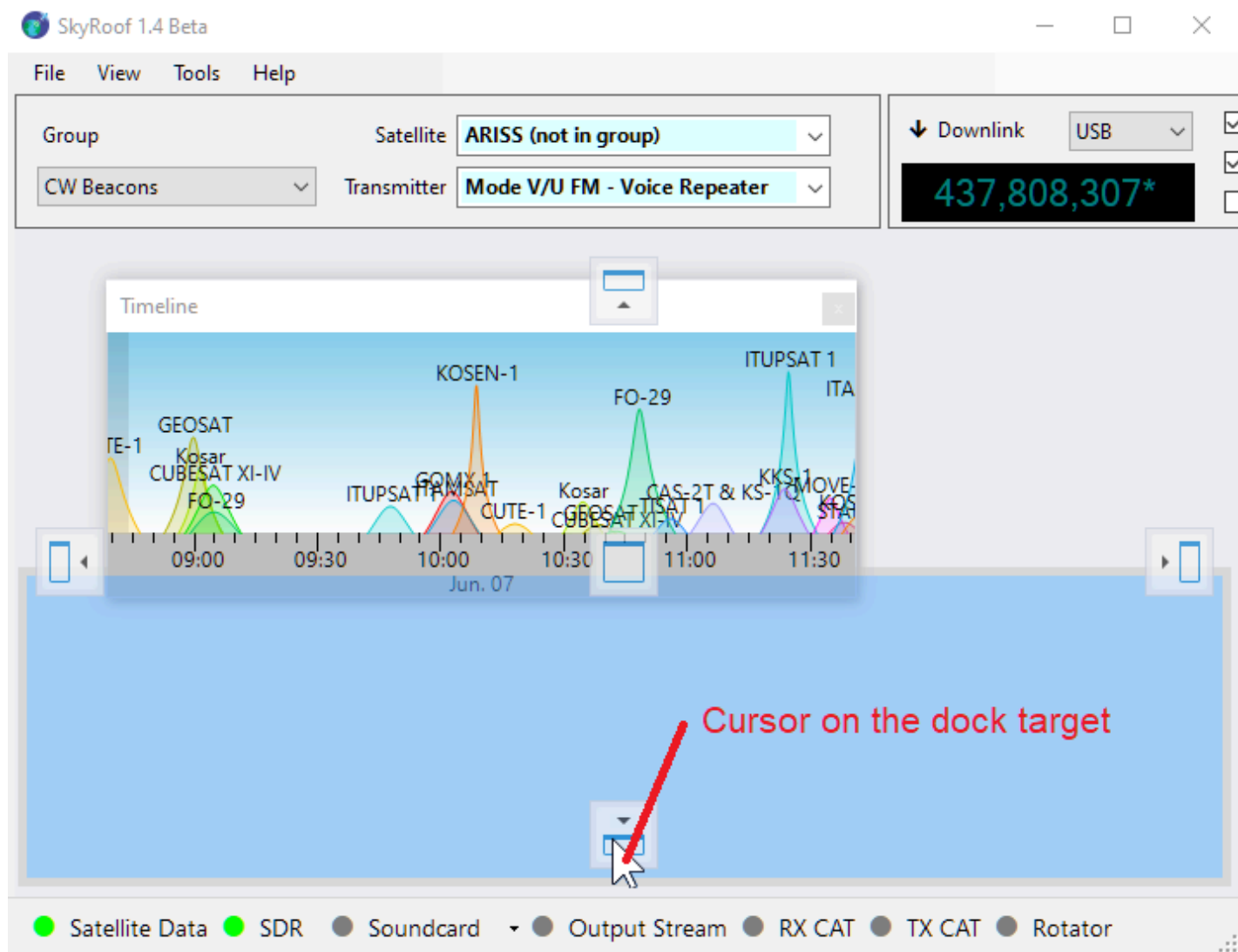
You will also notice a **dimmed rectangle** showing where the panel would be docked if released.

⚠ Important: This rectangle is only a visual preview. Do **not** try to align it with the dock target icon. Instead, focus on where your mouse cursor is.



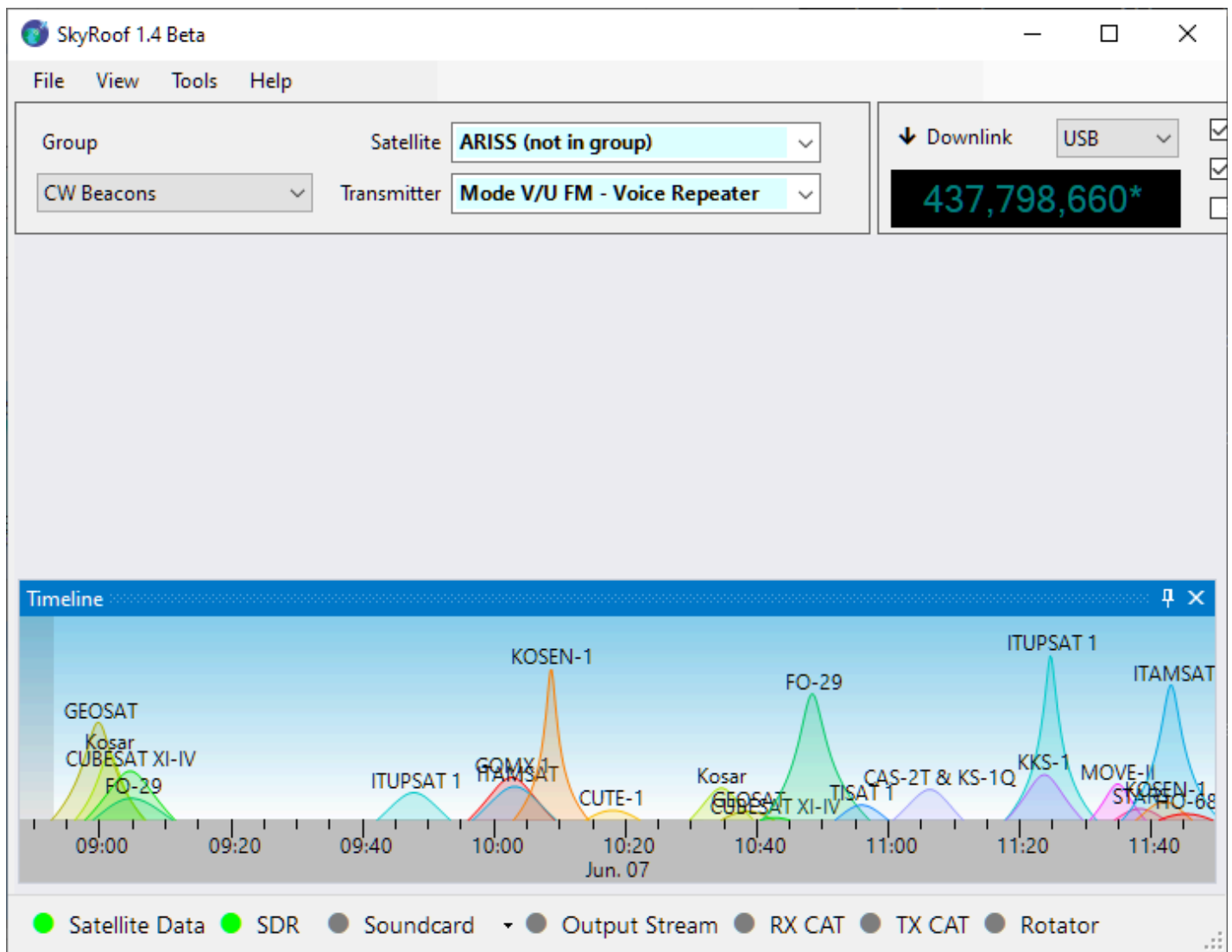
4. Position the Mouse Cursor Over a Dock Target Icon

Move your **mouse cursor** over one of the dock target icons (not the rectangle). The docking location will update automatically. The panel will only dock if the cursor is directly over the icon.



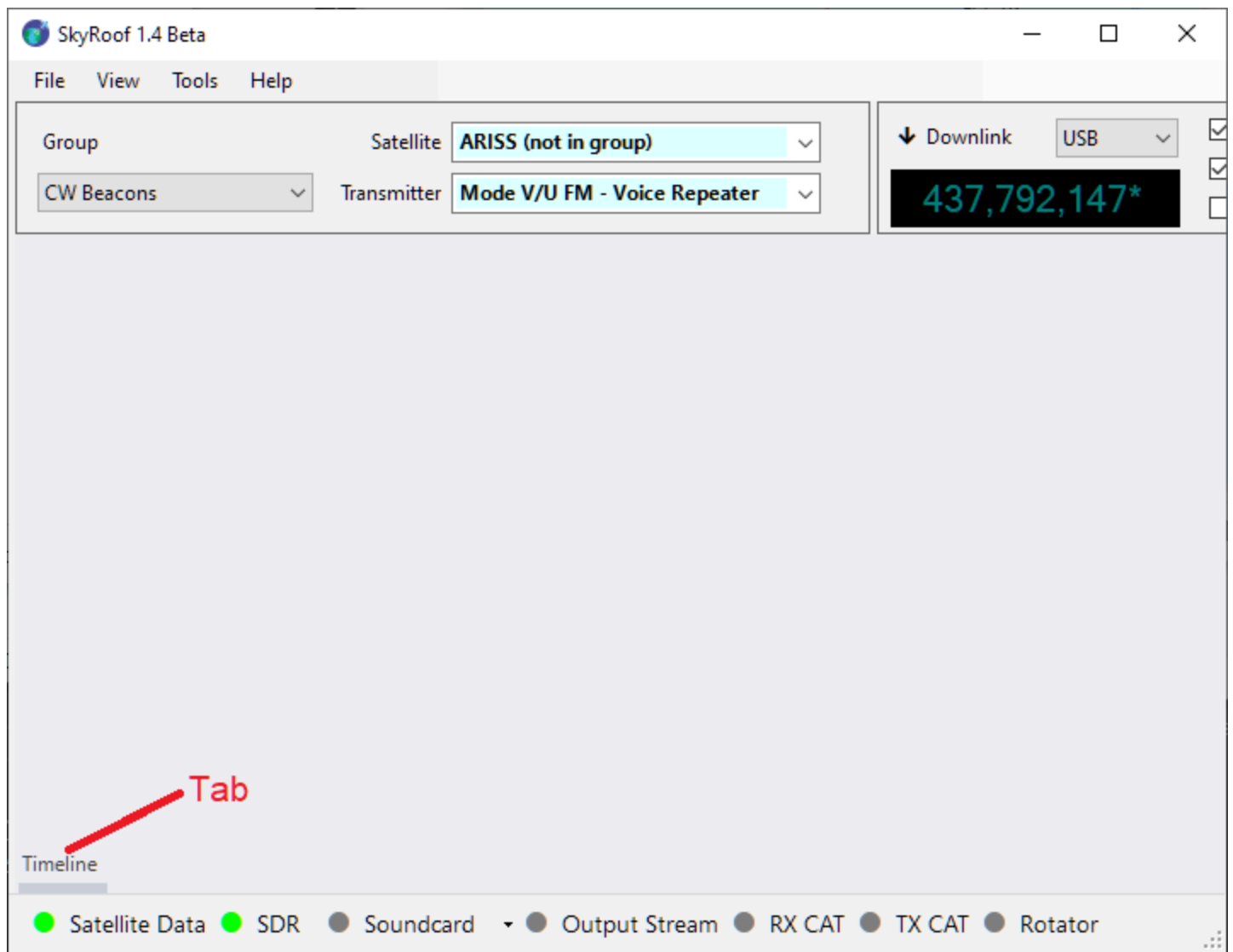
5. Release to Dock

Once the cursor is over your desired dock target icon, **release the mouse button**. The panel will snap into place — either at the side, in the center, or nested inside another panel, depending on the selected icon. Once docked, the panel becomes part of the main window layout, helping keep your workspace clean and organized.



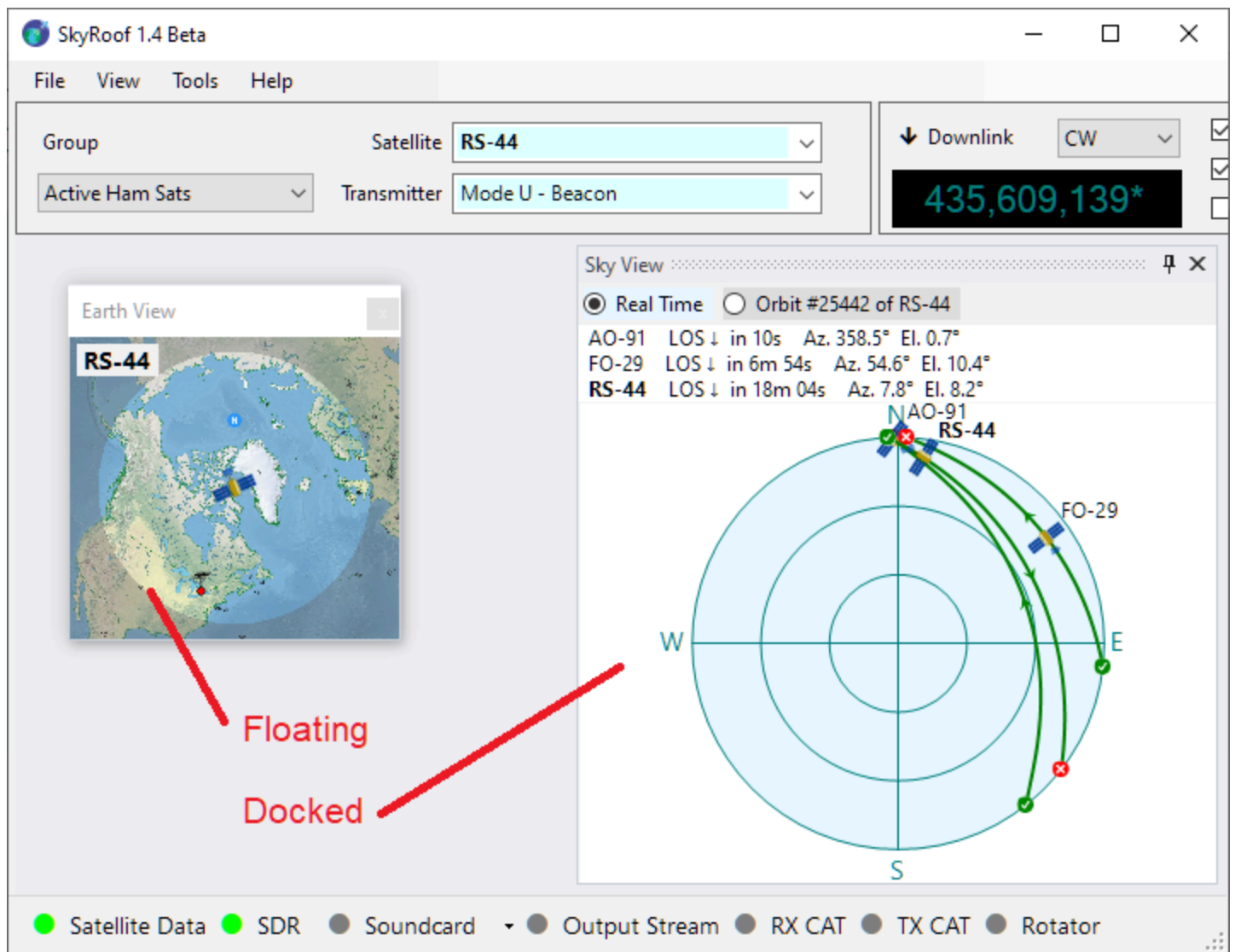
Auto-Hide

A panel can be switched to the **auto-hide mode**, which allows it to stay hidden until you click on its tab. In the screenshot below, notice the **"Timeline" tab** in the bottom-left corner. Clicking this tab temporarily slides the **Timeline** panel into view. To enable auto-hide mode, click the **Auto-Hide** button  located on the panel's title bar.

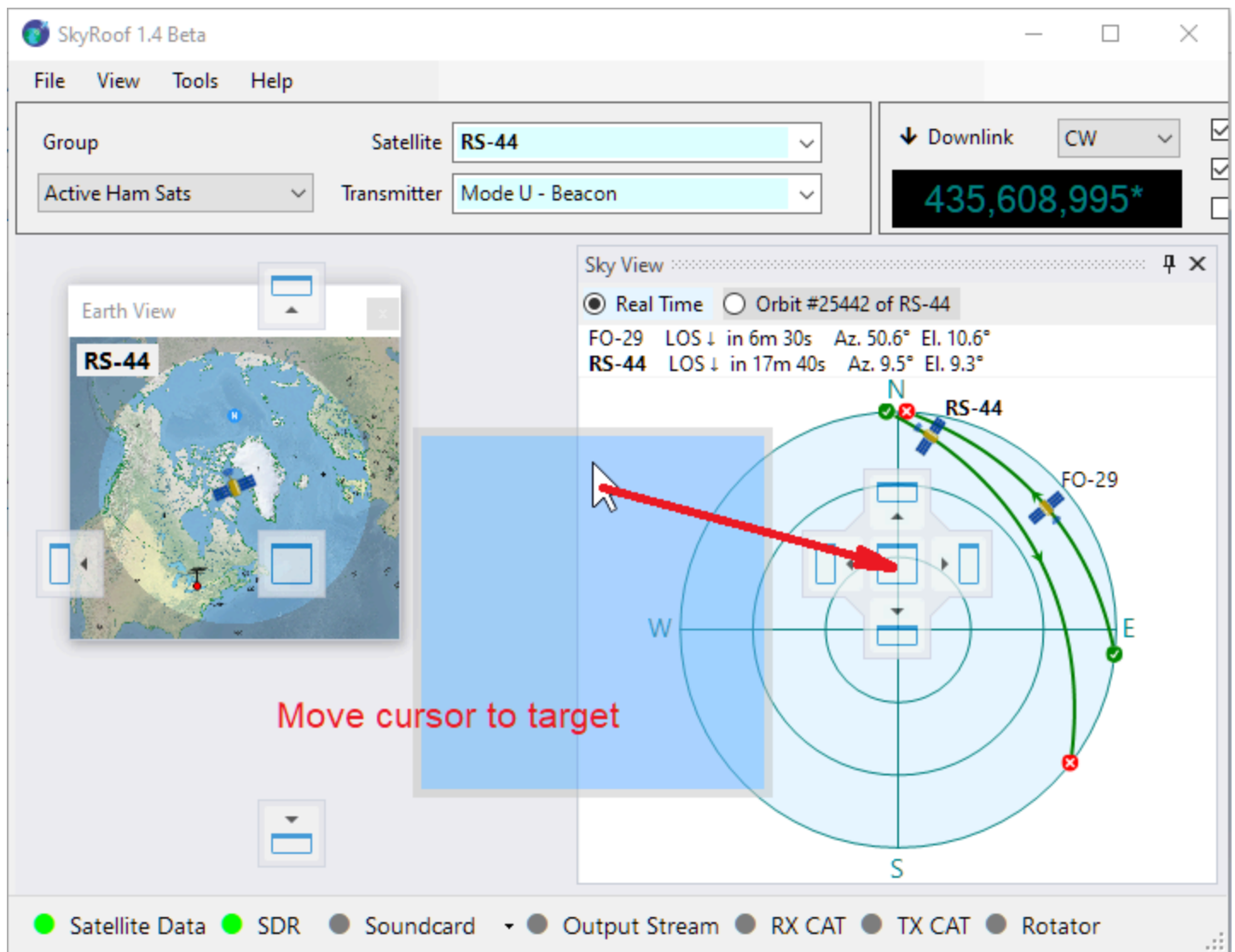


Tabbing

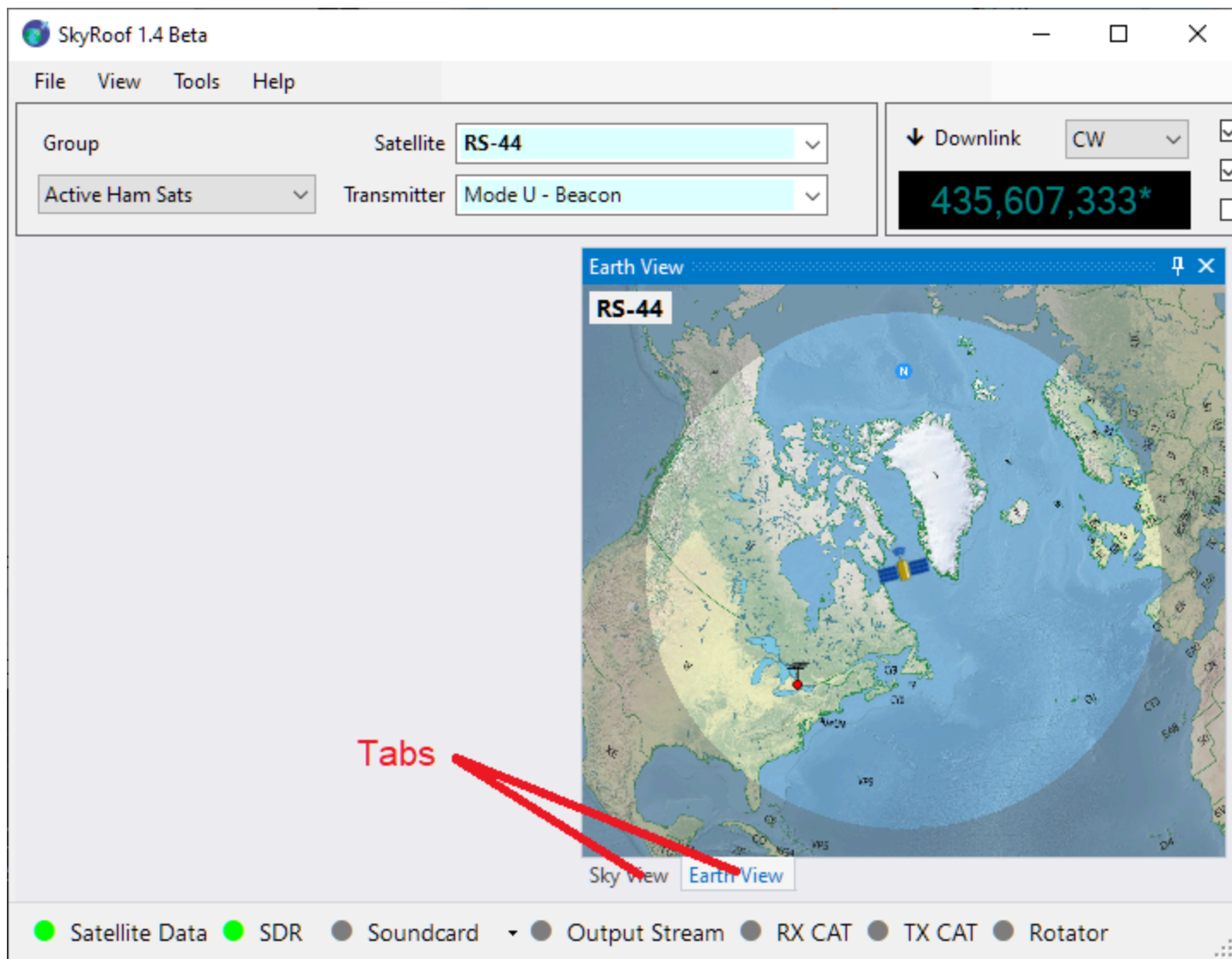
You can also organize panels as **tabs**.



If you drop one panel directly onto another, they will be grouped into a tabbed interface.



In the screenshot below the [Sky View Panel](#) and [Earth View Panel](#) have been combined into a tabbed view. You can switch between these panels by clicking on their respective tabs.



Setting Up SDR

Supported Radios

SkyRoof uses the [Soapy SDR](#) engine to interface with the SDR radios. Currently it supports:

- Airspy;
- AirspyHF+;
- SDRplay;
- RTL-SDR;
- HackRF;
- PlutoSDR;

NOTE

It may be possible to add support of other SDR devices to SkyRoof. Contact me if you have an unsupported SDR and are willing to do extensive testing.

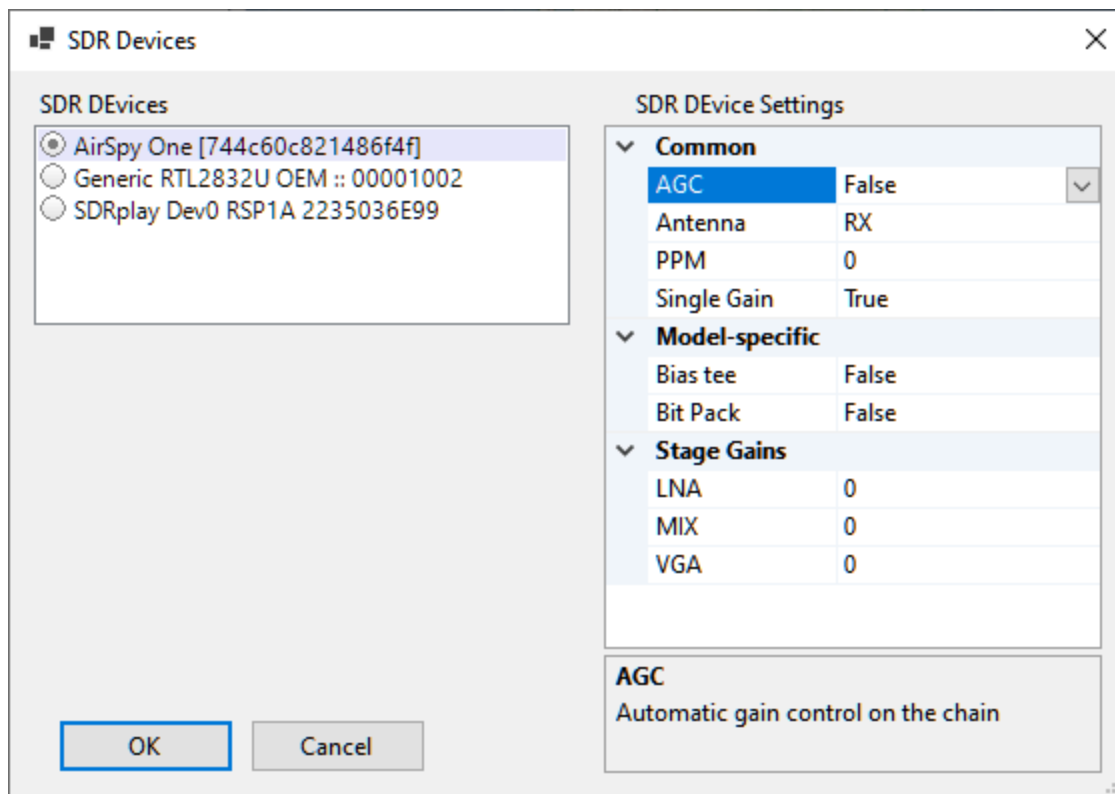
Installing The Drivers

Most of the SDR devices require the driver to be installed before you can start using them. Check the manufacturer's web site, or search on Google, for the driver installation instructions. At the time of this writing, the following instructions were available on the Web: [Airspy](#), [RTL-SDR](#), [SDRplay](#), [HackRF](#), [PlutoSDR](#).

Once you install the drivers and make your radio work with its native software, proceed to the next step.

Selecting an SDR device

Connect your SDR device to the computer, then click on **Tools / SDR Devices** in the main menu. This will open the **SDR Devices dialog**:



All active SDR devices are listed on the left panel. Click on the one that you want to use.

Configuring the device

The right panel shows all settings that the device driver understands. The setting names and descriptions (shown on the bottom panel) come from the driver, with two exceptions described below. For information about these settings see the documentation that comes with the radio.

The two settings, common to all radios, are:

- **PPM** - the correction factor for the SDR clock frequency, expressed in parts per million. This setting is important for the correct operation of the Doppler tracking algorithm, see the [Calibrating PPM Correction](#) section for details;
- **Single Gain** - when set to true (**and AGC is off**), the SDR gain is controlled by the **RF Gain** slider on the toolbar. This is the recommended setting. When it is set to false, the settings in the **Stage Gains** are applied to the individual stages of the SDR, and the gain slider is disabled.

Using an SDR with a Transverter

If you connect your SDR to the IF output of a VHF/UHF transverter (so the SDR is tuned to 28–30 MHz while it actually receives 144/432 MHz signals), enable the **SDR Offset** in the **Transverter** section of the Settings. SkyRoof will tune the SDR to the IF band while the

waterfall scale and the frequency display continue to show the actual satellite RF. See [Setting Up Transverter](#).

Using Remote SDR

SkyRoof can use SDR devices connected to a remote (or local) computer via the [SoapyRemote](#) driver. To enable remote access to SDR:

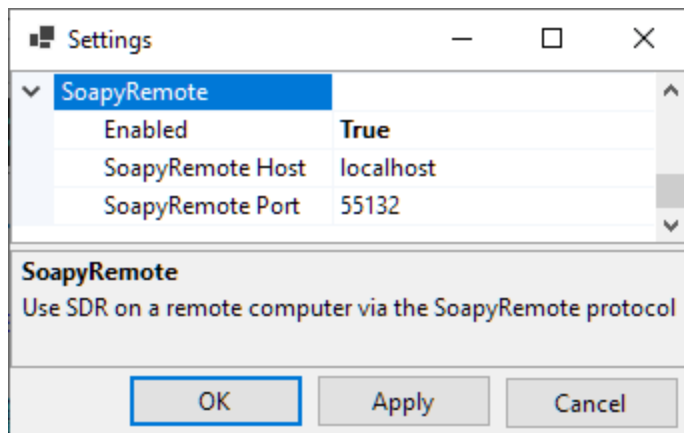
- **On the remote computer:**

- install SoapySDR which is available as part of [PothosSDR](#);
- run the remote server that comes with SoapySDR:

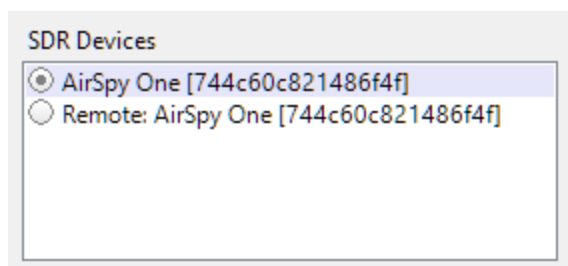
```
SoapySDRServer.exe --bind
```

- **In SkyRoof:**

- enable **SoapyRemote** in the **Settings** dialog;
- enter the **host** name of the remote computer, or leave "localhost" if the radio is on the same computer.



- Open the **SDR Devices** dialog and select the remote SDR device from the list. If the device and remote server are on the localhost, you will have a choice between a direct connection to the radio and a connection via the server.



Calibrating PPM Correction

Motivation

The clock frequency of an SDR, as it comes from the factory, is rarely accurate. Typical errors are in the range of a few PPM (parts per million), which translates to a tuning error of 1-2 KHz on the 70 cm band. For accurate tracking of the satellite signals this error must be calibrated out. The calibration process is simple, we just find a signal of known frequency, check on what frequency it appears on the waterfall, and compute the PPM correction factor from the difference between the two.

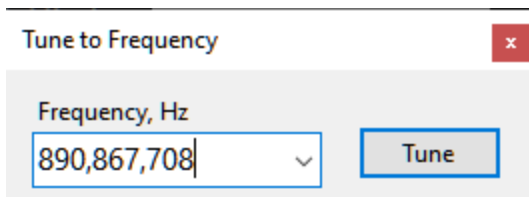
There is plenty of signals on the air that may be used for calibration, if one knows what to look for. One of such signals is the [FCCH channel](#) of a [GSM](#) downlink. This channel is located 67,708 Hz above the center frequency of a GSM channel, and the accuracy of its frequency is claimed to be better than 0.05 PPM.

3-rd Party Software

For the RTL-SDR dongles you can use the [Kalibrate](#) utility that performs such calibration automatically. For other radios follow the steps below.

Steps

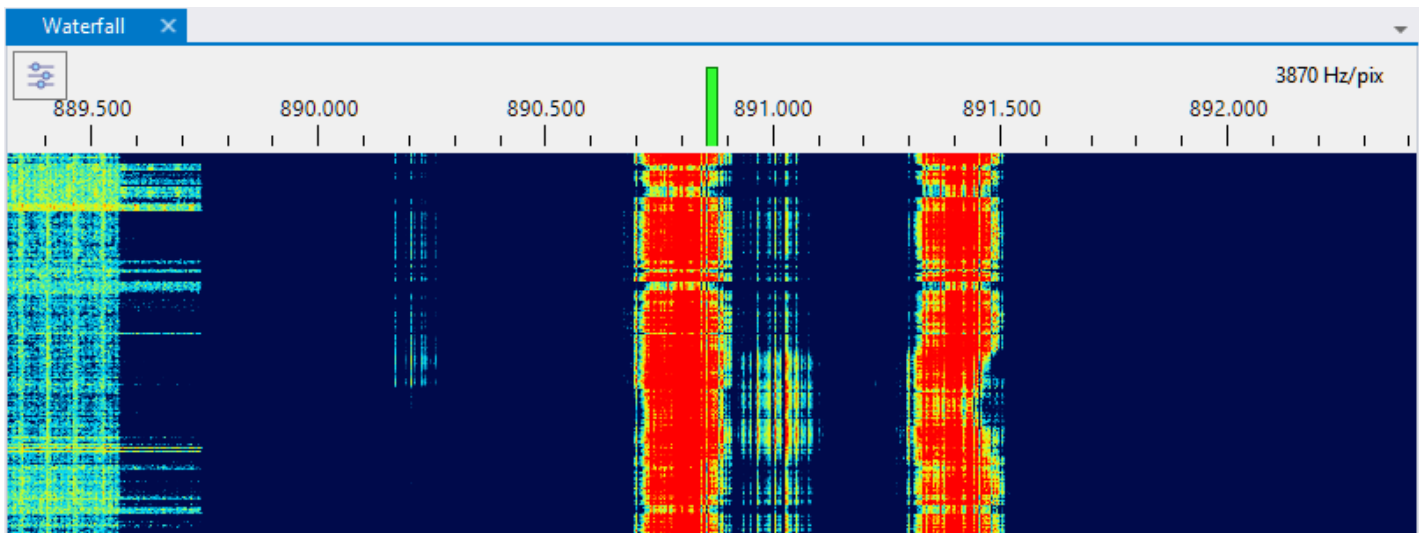
1. Find a strong GSM signal, or any other signal of known frequency. In my area one of such signals is present on 890.8 MHz.
2. Click on the Downlink frequency display in the [Frequency Control](#) panel on the toolbar to open the frequency entry dialog:



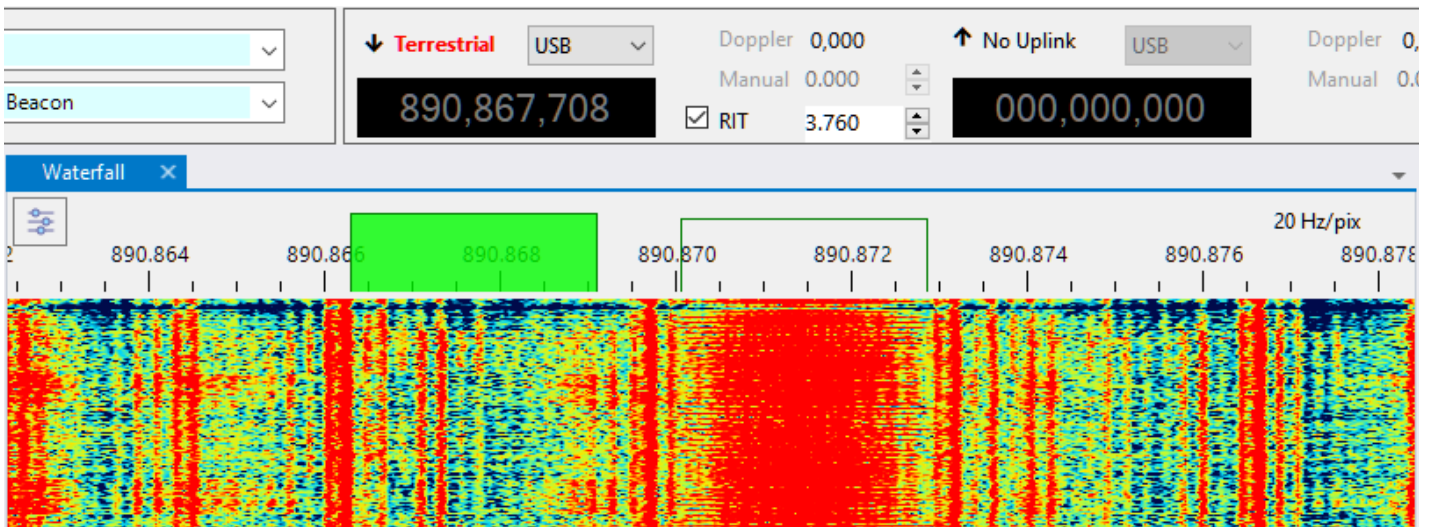
3. Enter the frequency of the channel plus the FCCH offset:

$$890,800,000 + 67,708 = 890,867,708 \text{ Hz}$$

4. Click on the Tune button in the dialog and verify that the SDR is tuned to the desired frequency:



5. Zoom in by spinning the mouse wheel over the waterfall display:
6. Find the FCCH signal. In the screenshot below it is about 4 kHz above the expected frequency:



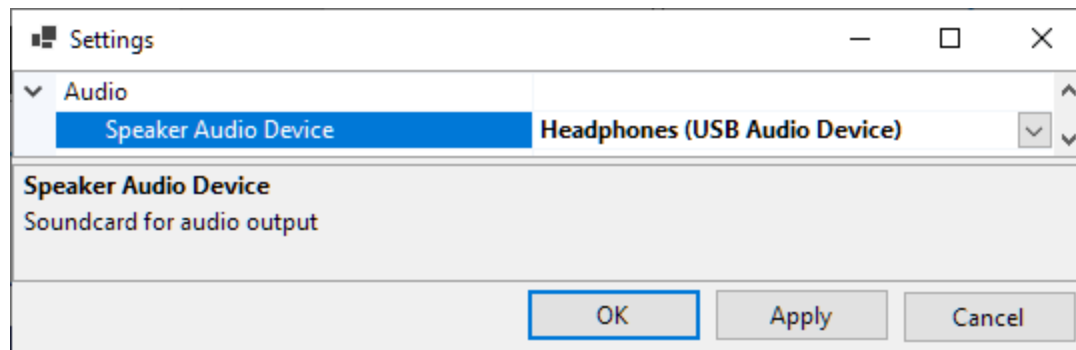
7. Now let us measure the offset between the receiver frequency (the center of the green rectangle that represents the receiver passband) and the FCCH frequency. Tick the **RIT** checkbox on the **Frequency Control panel** and adjust the RIT offset until the RIT passband (the clear rectangle) aligns with the signal. You can tune RIT in many different ways, as described in the [Frequency Control](#) and [Frequency Scale](#) sections. For now, just use the up/down buttons in the RIT offset box, or spin the mouse wheel over that box.
8. Compute the PPM correction. The frequency error measured in the previous step is 3,760 Hz, so the PPM is:

$$3,760 / 890,867,708 * 1e6 = 4.22 \text{ PPM}$$

9. Now enter this value in the [SDR Devices dialog](#), and you are done.

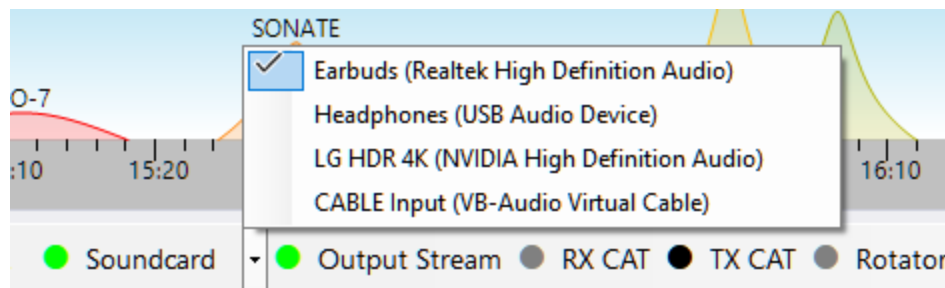
Setting Up Audio

Click on **Tools / Settings** in the main menu to open the [Settings window](#):



- **Speaker Audio Device** - select the audio device that will be used to output the audio received with SDR;
- **FM Squelch** enable or disable squelch in the FM mode.

The output to the soundcard can be toggled by clicking on the **Soundcard** label on the status bar. A drop-down list next to **Soundcard** allows switching between the audio devices:

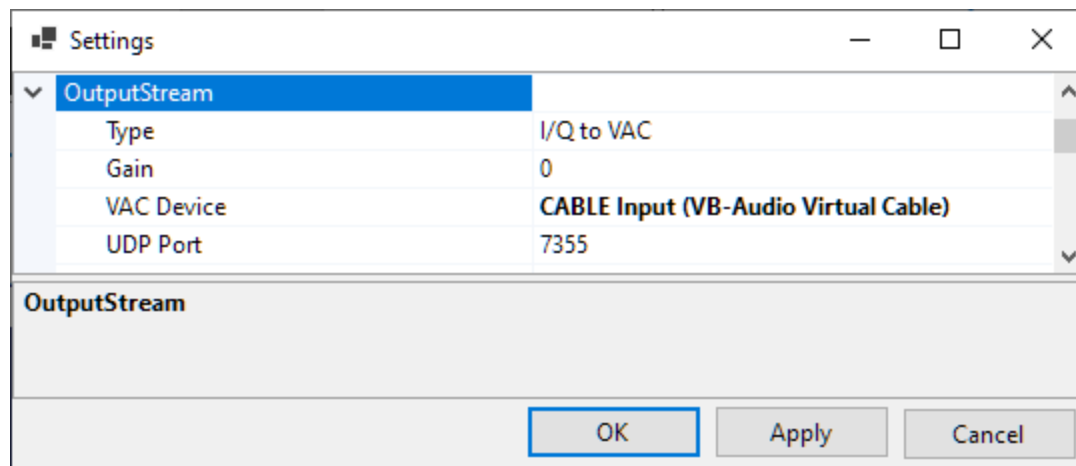


Setting Up Output Stream

SkyRoof can optionally send the raw I/Q data or demodulated audio, either to a Virtual Audio Cable (VAC) or as a stream of UDP packets. The data are sent as 32-bit floating point values in the IEEE 754 format. The sampling rate is 48 kHz in all streaming modes.

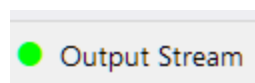
Configuring

Click on **Tools / Settings** in the main menu to open the [Settings window](#):



- **Type** - select the stream type:
 - I/Q to VAC;
 - Audio to VAC;
 - I/Q to UDP;
 - Audio to UDP.
- **Gain** - gain or attenuation, in dB, that will be applied to the streamed data;
- **VAC Device** - the Virtual Audio Cable device to use;
- **UDP Port** - the UDP port number to use.

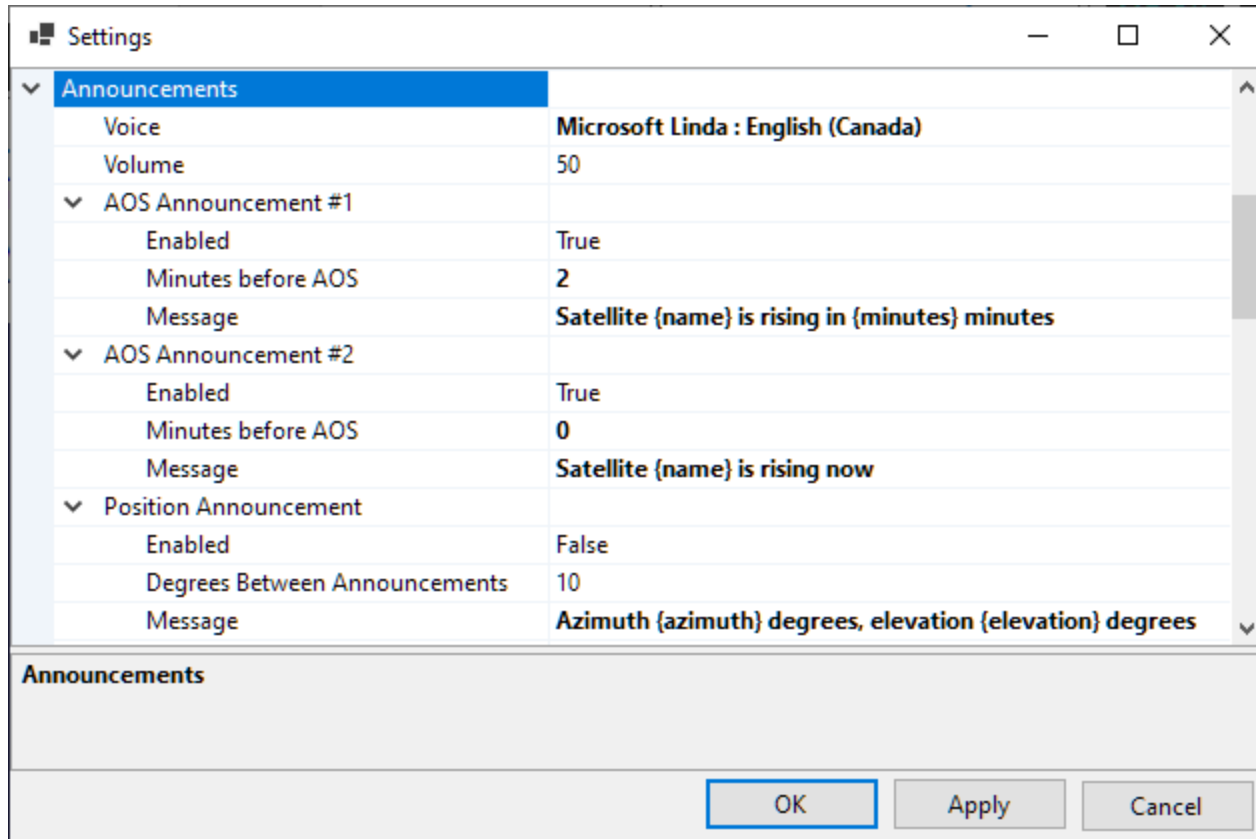
Output streaming can be turned on and off by clicking on the **Output Stream** label on the status bar:



Setting Up Voice Announcements

SkyRoof can make voice announcements of the the satellite AOS events and position changes. Up to two AOS announcements may be enabled.

Click on **Tools / Settings** in the main menu to open the [Settings window](#):



The screenshot shows a 'Settings' window with a tree view on the left and a corresponding configuration table on the right. The 'Announcements' section is expanded in the tree view. The configuration table has the following settings:

Setting	Value
Voice	Microsoft Linda : English (Canada)
Volume	50
AOS Announcement #1	
Enabled	True
Minutes before AOS	2
Message	Satellite {name} is rising in {minutes} minutes
AOS Announcement #2	
Enabled	True
Minutes before AOS	0
Message	Satellite {name} is rising now
Position Announcement	
Enabled	False
Degrees Between Announcements	10
Message	Azimuth {azimuth} degrees, elevation {elevation} degrees

At the bottom of the window, there are three buttons: 'OK', 'Apply', and 'Cancel'.

- **Voice** - select one of the voices available on your system. To install a new voice package in Windows, go to **Settings > Time & language > Speech** and then select **Add voices** to download and install the desired voice package.
- **Volume** - set the volume between 1 and 100;
- **Enable** - enable or disable the announcement;
- **Minutes Before AOS**: enter 0 to 5 minutes;
- **Degrees Between Announcements**: 1° to 30°. Satellite position is announced when the angular distance between the previous and current positions exceeds this value;
- **Message** - enter the announcement message. For the satellite name enter {name}, for the number of minutes before AOS enter {minutes}, for the azimuth and elevation enter {azimuth} and {elevation} respectively.

Setting Up CAT Control

SkyRoof uses an external program, either `skycatd.exe` from SkyCAT or `rigctld.exe` from HamLib, to control the transceiver. In both cases the CAT control commands are sent using the TCP protocol, so the radio may be moved to a remote computer and controlled via the network if desired.

SkyCAT

SkyCAT is specifically designed for the best satellite tracking experience. It has an open architecture where support of the new radio models is added by creating the command definition files with the corresponding commands. Check the [SkyCAT web site](#) for the list of currently supported radios. If your radio model is not yet supported, either create a command definition file for it using the instructions on the SkyCAT web site, or use `rigctld.exe` instead (see below).

Using skycat.exe

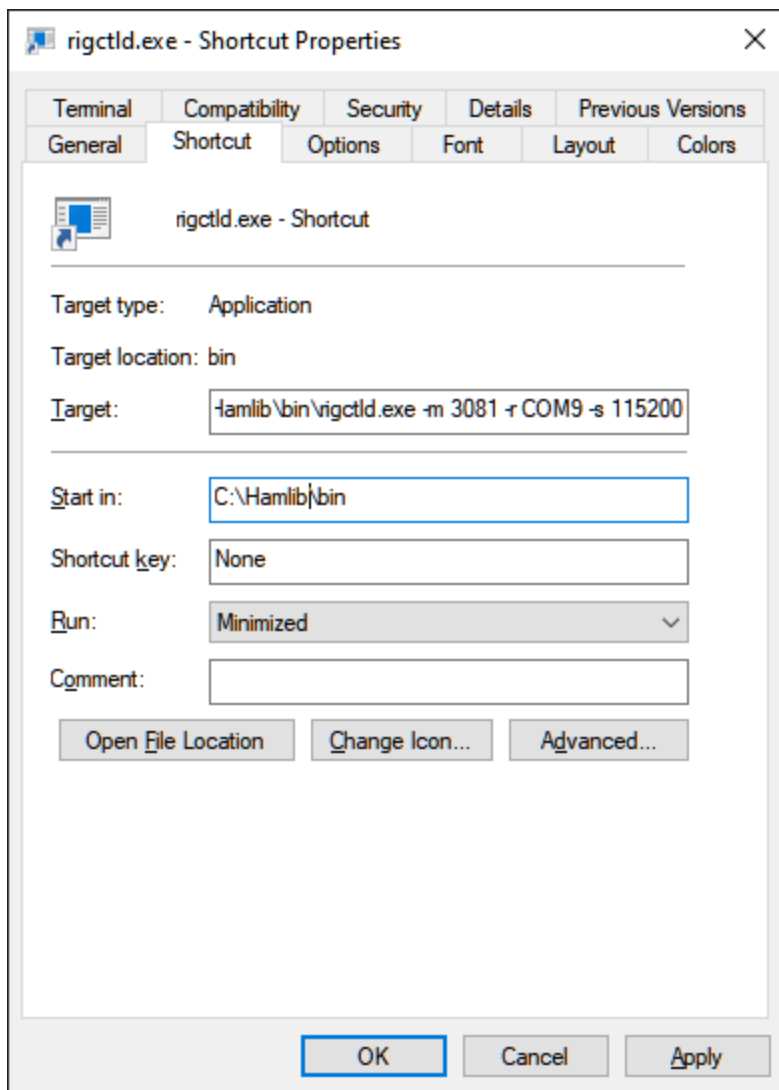
If your radio is supported by SkyCAT, use `skycatd.exe`, a command line program that comes as part of the SkyCAT package. To start using it, follow the [setup instructions](#) on the SkyCAT web site.

The latest command definition files are available [here](#). Update the file for your radio before you proceed.

Using rigctld.exe

If a SkyCAT command definition file for your transceiver is not yet available, use **`rigctld.exe`**, a HamLib-based CAT control daemon. Note, however, that some commands may not work properly with `rigctld.exe`.

1. Download **`hamlib-w64-4.5.5.exe`** [from GitHub](#). Other versions may not work correctly.
2. Run the downloaded file to install HamLib, note the folder where it is installed.
3. Create a shortcut to start `*rigctld.exe`, with command line arguments:



The arguments on the command line must be tailored for your specific radio and COM port settings. Refer to the [rigctld documentation](#) for a complete description of the arguments.

Assuming that HamLib is installed in the default location, here is an example string for the shortcut:

```
"C:\Program Files\hamlib-w64-4.5.5\bin\rigctld.exe" -m 3081 -r COM9 -s 115200
```

In the string above the following arguments are used:

- **-m 3081** - the radio model; 3081 is the Id of IC-9700 (see the [list of id's](#));
- **-r COM9** - the COM port used by the radio. In this case, the USB connection to IC-9700 creates two virtual COM ports, COM9 and COM10. The port with the lower number is used for CAT;
- **-s 115200** - use the highest available COM port speed;

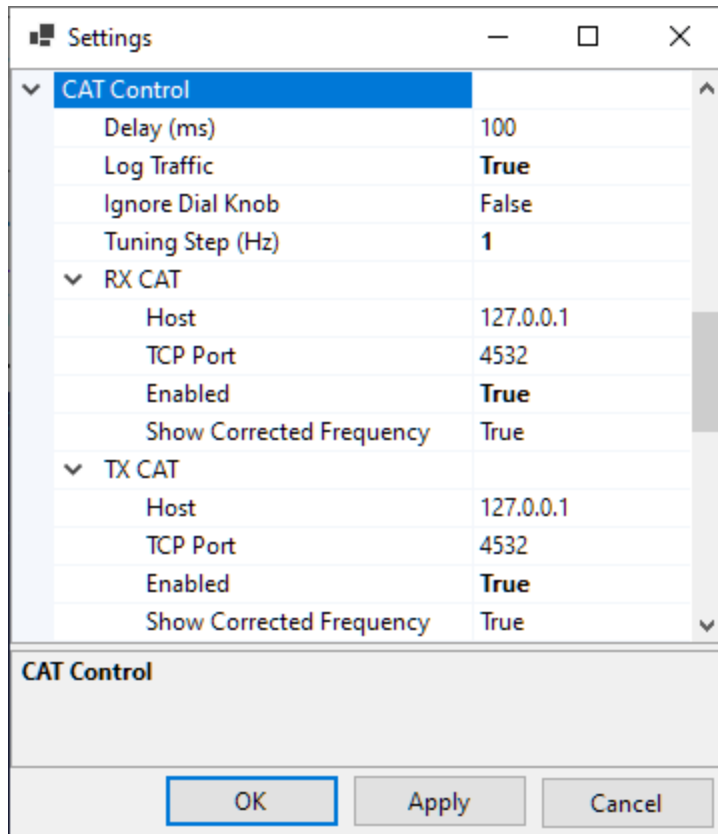
- **-vvvvv** - optional, writes detailed information to the console window. Useful for troubleshooting.

4. Run rigctld.exe using this shortcut before you enable CAT control in SkyRoof.

Settings

The CAT Control settings in the [Settings dialog](#) are the same for skycatd.exe and rigctld.exe.

Click on **Tools / Settings** in the main menu to open the **Settings dialog**:



- **Delay** determines how often SkyRoof sends commands to the radio. The default delay of 100 ms is good in most cases. Increase the delay if your radio's CAT interface is slow;
- **Log Traffic** should be set to False and enabled only for debugging;
- **Ignore Dial Knob** - by default, CAT control allows you to change the frequency both in the program and by spinning the dial knob. If for some reason this causes trouble, change this setting to True, so that the dial knob rotation is ignored.

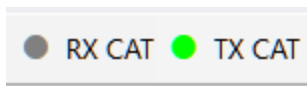
The two sections in the Settings, **RX CAT** and **TX CAT**, allow you to use either the same radio for RX and TX, or two different radios. You can also enable only one of those, or disable both. The recommended configuration is to use an SDR for reception and a transceiver for transmission, in this case RX CAT should be disabled.

To use the same radio for RX and TX, set **Host** and **TCP Port** to the same values in both sections.

To use two different radios, create a second shortcut for the second radio, and specify a different port number on the command line. Enter this port number in the settings as well, and run two instances of **rigctld.exe** using both shortcuts.

The settings in the RX and TX sections are:

- **Host** - should be "127.0.0.1" or "localhost" if skycatd or rigctld is running on the same computer as SkyRoof. It may be changed to a different address for remote control;
- **TCP Port** - 4532 is the default port used by skycatd and rigctld. Use a different port in one of the sections to control different radios for RX and TX;
- **Enabled** - enable or disable CAT. Another way to toggle CAT is to click on the CAT labels on the status bar:



- **Show Corrected Frequency** - The SkyRoof can display either the nominal frequency of the satellite transmitter, or the frequency with all corrections applied. Another way to toggle this setting is via the right-click menu on the frequency display widget on the toolbar.

Using CAT with a Transverter

If your transceiver does not understand VHF/UHF frequencies (e.g., a "dumb" HF rig connected via a transverter, or an HF rig such as the IC-7300), enable the **RX CAT Offset** and / or **TX CAT Offset** in the **Transverter** section of the Settings. SkyRoof will then send the IF frequency (e.g., 29.950 MHz) to the radio instead of the satellite RF (e.g., 145.950 MHz). Modern rigs with built-in XVRT support (e.g., the K3S) do their own conversion internally and should leave these offsets **disabled**.

When a CAT offset is enabled but no transverter band covers the current frequency, the **RX CAT** or **TX CAT** label on the status bar turns **yellow** and SkyRoof skips the CAT command rather than send a frequency the radio cannot tune. Hover over the label to see the explanation.

See [Setting Up Transverter](#).

Model-Specific Notes

IC-9700

- set **CI-V USB Port** in the transceiver menu to **Unlink from REMOTE**;
- note that the radio is used in the Dual Watch mode, not in the Sat mode, so the upper frequency on the transceiver screen is uplink, the lower one is downlink.

IC-991A

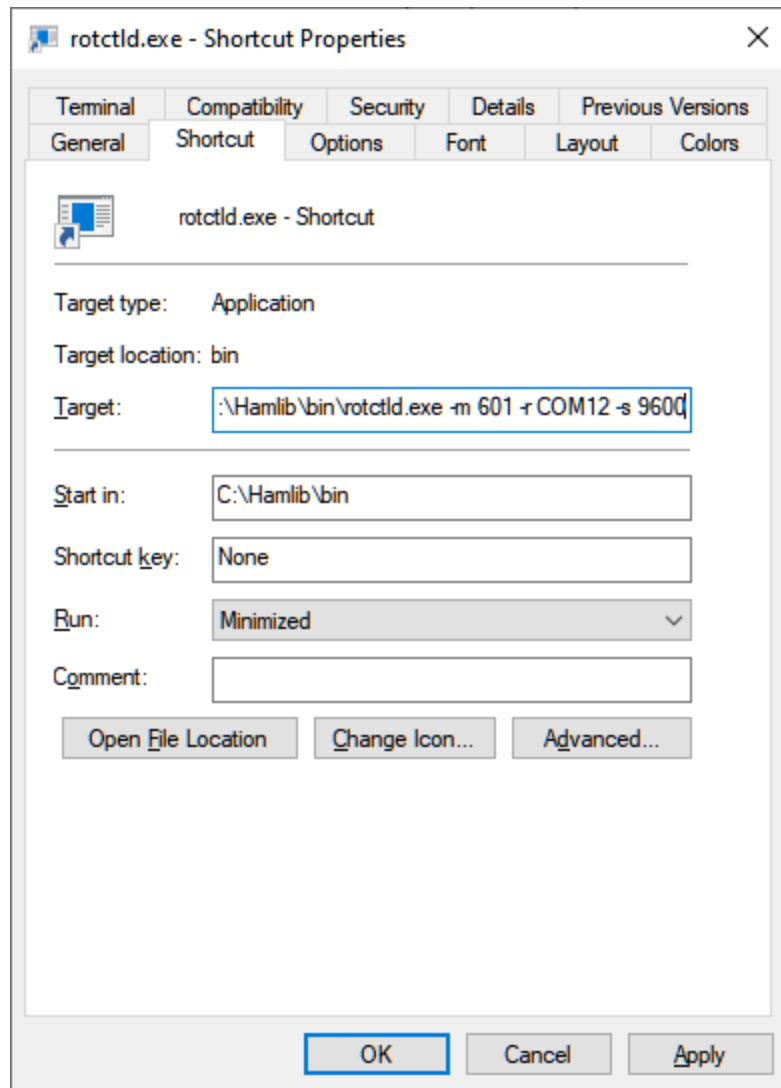
- set **CAT RTS** in the transceiver menu to Disabled.

Setting Up Rotator Control

rotctld.exe

SkyRoof uses **rotctld.exe**, a HamLib-based rotator control daemon, to control the antenna rotator. See the [Setting Up CAT Control](#) section for the instructions how to download and install HamLib.

Create a shortcut to start **rotctld.exe*, with command line arguments:



The arguments on the command line must be tailored for your specific rotator and COM port settings. Refer to the [rotctld documentation](#) for a complete description of the arguments.

Assuming that HamLib is installed in the default location, here is an example string for the shortcut:

```
"C:\Program Files\hamlib-w64-4.5.5\bin\rotctld.exe" -m 601 -r COM12 -s 9600
```

In the string above the following arguments are used:

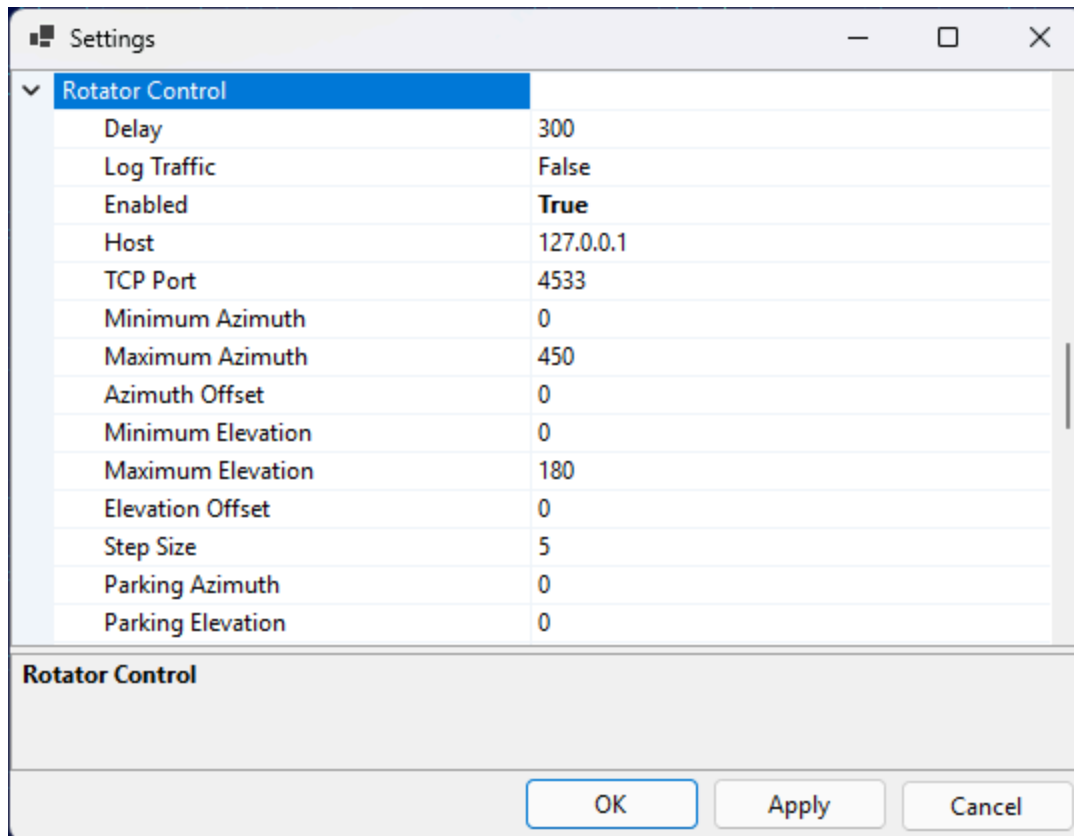
- **-m 601** - the rotator model is Yaesu GS-232A;
- **-r COM12** - the COM port used by the rotator;
- **-s 9600** - the COM port speed.

Run rotctld.exe using this shortcut before you enable rotator control in SkyRoof.

The list of rotator models supported by rotctld.exe is available on the [HamLib web site](#). Note that one of the models is **PSTRotator**. When this model is selected, rotctld.exe just passes the rotation commands to the PST Rotator software.

Settings

Click on **Tools / Settings** in the main menu to open the [Settings dialog](#):



The screenshot shows the 'Settings' dialog box with the 'Rotator Control' tab selected. The dialog has a title bar with 'Settings' and standard window controls. The 'Rotator Control' tab is expanded, showing a list of settings. Below the list is a section labeled 'Rotator Control' with a description area, and at the bottom are 'OK', 'Apply', and 'Cancel' buttons.

Setting	Value
Delay	300
Log Traffic	False
Enabled	True
Host	127.0.0.1
TCP Port	4533
Minimum Azimuth	0
Maximum Azimuth	450
Azimuth Offset	0
Minimum Elevation	0
Maximum Elevation	180
Elevation Offset	0
Step Size	5
Parking Azimuth	0
Parking Elevation	0

Rotator Control

OK Apply Cancel

- **Delay** determines how often SkyRoof sends commands to the rotator. The default delay is 300 ms, but you can set it to a lower value, such as 100 ms, without adverse effects;
- **Log Traffic** should be set to False and enabled only for debugging;
- **Enabled** - enable or disable rotator control. Another way to toggle the rotator control is to click on the Rotator label on the status bar;
- **Host** - should be "127.0.0.1" or "localhost" if rotctld is running on the same computer as SkyRoof. It may be changed to a different address for remote control;
- **TCP Port** - 4533 is the default port used by rotctld;

- **Minimum Azimuth, Maximum Azimuth, Minimum Elevation, Maximum Elevation** - specify the range of azimuth and elevation values your rotator accepts;
- **Azimuth Offset, Elevation Offset** - if your rotator is not perfectly calibrated, these settings allow you to apply a correction;
- **Step Size** - to prevent the rotator from starting and stopping too often, change the bearing only when the required change is greater than the step size. The default value is 5 degrees;
- **Parking Azimuth, Parking Elevation** - specify the parking position for your antenna.

If your rotator does not control elevation, set the MinimumElevation and MaximumElevation to the same value. With such settings, wrong elevation will not be considered a bearing error. Note that the bearing error is indicated with a pink color on the [Rotator Control](#) panel.

See Also

- [Smart Antenna Rotation](#)

Setting Up Transverter

If you receive on an HF SDR connected to a VHF/UHF transverter, or transmit with an HF transceiver connected to a transverter, the SDR and the rig "see" the satellite signal at the transverter's **IF frequency** (typically 28–30 MHz), not at the actual RF frequency (144 MHz, 432 MHz). SkyRoof can apply the LO offset for you so that:

- the **SDR** is tuned to the IF that the transverter outputs, while the waterfall scale and the frequency display continue to show the RF frequency of the satellite;
- the **transceiver** (via CAT) receives the IF frequency that it can actually tune, instead of the VHF/UHF frequency that it cannot.

The three offsets — for the SDR, for the RX CAT channel and for the TX CAT channel — are independently enabled, so you can mix-and-match: a smart VHF/UHF rig with an SDR-on-transverter setup, a dumb HF rig with the SDR on the antenna directly, or any other combination.

NOTE

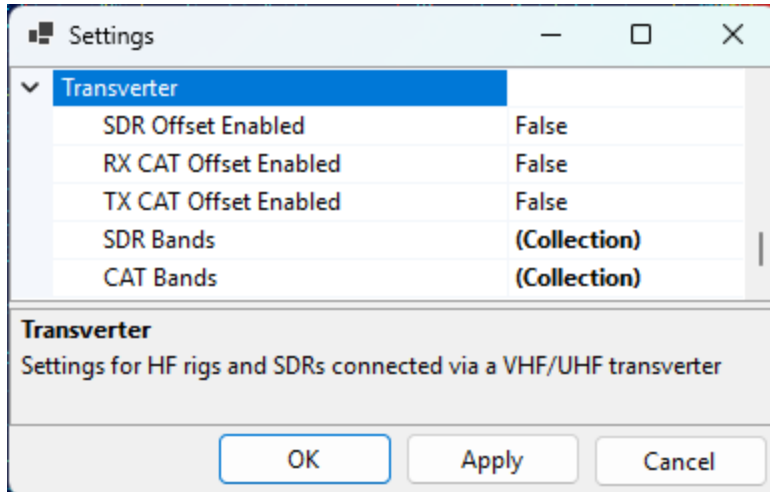
Frequencies are **always** displayed at the actual RF (144/432 MHz). The IF translation is applied internally by SkyRoof when it tunes the SDR or sends a CAT command.

When You Need This

- **HF transceiver + transverter, no built-in XVRT support** (e.g., IC-735, IC-7300): enable **RX CAT** and **TX CAT** offsets. SkyRoof sends 29.950 MHz to the radio when the satellite is on 145.950 MHz.
- **Modern transceiver with built-in XVRT offset** (e.g., K3S + Elecraft XV): the radio already converts internally. Leave the **CAT** offsets **disabled** so SkyRoof sends the RF frequency, which the rig converts itself.
- **HF SDR on the transverter IF output** (e.g., Airspy HF+, RTL-SDR with an upconverter on HF range, IC-R7000 IF tap): enable the **SDR** offset. The SDR is tuned to the IF band, but the waterfall and frequency display continue to show the actual satellite RF.
- **VHF/UHF SDR on the antenna directly, no transverter**: leave all three offsets disabled; SkyRoof behaves exactly as without this feature.

Settings

Click on **Tools / Settings** in the main menu to open the [Settings dialog](#), then expand the **Transverter** section:

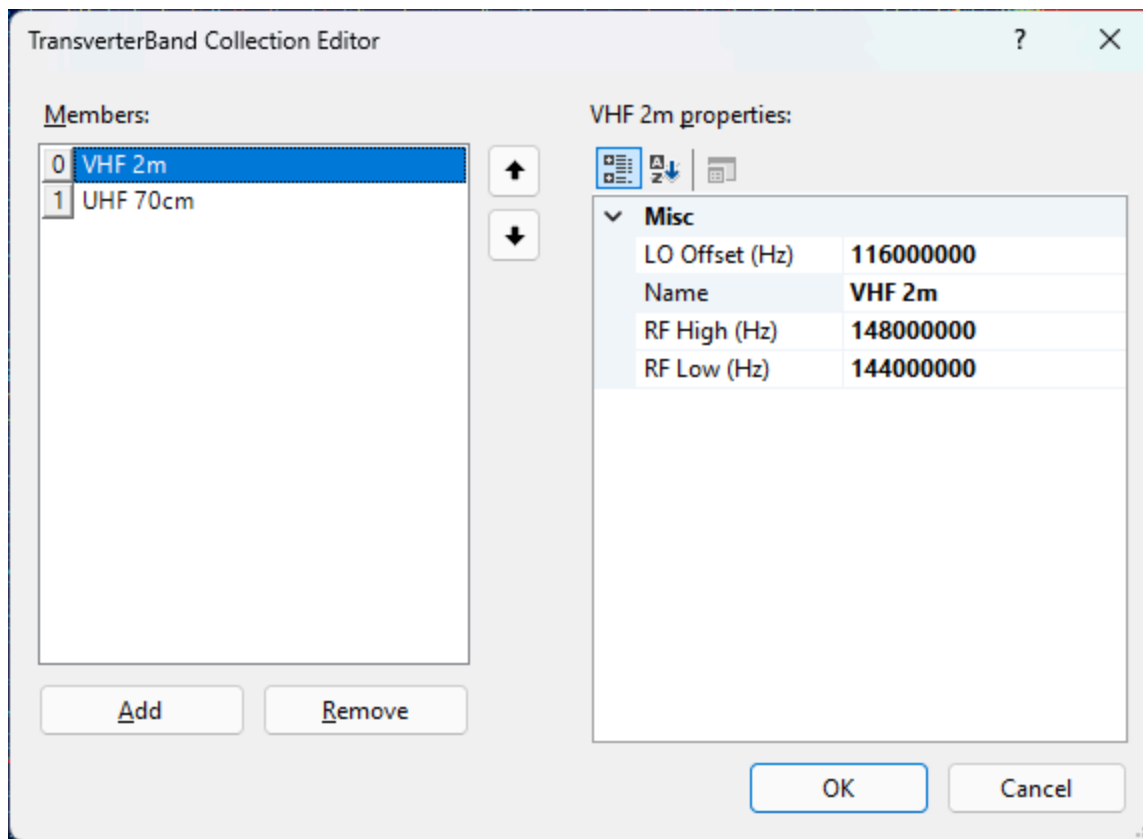


- **SDR Offset Enabled** — when on, the SDR is tuned to the IF of the matching SDR transverter band, not to the satellite RF;
- **RX CAT Offset Enabled** — when on, the RX frequency sent to the radio is offset by the LO of the matching CAT band;
- **TX CAT Offset Enabled** — same, for the TX frequency.
- **SDR Bands** — list of bands for the SDR path;
- **CAT Bands** — list of bands for the CAT path.

The two band lists are separate because the SDR transverter and the transmit transverter often use different local oscillators, and because the CAT path's RF range may be narrower than the SDR path (the SDR sees the entire transverter passband, but the radio's HF receiver can only tune a slice of it).

Editing Bands

Click the ... button in the **SDR Bands** or **CAT Bands** row to open the collection editor. You can **add new bands, edit the existing ones, or delete bands**:



Each band has:

- **Name** — a free-text label (e.g., **VHF 2m**, **UHF 70cm**, **23 cm**);
- **RfLow** / **RfHigh** — RF frequency range in Hz, inclusive;
- **LoOffset** — local oscillator offset in Hz. $IF = RF - LO$.

When SkyRoof needs the IF for an active RF, it looks up the first band whose [**RfLow**, **RfHigh**] range covers that RF and subtracts its **LoOffset**. If no band matches, the offset is zero (SDR case) or the CAT command is skipped with a warning (CAT case — see below).

Restoring the Defaults

If you delete entries from a band list and want to bring back the seeded defaults, right-click on the **SDR Bands** or **CAT Bands** row and choose **Reset**. This replaces the entire list with the default entries. An empty list, on its own, is preserved across sessions — SkyRoof will not "helpfully" re-seed it on the next launch.

Default Bands

Default SDR Bands

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	144.000	148.000	116.000	28.0–32.0 MHz

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
UHF 70cm	432.000	438.000	407.000	25.0–31.0 MHz

The SDR band table covers the full ham VHF allocation and the satellite portion of the UHF allocation. The IF spans (4 MHz and 6 MHz) exceed the SDR's maximum bandwidth of 3.1 MHz, so the SDR cannot show the entire transverter passband at once — see [Behavior at Runtime](#) below for how SkyRoof handles this.

Alternative SDR Bands (10.7 MHz IF)

For receivers that use 10.7 MHz as the IF output (e.g., the IC-R7000 wideband communications receiver), edit the SDR bands to use these LO values instead:

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	144.000	148.000	135.300	8.7–12.7 MHz
UHF 70cm	432.000	438.000	424.300	7.7–13.7 MHz

Default CAT Bands

Name	RF Low (MHz)	RF High (MHz)	LO (MHz)	IF Range
VHF 2m	145.800	146.000	116.000	29.8–30.0 MHz
UHF 70cm AO-7 uplink	432.000	434.000	404.000	28.0–30.0 MHz
UHF 70cm	435.000	437.000	407.000	28.0–30.0 MHz

The CAT RF ranges are narrower than the SDR ranges so the resulting IF lands inside a typical HF rig's tuning range (28–30 MHz). The AO-7 mode U-V uplink (432–434 MHz) is split into its own entry because it uses a different LO than the rest of the 70 cm band.

Behavior at Runtime

When **SDR Offset Enabled** is on:

- The SDR is retuned to the IF center of the active satellite's band on the first activation and whenever the satellite crosses to a different band. The waterfall history resets on each such retune;
- Within a band, only the audio slicer offset moves as the Doppler-corrected signal drifts; the SDR center stays put and the waterfall keeps scrolling;

- If the transverter band is wider than the SDR's bandwidth (the default 4 MHz VHF and 6 MHz UHF bands are), the SDR center is repositioned as needed to keep the Doppler-corrected signal visible, while keeping the SDR passband within the transverter band limits;
- The frequency scale, the satellite labels and the pass-visibility check always work in RF, so what you see on screen is the satellite's actual frequency regardless of where the SDR is physically tuned.

When **RX CAT** or **TX CAT** offset is enabled but the active RF does not fall into any CAT band, SkyRoof **skips the CAT frequency command** rather than sending a frequency the radio cannot tune. A warning is written to the log, and:

- the corresponding **RX CAT** or **TX CAT** label on the status bar turns **yellow**;
- the tooltip on that label explains the out-of-band condition, naming the offending frequency.

This is the signal that you need to either disable the CAT offset (if you are listening to a non-ham satellite) or add a band entry to the CAT list (if you want to extend coverage).

Frequency Display Tooltips

When any of the three offsets is enabled, the tooltip on the downlink and uplink frequency display in the [Frequency Control](#) panel includes extra lines:

- **SDR Transverter IF: <value> MHz** — only on the downlink, when the SDR offset is enabled and an SDR band matches;
- **CAT Transverter IF: <value> MHz** — on the downlink when RX CAT offset is enabled, and on the uplink when TX CAT offset is enabled, when a CAT band matches.

This lets you confirm at a glance what IF frequency is being sent to the SDR and to the radio.

Common Configurations

K3S with XVRT + SDR on the transverter IF

- **SdrOffsetEnabled = true**, both CAT offsets **false**.
- SkyRoof tunes the SDR to the IF band and keeps the satellite visible there.
- SkyRoof sends 435.835 MHz to the K3S (unchanged); the rig applies its own XVRT conversion internally.

IC-7300 (no transverter awareness) + SDR on the transverter IF

- `SdrOffsetEnabled = true, RxCatOffsetEnabled = true, TxCatOffsetEnabled = true.`
- Downlink 435.835 MHz → SDR shows 28.835 MHz IF; IC-7300 commanded to 28.835 MHz.
- Uplink 145.965 MHz → IC-7300 commanded to 29.965 MHz.
- The IC-7300 covers 1.8–54 MHz continuously, so both IFs are inside its tuning range.

SDR on the antenna, smart rig

- All offsets disabled. Behaviour is identical to a SkyRoof setup without any transverter.

AO-7 mode U-V

- Uplink 432.xxx MHz → matches the **UHF 70cm AO-7 uplink** CAT band → CAT LO = 404 MHz → IF $\approx$ 28.xxx MHz.
- Downlink 145.xxx MHz → matches the SDR's **VHF 2m** band → SDR LO = 116 MHz → IF $\approx$ 29.xxx MHz.

Non-ham telemetry satellite (e.g., 400.550 MHz)

- No CAT or SDR band covers 400 MHz → both offsets resolve to zero → SDR tunes directly to 400.550 MHz, RF is sent to the radio (if it can tune there) → no transverter behaviour.

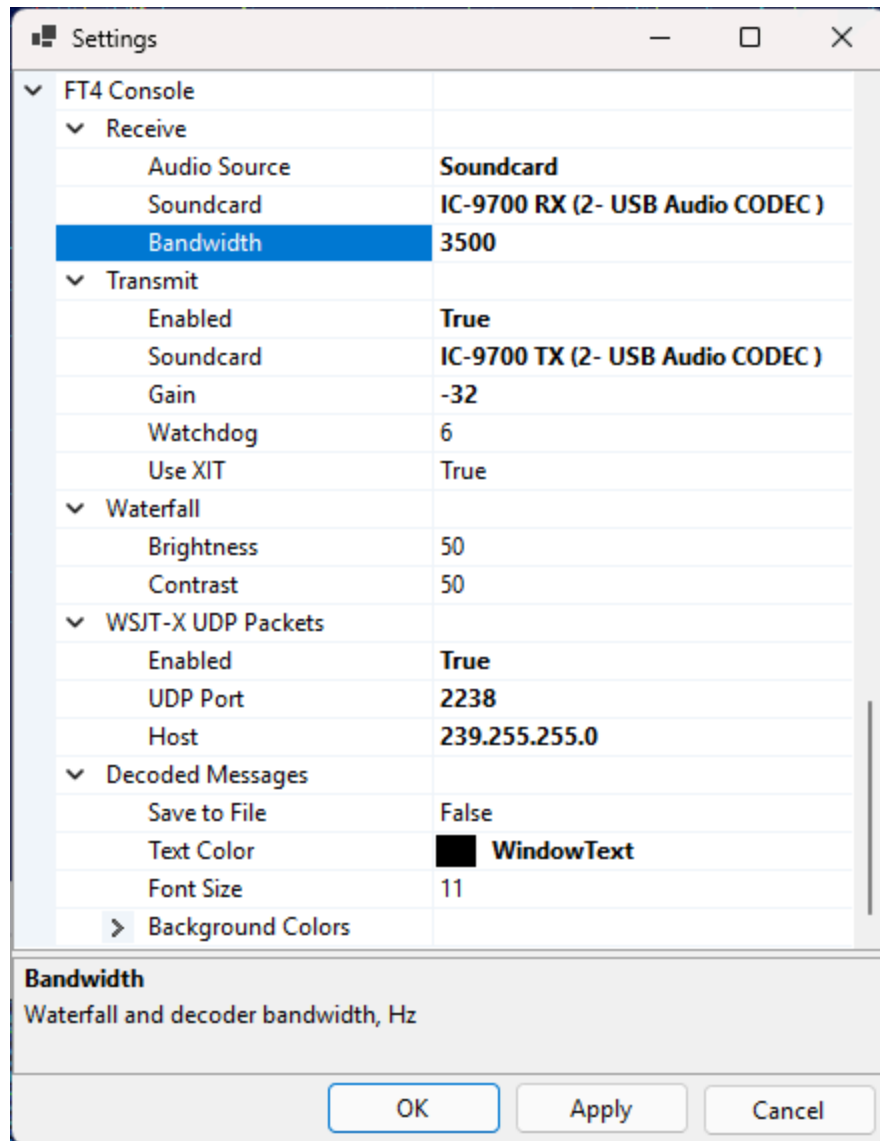
See Also

- [Setting Up SDR](#)
- [Setting Up CAT Control](#)
- [Frequency Control](#)

Setting Up FT4 Console

FT4 Console Settings

To configure FT4 Console, click on Tools / Settings to open the Settings dialog and scroll down to the FT4 Console section:



Receive

- **Audio Source** - either SDR or Soundcard. SDR is the default, but it is also possible to receive FT4 from the transceiver via a soundcard;
- **Soundcard** - if Soundcard is selected as audio source, this field should contain the soundcard to use. Windows allows you to rename the soundcards, so you can change the default name, such as "USB Audio CODEC" to something that is easy to recognize, "IC-9700 RX" in my case;

- **Bandwidth** - the receiver bandwidth, in Hertz. Affects both the waterfall display and decoding. See **Receiver Bandwidth** below.

Transmit

- **Enabled** - by default, the transmit function is disabled. Set this field to true if you want to transmit FT4;
- **Soundcard** - the soundcard that sends the FT4 audio to the transmitter;
- **Gain** - the FT4 output audio is multiplied by this factor (in dB). The default is -30 dB, see the **TX Gain** section below for the instructions how to find the correct value;
- **Watchdog** - the maximum transmit time. After the specified number of minutes, if the user has not performed any actions, the transmissions automatically stop;
- **Use XIT** - offset the transmitter frequency relative to the receiver depending on the transmitted audio tone. See **Using XIT** below for details.

Waterfall

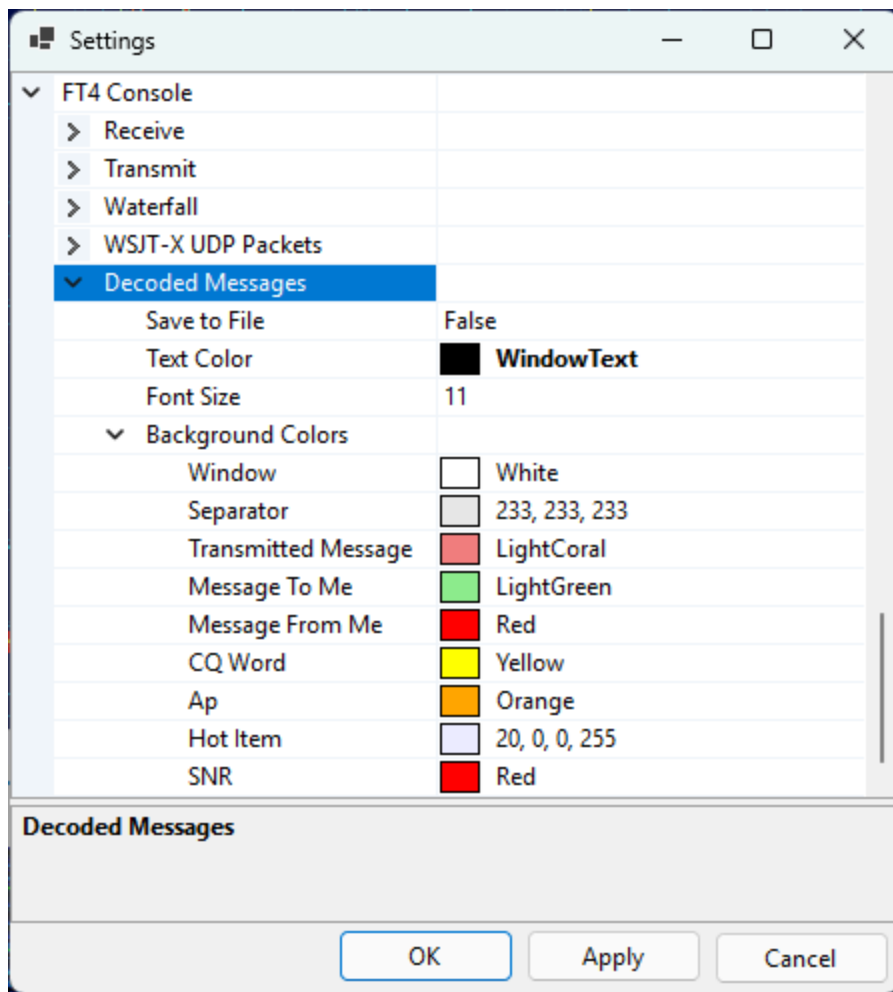
- **Brightness** - waterfall brightness, 0 to 100;
- **Contrast** - waterfall contrast, 0 to 100;

WSJT-X UDP Packets

- **Enabled** - enable sending the UDP packets with information about decoded messages and QSO. The format of the messages is the same as in WSTT-X;
- **UDP Port** - the port to which the messages will be sent;
- **Host** - this could be a Unicast or Multicast address. See **UDP Packets** below.

Decoded Messages

- **Save to File** - when set to true, all received and transmitted messages will be saved to a file in the [Data folder](#), FT4 sub-folder;
- **Text Color** - the text color in the Decoded Messages window;
- **Font Size** - the font size in the Decoded Messages window;
- **Background Colors** - the background colors of the messages of different types:



RX Signal Strength

When FT4 is decoded from the SDR, the signal strength is adjusted automatically, but when the signals are coming from a soundcard, the amplitude of the input audio must be adjusted so that the signal strength bar is at about 50%:



This could be done in several ways.

Option 1: Transceiver Menu

In IC-9700 press **Menu / Set / Connectors / USB AF/IF Output** and adjust **AF Output Level**.

Option 2: Windows Audio Settings

In Windows 11 open **Settings / System / Sound**, select your soundcard, and adjust the **Input Volume** setting:

System > Sound > Properties



IC-9700 RX

2- USB Audio CODEC

Rename

Provider

(Generic USB Audio)

Driver date

2025-11-25

Driver version

10.0.26100.7309

Check for driver updates

General

Audio

Allow apps and Windows to use this device for audio

Don't allow

Input settings

Format

1 channels, 16 bit, 48000 Hz (DVD Quality) ▾

Input volume

 40 

Microphone test

Start the test, then talk or play audio at a normal level for a few seconds

Volume: 3%

Start test

^

Test mode for microphone audio processing

Choose different modes to hear how recorded audio will sound

Default ▾

 Audio enhancements

Select an enhancement available for your device

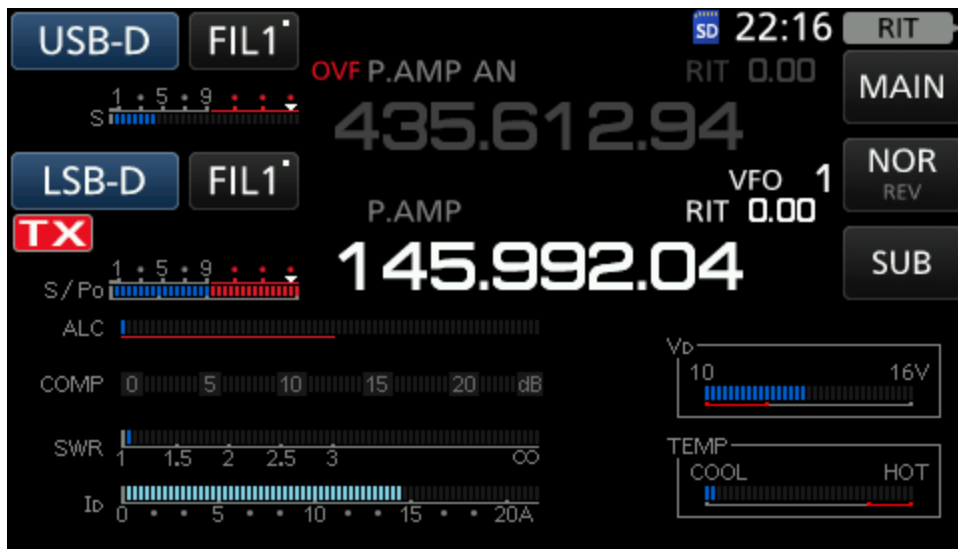
Off ▾

TX Gain

46 / 121

The amplitude of FT4 audio output must be set carefully to ensure the desired output power and to prevent transmission of audio harmonics. To adjust the output amplitude, click on the **Tune** button in FT4 Console to start transmission and adjust **TX Gain slider**. Use the meters in the transceiver to ensure that the radio delivers full output power, and that the ALC level is around zero.

In IC-9700 you can display all meters in the satellite mode by pressing **Menu / Meter**, as shown in the screenshot below. **S/Po** must be at the maximum, and **ALC** at the minimum. For IC-9700 the optimal TX gain is about -32 dB.



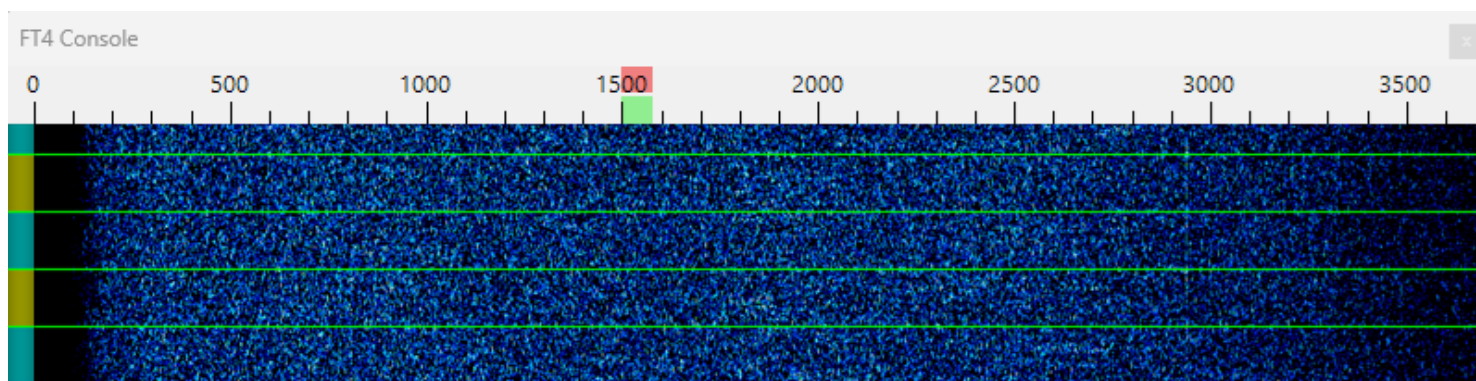
Once the audio output level is set, adjust the output power of your radio according to your operating needs. In most cases only a few watts are required for satellite contacts if a good antenna is used.

Receiver Bandwidth

The **FT4 Console / Receive / Bandwidth** parameter in the Settings dialog controls the bandwidth of the FT4 decoder, as well as the frequency segment shown on the waterfall display.

When decoding from an SDR, set it to 5000 Hz, the bandwidth of the SDR receiver in the USB-D and LSB-D modes;

When decoding from a soundcard, set this parameter to match the receiver bandwidth of your radio. Make sure that the dark area on the waterfall is not wider than 200-300 Hz at each side, as in the screenshot below:



In most transceivers the RX bandwidth is around 3000 Hz, even in the Data mode. IC-9700 may be configured for a bandwidth of 3600 Hz in USB-D and LSB-D modes.

Using XIT

Keep the **Transmit / Use XIT** option set to true unless your radio is unable to change the TX frequency when transmitting.

When this option is enabled, the audio tone sent to the transmitter is always in the range of 1000 Hz to 2000 Hz. To transmit signals outside of this range, the transmitter frequency is changed up or down. For example, to transmit on 3200 Hz, the transmitter is tuned 2000 Hz up (or down for inverting transponders), and the audio tone sent to the radio is 1200 Hz.

The XIT mode serves two purposes:

- it prevents the odd audio harmonics from being transmitted since they are outside of the transmitter passband;
- it ensures that you can transmit on any audio frequency, even outside of the transmitter passband.

UDP Packets

SkyRoof can send UDP packets with information about decoded messages and completed QSO, in the same format as WSJT-X. There is a number of programs that listen to such messages and use them to write QSO to the log or to update Worked statistics. Some even reply to these messages with callsign status information, so that SkyRoof can color-highlight the callsigns according to their Needed status.

This option is disabled by default. You can enable it in the Settings dialog.

Both Unicast and Multicast messaging is supported. In the Unicast mode the messages can be sent to only one program. If you start another program that listens on the same UDP port, that program fails with the Port in Use error. In the Multicast mode multiple programs can receive messages on the same port. Use the multicast mode if possible, or use unicast with older programs that do not support multicast.

UDP packets are sent in the multicast mode if the IP address is in the range of 239.0.0.0 - 239.255.255.255, otherwise they are sent as unicast.

The default settings are:

Unicast:

- IP Address: 127.0.0.1
- Port: 7311

Multicast:

- IP Address: 239.255.255.0
- Port: 2238

Satellites and Groups

Click on **Tools / Satellites and Groups** in the main menu to open the **Satellites and Groups** window:

Satellites and Groups

Satellites

Status: ☒ Alive ☒ Future ☐ Re-Entered Bands: ☒ VHF ☒ UHF ☒ Other

Radio: ☒ Transponder ☒ Transceiver ☒ Transmitter Service: ☒ Ham ☒ Non-Ham

Search: Updated 2025-05-26

Name	NORAD ID	Launched	Service
Bufeng-1A	44312		
BUGSAT-1	40014	2014-06-19	Amateur
BUZZER-1	98693	2025-01-14	
CAKRA-1	98861	2024-08-16	
CALIPSO	29108	2006-04-28	Meteorological
CANX-4	40055	2014-06-30	Amateur
CANX-5	40056	2014-06-30	Amateur
CANYVAL-C 1U	99732	2021-03-22	
CANYVAL-C 2U	99731	2021-03-22	
CAPE-1	31130	2007-04-17	Amateur
CARBONITE-1	40710	2015-07-10	

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Satellite Groups

- Active Ham Sats
 - AO-73
 - RS-44
 - MO-122
 - FO-29
 - AO-7
 - AO-91
 - ARISS
 - JO-97
 - SO-50
 - SO-124
 - SONATE
 - AO-123
- CW Beacons
 - KKS-1
 - CUTE-1
 - KOSEN-1
 - Veronika
 - SEEDS II

OK Cancel

Exploring the satellite data

This window is a great tool for browsing the information about the satellites that is available in SkyRoof. For a detailed description of the data see the [Satellite Data](#) section.

Highlighting

The satellites are highlighted based on their properties as described in the [Satellite Highlighting](#) section.

Filtering

Using the checkboxes at the top of the left panel, the satellites may be filtered by:

- **Status** - Alive, Future or Re-Entered. If you cannot find some satellite in the list, tick the Re-Entered checkbox, maybe this satellite has already re-entered the atmosphere;
- **Bands** - show only the satellites that have at least one transmitter working in the VHF (2m), UHF (70cm) or in any other band;

- **Radio** - the radio type: linear transponder, FM transceiver or a telemetry/beacon transmitter;
- **Service** - Ham or non-Ham, as marked in the database.

Searching

Use the **Search** box to search the satellites by name, callsign or NORAD Id. The search is case-insensitive, punctuation is ignored. If a satellite has multiple names, any name may be used. For example, the RS-44 satellite is found by entering "RS-44", "rs44" or "DOSAAF-85" in the search box.

The numbers on the bottom bar show the total number of satellites in the database and the number of those that match the filters and search string.

Viewing Details

Right-click on a satellite and click on **Satellite Details**, or press **Ctrl-D**, to open the [Satellite Details window](#).

Renaming Satellites

To rename a satellite, press **F2**, or right-click on the satellite and click on **Rename** in the popup menu.

Editing Satellite Groups

The right panel of the window shows the satellite groups. See the [Creating Satellite Groups](#) section for information about creating and editing the satellite groups. The following editing commands are available:

- the **[+]** button creates a new group. Press **F2**, or use the popup menu, to rename the group;
- the **[>]** and **[<]** buttons add or remove the satellites to/from the group;
- drag-and-drop from the satellite list to the group adds the satellite to the group;
- drag-and-drop of a satellite between the groups moves it to another group;
- drag-and-drop with the **Ctrl** key down adds a copy of the satellite to another group;
- drag-and-drop re-orders the groups, or the satellites in the group;
- the Delete key deletes the group, or the satellite from the group;
- the popup menu of the group or satellite is an alternative way of accomplishing the same tasks.

Satellite Details Window

The Satellite Details window is available via the right-click menu in many panels, including the [Satellites and Groups window](#), [Current Group panel](#) and [Frequency Scale](#).

Satellite Details

RS-44

a.k.a. BREEZE-KM R/B, DOSAAF-85, RS44

Names

SatNOGS	RS-44
SatNOGS Alt	RS-44, RS-44
JE9PEL	DOSAAF-85, RS-44
Callsigns	RS44
LoTW	RS-44
AMSAT	RS-44

Orbit

TLE	2025-05-26 12:36 (Space-Track.org)
Period, min	112
Inclination, deg.	82
Elevation, km	1307
Footprint, km	7550

SatNOGS Database

sat_id	UFYD-5782-6372-2920-6054
norad_cat_id	44909
norad_follow_id	
status	alive
decayed	
launched	2019-12-26
deployed	
operator	None
countries	RU

Transmitters

Transmitter	Downlink	Uplink
SatNOGS		
Doka-B	435,215.0	
Mode U	435,315.0	
Mode U - Beacon	435,605.0	
Mode U Audio	435,660.0	
Mode V/U - Transponder	435,670.0 - 435,6...	145,935.0 - 145,9...
JE9PEL		
SSB CW	435.670-435.610	145.935-145.995
SSB CW	435.670-435.610	145.935-145.995

[Image](#) [SatNOGS](#)

Close

The transmitters are highlighted according to the downlink band. Ham transmitters are bold, inactive ones are grayed.

The blue links at the bottom of the window open the web pages with extra information about the satellite.

See the [Satellite Data](#) section for the description of the satellite data available in SkyRoof.

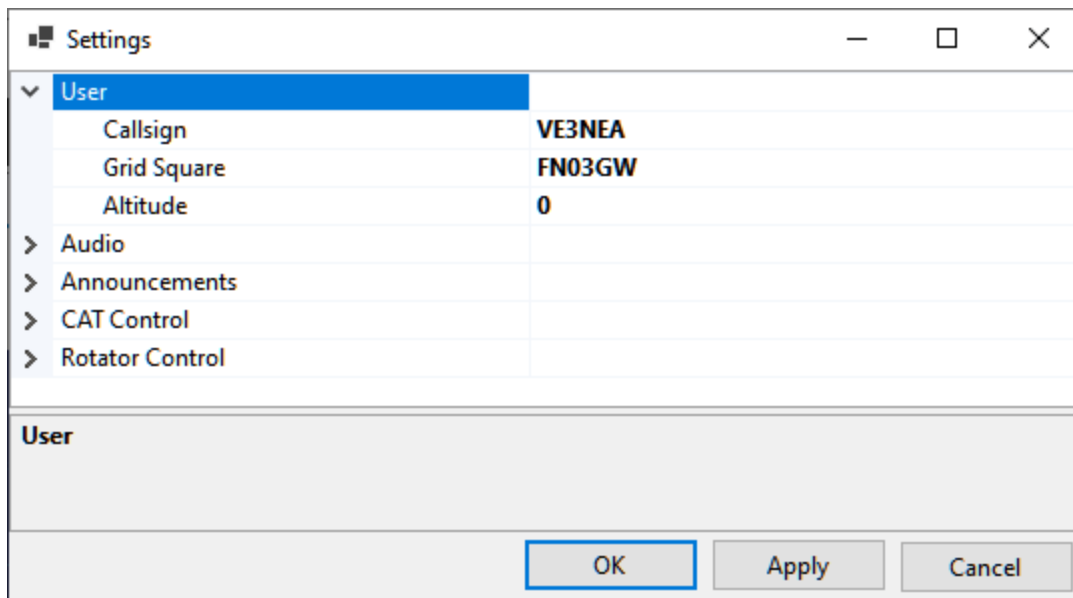
Satellite Highlighting

The satellite entries in various windows and panels, such as [Satellites and Groups window](#), [Satellites Details window](#) and [Current Group panel](#), are highlighted according to their properties:

- **Cyan background** - the satellite has at least one transmitter in the UHF band;
- **Yellow background** - the satellite has at least one transmitter in the VHF band;
- **Bold text** - the satellite has the Amateur Service (Ham) flag in the [database](#);
- **Grayed text** - the satellite is not Alive;
- **Striked-out** - the orbit elements (TLE) are not available for this satellite.

Settings window

Click on **Tools / Settings** in the main menu to open the Settings window.



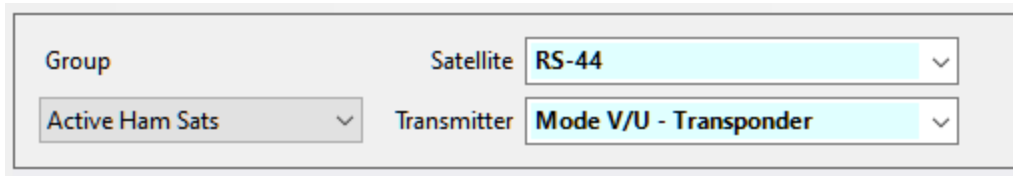
This dialog allows you to edit the settings of SkyRoof. In particular, here you can change the **Callsign**, **Grid Square** and **Altitude** that you entered on the first run of the program.

For the description of other settings see these sections:

- [Setting Up Audio](#)
- [Setting Up Voice Announcements](#)
- [Setting Up CAT Control](#)
- [Setting Up Rotator Control](#)
- [Setting Up Transverter](#)

Satellite Selector

Satellite Selector is the panel on the toolbar where you can select the satellite group, the satellite within the group, and the transmitter of the satellite:



The image shows a 'Satellite Selector' panel with three dropdown menus. The first dropdown is labeled 'Group' and has 'Active Ham Sats' selected. The second dropdown is labeled 'Satellite' and has 'RS-44' selected. The third dropdown is labeled 'Transmitter' and has 'Mode V/U - Transponder' selected. Each dropdown has a small downward arrow on its right side.

Move the mouse cursor over the **Satellite** or **Transmitter** drop-down box to see detailed information about the selected item on the mouse tooltip.

See the [Creating Satellite Groups](#) section for the instructions how to create and edit the groups.

Satellite Photo

The **Satellite Photo** widget on the toolbar shows a thumbnail image of the currently selected satellite:



The image is taken from the [SatNOGS](#) satellite database. Hover the mouse cursor over the thumbnail to see the satellite name as a tooltip.

The widget is empty when the selected satellite has no image available in the database.

Image Cache

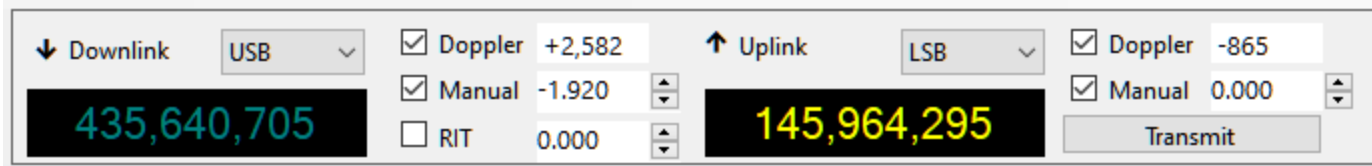
Downloaded thumbnails are cached on disk so they only need to be fetched once. The cache is kept in the `sat_images` sub-folder of the [Data Folder](#). You may delete this folder at any time to force SkyRoof to re-download the images.

Visibility

The widget hides itself automatically when the main window is too narrow to fit all of the other toolbar widgets. Resize the main window wider to bring it back into view.

Frequency Control

Frequency Control is the panel on the toolbar that allows you to read and control the frequencies of the SDR receiver, external receiver and external transmitter:

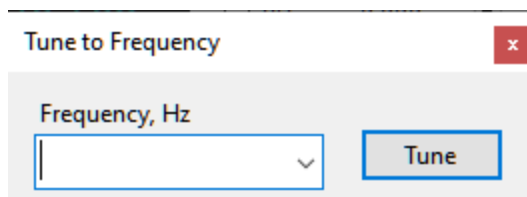


Downlink

The left hand part of the panel represents the receiver settings that apply to both SDR and external radio.

Label

When the receiver is tuned to a downlink transmitter of some satellite, the label "Downlink" appears; when it is tuned to a terrestrial station, the label "Terrestrial" is displayed. To tune to a downlink, select some satellite in the [Satellite Selector](#) panel, or select a different transmitter from the drop-down list, or click on the satellite name in any panel. To tune to a terrestrial signal, click on it on the [Waterfall Display](#) or on the [Frequency Scale](#), or click on the downlink frequency display and enter the frequency in the **Tune to Frequency** window:



Mode

Select the mode manually for every satellite transmitter that you are using. Your selection is remembered and restored when the transmitter is selected again.

The **Mode** selected in the drop-down box applies to the SDR receiver, if it is enabled, and to the external receiver, if RX CAT is enabled. To enable or disable the SDR or RX CAT, click on the corresponding label on the status bar.

Frequency Display

The frequency display shows either the nominal frequency of the downlink, or the frequency with all corrections applied. Right-click on the display to switch between the two frequencies.

The mouse tooltip of the frequency display shows both frequencies and some other details.

When RX CAT is enabled and working properly, the frequency is shown in a bright color, otherwise the display is dimmed. The color depends on the band: yellow/olive for VHF, cyan/teal for UHF, white/gray for all other bands.

Doppler

The **Doppler** box shows the current Doppler offset of the downlink signal. This value is not editable, but Doppler correction may be enabled or disabled using the checkbox. See the [Doppler Tracking](#) section for a detailed discussion of Doppler offset calculation and tracking.

Manual

The manual correction of the downlink frequency. The frequencies of the satellite downlink signals usually differ from the nominal values in the database, for different reasons, by a few hundred Hertz and up to a couple of kilohertz. This difference is pretty stable, so it is enough to enter the correction once to have the receiver accurately tuned. SkyRoof remembers the manual correction for each satellite.

The value of the manual correction may be entered in the **Manual** box by clicking on the up/down buttons, or by spinning the mouse wheel over the box, or by typing the value directly. However, it is more convenient to adjust the correction visually, using the mouse on the [Frequency Scale](#).

The checkbox allows you to disable the manual correction if necessary.

RIT

The RIT function is useful when listening to a conversation of two stations that are not exactly on the same frequency, or when your CQ is answered off the frequency.

The RIT offset may be entered in the RIT box, but it is more convenient to control it on the [Frequency Scale](#).

Use the checkbox, or the commands on the **Frequency Scale**, to toggle RIT.

Uplink

The Uplink part of the panel is similar to the Downlink part described above. It is enabled only if the selected satellite transmitter has an uplink. The bright color of the frequency display means that TX CAT is enabled and working properly. The **Transmit** button switches the external radio between the RX and TX modes.

The **Manual Correction** setting of the uplink allows you to align your transmit and receive frequencies. See the [Frequency Scale](#) section for details.

Dial Knob

The dial knob of the transceiver can be used to tune the frequency when CAT control is enabled and the **Ignore Dial Knob** option is set to **false**.

When both RX CAT and TX CAT are enabled, the dial knob controls the receiver frequency.

NOTE

When the radio is in the SAT mode, the NOR/REV switch should be in the NOR position for correct tuning with the dial knob.

Gain Control

The Gain Control panel on the toolbar has two sliders to control the RF and AF gain:

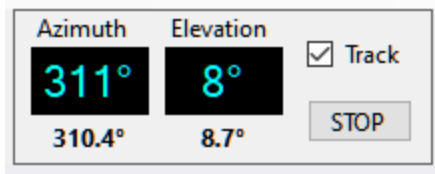


The **RF Gain** slider is enabled only if the **Single Gain** setting is set to **true** in [SDR settings](#).

To adjust the gain, click on the slider, or spin the mouse wheel over it, or drag the thumb control.

Rotator Control

The Rotator Control panel on the status bar shows the current position of the selected satellite and the antenna bearing, if the rotator control function is enabled:

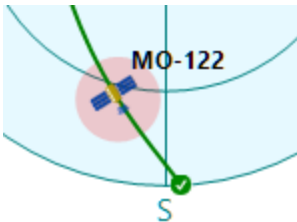


Display

The large Azimuth and Elevation display shows the satellite location, the small numbers below it show the antenna bearing.

The satellite location is dimmed when the rotator control function is disabled. Click on **Rotator** on the status bar to enable or disable this function.

When rotator control is enabled, the current antenna bearing is marked on the [Sky View panel](#) with a red spot:



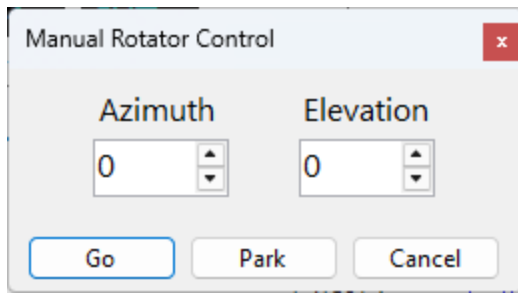
Tracking

When rotator control is enabled but the **Track** checkbox is not ticked, the panel only displays the antenna bearing but does not attempt to change it. Tick the **Track** checkbox to start tracking. Note that the check box is cleared when you switch to another satellite.

In the satellite tracking mode, the antenna bearing turns pink if it differs from the satellite position by more than 1.5 the **Step Size** setting entered in the [rotator settings](#).

Manual Control

Click on the satellite position display to open the **Manual Rotator Control** window:



- click on **Go** to rotate the antenna to the specified azimuth and elevation;
- click on **Park** to rotate the antenna to the parking position.

Stopping

To stop antenna rotation, either manual or due to the satellite tracking, click on the **Stop** button.

See Also

- [Smart Antenna Rotation](#)

Current Group

The Current Satellite Group panel shows the list of the satellites in the currently selected group:

Current Satellite Group ✕			
Group: Active Ham Sats			
Name	NORAD ID	Next Pass	Max
✕ AO-7	7530	Now	35°
✕ SONATE	59112	18m 43s	12°
✓ AO-73	39444	19m 09s	8°
✓ JO-97	43803	3h 13m 07s	0°
✓ SO-50	27607	3h 27m 43s	10°
✓ RS-44	44909	3h 34m 09s	7°
✕ MO-122	60209	3h 46m 33s	24°
✓ AO-123	61781	4h 27m 46s	14°
✕ FO-29	24278	4h 42m 45s	7°
SO-124	62690	4h 48m 08s	1°
✕ AO-91	43017	4h 52m 10s	17°
✓ ISS	25544	13h 33m 08s	8°

The green and red icons indicate real-time satellite status (active or inactive) according to the AMSAT web site. See [Satellite Data](#) for details.

Click on a satellite to select it.

Move the mouse cursor over the satellite name to see the mouse tooltip.

Right-click on the satellite and click on **Satellite Details** in the popup menu to open the [Satellite Details Window](#).

Satellite Details

The Satellite Details panel shows information about the currently selected satellite:

Satellite Details	
RS-44	
a.k.a. BREEZE-KM R/B, DOSAAF-85, RS44	
▼ Names	
SatNOGS	RS-44
SatNOGS Alt	RS-44, RS-44
JE9PEL	DOSAAF-85, RS-44
Callsigns	RS44
LoTW	RS-44
AMSAT	RS-44
▼ Orbit	
TLE	2025-05-26 12:36 (Space-Track.org)
Period, min	112
Inclination, deg.	82
Elevation, km	1313
Footprint, km	7566
▼ SatNOGS Database	
sat_id	UFYD-5782-6372-2920-6054
norad_cat_id	44909
norad_follow_id	
status	alive
decayed	
launched	2019-12-26
deployed	
operator	None
countries	RU
telemetries	
updated	2022-08-01 17:59
citation	CITATION NEEDED - https://xkcd.com/1105/
is_frequency_violator	False
associated_satellites	NFQU-3521-4316-6654-7962
Image SatNOGS	

This panel displays the same information as the upper portion of the [Satellite Details window](#). The difference is that, unlike the window, the panel may be docked anywhere in the user interface.

See the [Satellite Data](#) section for information about available data.

Satellite Transmitters

The Satellite Transmitters panel shows the list of transmitters carried by the currently selected satellite. It is similar to the bottom part of the [Satellite Details window](#) but is dockable for permanent visibility:

Satellite Transmitters		
RS-44		
Transmitter	Downlink	Uplink
SatNOGS		
Doka-B	435,215.0	
Mode U	435,315.0	
Mode U - Beacon	435,605.0	
Mode U Audio	435,660.0	
✓ Mode V/U - Transponder	435,670.0 - 435,610.0	145,935.0 - 145,995.0
JE9PEL		
SSB CW	435.670-435.610	145.935-145.995
SSB CW	435.670-435.610	145.935-145.995

Click on the transmitter in the **SatNOGS** section to select it. Move the mouse cursor over the transmitter name to see the details on the mouse tooltip.

Satellite Passes

The Satellite Passes panel shows the list of predicted satellite passes over your location:

Satellite Passes			
☐ Current Satellite ☒ Group ☐ All VHF/UHF			
JO-97 #35512	in 3m 58s		^
2025-05-29 12:22:24 to 12:31:46	9 min 12°		
SO-124 #2049	in 29m 18s		
2025-05-29 12:47:44 to 12:58:52	11 min 57°		
ARISS #51229	in 31m 05s		
2025-05-29 12:49:31 to 12:58:14	8 min 11°		
RS-44 #25329	in 42m 41s		
2025-05-29 13:01:07 to 13:22:50	21 min 35°		
SO-50 #20761	in 44m 51s		
2025-05-29 13:03:17 to 13:06:42	3 min 1°		
AO-7 #31245	in 54m 06s		
2025-05-29 13:12:32 to 13:18:53	6 min 1°		
AO-73 #62197	in 1h 59m 21s		
2025-05-29 14:17:47 to 14:27:10	9 min 11°		
SONATE #6868	in 2h 3m 26s		
2025-05-29 14:21:52 to 14:28:31	6 min 5°		

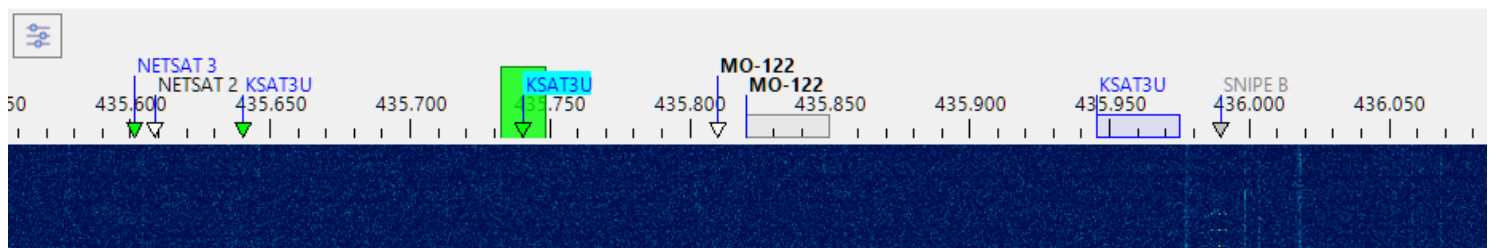
The radio buttons at the top allow you to view either the passes of the selected satellite, of all satellites in the currently selected group, or all satellites carrying a VHF or UHF downlink transmitter and thus visible on the [Frequency Scale](#).

Click on a pass to make the satellite selected and to view its trajectory on the [Sky View panel](#).

Move the cursor over the satellite name to view extra information on the mouse tooltip, or right-click to open the [Satellite Details window](#).

Frequency Scale

The frequency scale appears on the [Waterfall Display](#) panel, above the waterfall:



Satellite Transmitters

The Doppler-corrected frequencies of the satellite transmitters are marked on the frequency scale with small triangles, labeled with the satellite names:

- **green triangles** - the satellites that are currently above the horizon;
 - **white triangles** - the satellites that will rise in the next 5 minutes;
 - **gray triangles** - the satellites that are already below the horizon, but whose signals are still may be visible on the waterfall;
-
- **blue rectangles** - the transponder segments of the satellites above the horizon;
 - **gray rectangles** - the transponder segments of the satellites below the horizon.

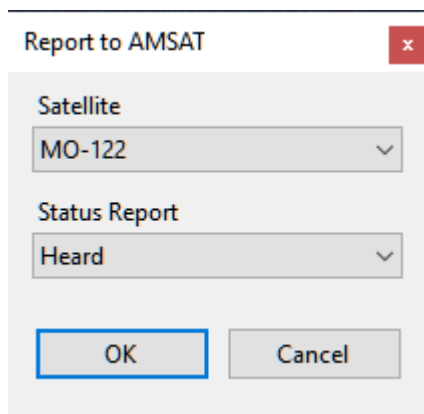
The names of the satellites that belong to the current group are shown in bold;

The current transmitter has its satellite name on the light blue background.

Move the mouse cursor over a satellite name to see the details on the mouse tooltip.

Right-click on a satellite name to open the popup menu with these commands:

- **Select Transmitter** - if the satellite has more than one transmitter on the same frequency, this command is enabled and allows you to set one of the transmitters as selected;
- **Add to Group** - add the satellite to one of the existing groups;
- **Report to AMSAT** - open the dialog to report your observation of this satellite to [AMSAT Live OSCAR Satellite Status Page](#) [↗](#)



- **Satellite Details** - opens the [Satellite Details window](#)

The frequency scale does not show all satellites at all times, only those that are currently available at your location. The satellite labels appear about 5 minutes before the AOS, and stay for a while after LOS, so that the existing signal traces could be identified, but then they disappear. This also applies to the linear satellites and their transponder segments. When the segment is not visible, you cannot tune within it using the mouse controls. The operator tunes in the transponder segment to tune to some station, or to find a clear space to send CQ. If the satellite is below the horizon, this, of course, cannot be done, and if you are still using the tuning commands, the program assumes that you are trying to do something else, e.g., tune to a terrestrial signal.

SDR Receiver

The passband of the SDR receiver is shown on the frequency scale as a green rectangle. To tune the receiver:

- click on one of the satellite labels to start tracking satellite's transmitter;
- click within a blue transponder segment to select the transmitter and set the transponder offset;
- click anywhere on the frequency scale to tune to a terrestrial signal;
- drag the green rectangle to another frequency with a mouse;
- spin the mouse wheel on or near the green rectangle. Hold the **Alt** key down to increase the tuning speed.

The effect of tuning depends on the transmitter selection:

- when tuned to a terrestrial signal, tuning the SDR receiver just changes the receiver frequency;
- when a satellite transmitter is tracked, tuning adjusts the **Manual Offset** of the satellite (see below);
- if a satellite transponder is selected, tuning changes the receiver offset within the transponder segment.

Another way to tune the SDR receiver is to use the [Frequency Control](#) on the toolbar.

If [RX CAT](#) is enabled, tuning the SDR receiver also tunes the external radio to the same frequency.

i NOTE

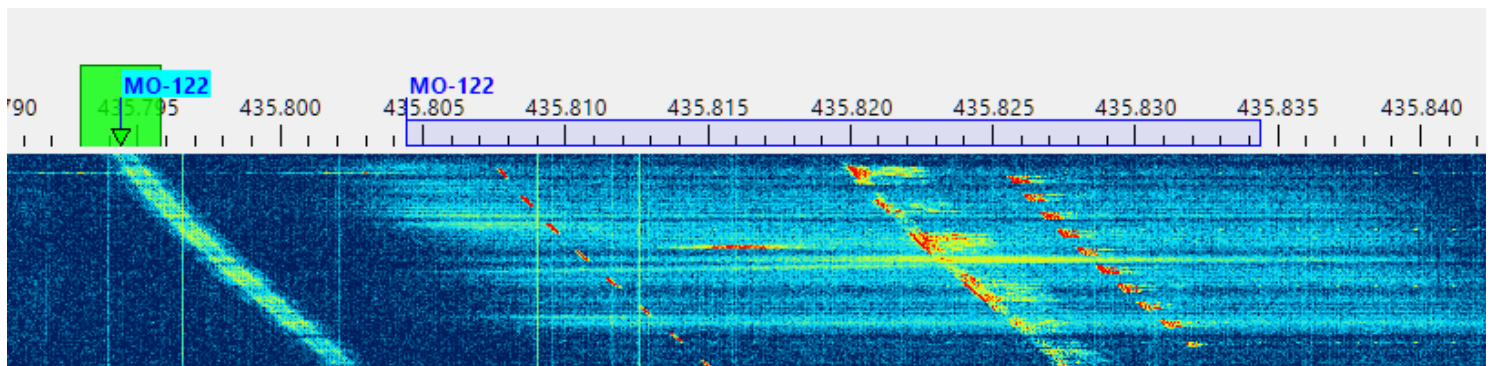
Get a mouse with a free-spinning wheel, such as Logitech MX Master 3S, this makes tuning much easier.



Manual Offset

The Manual Offset setting compensates for the transmitter frequency error, see [Frequency Control](#) and [Doppler Tracking](#) for details.

The offset value is usually the same for all transmitters of a satellite, so you can adjust it for some non-transponder transmitter before using the transponder. Most satellites with a transponder also carry a telemetry or beacon transmitter that you can use to set the manual offset. The screenshot below shows the telemetry signal of the MO-122 satellite, and the green rectangle perfectly aligned with its frequency by adjusting the Downlink Manual Offset:



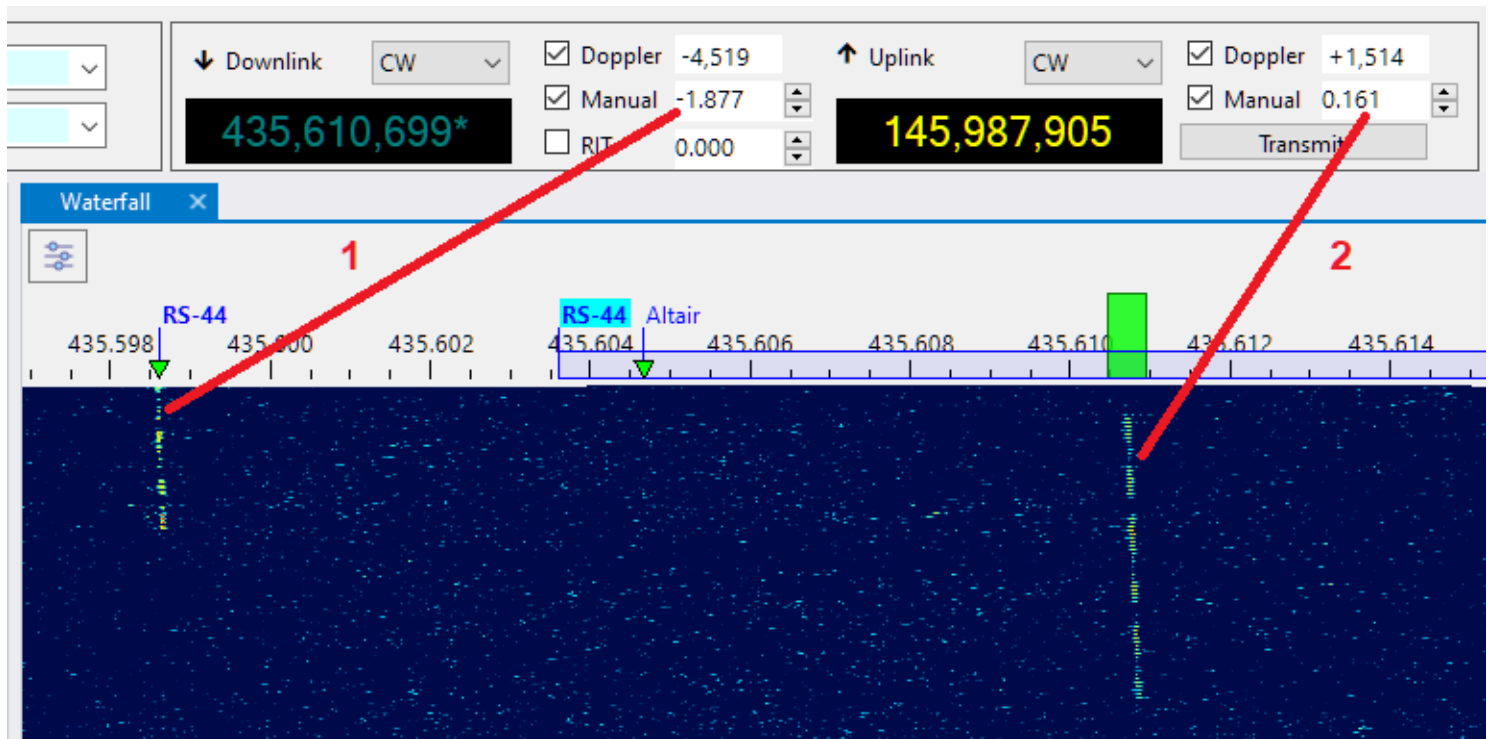
Aligning The Uplink and Downlink Frequencies

The screenshot below shows how to set Manual Correction for the uplink and downlink for a linear transponder, such as RS-44.

1. Select the beacon transmitter of the satellite and adjust the Downlink Manual Correction setting to align the transmitter label and its signal trace on the waterfall. For

RS-44 the required correction is about -1900 Hz. If your offset is significantly different from this value, this means that your SDR requires [PPM calibration](#).

2. Select the transponder transmitter and find a clear frequency within the transponder segment. Send a sequence of dots and adjust the Uplink Manual Correction to align the center of the green rectangle with the trace of your signals coming from the satellite.

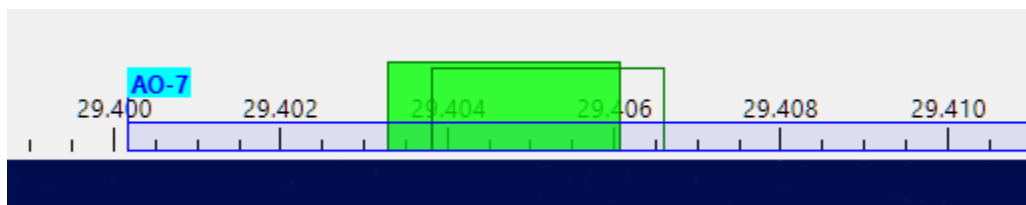


These adjustments need to be done only once. They stay the same, within a few tens of Hertz, between the satellite passes.

When TX CAT is enabled, RX CAT is disabled and the Ignore Dial Knob is set to false in the Settings, it is possible to adjust the Uplink Manual Offset using the dial knob on the radio.

RIT

While the RIT function could be controlled using the [Frequency Control](#) panel, it is more convenient to do this on the frequency scale:



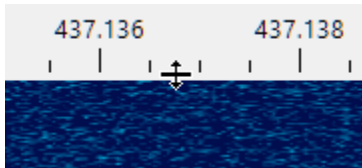
When RIT is enabled, the clear rectangle shows the current receiver passband, while the green rectangle stays on the main frequency.

- spin the mouse wheel on or near the green rectangle while holding the **Ctrl** key down: this enables RIT and tunes its offset;

- spin the mouse wheel on or near the green rectangle WITHOUT holding the **Ctrl** key down: this disables RIT and tunes the main frequency;
- right-click on or near the green rectangle to turn RIT on and off.

Resizing

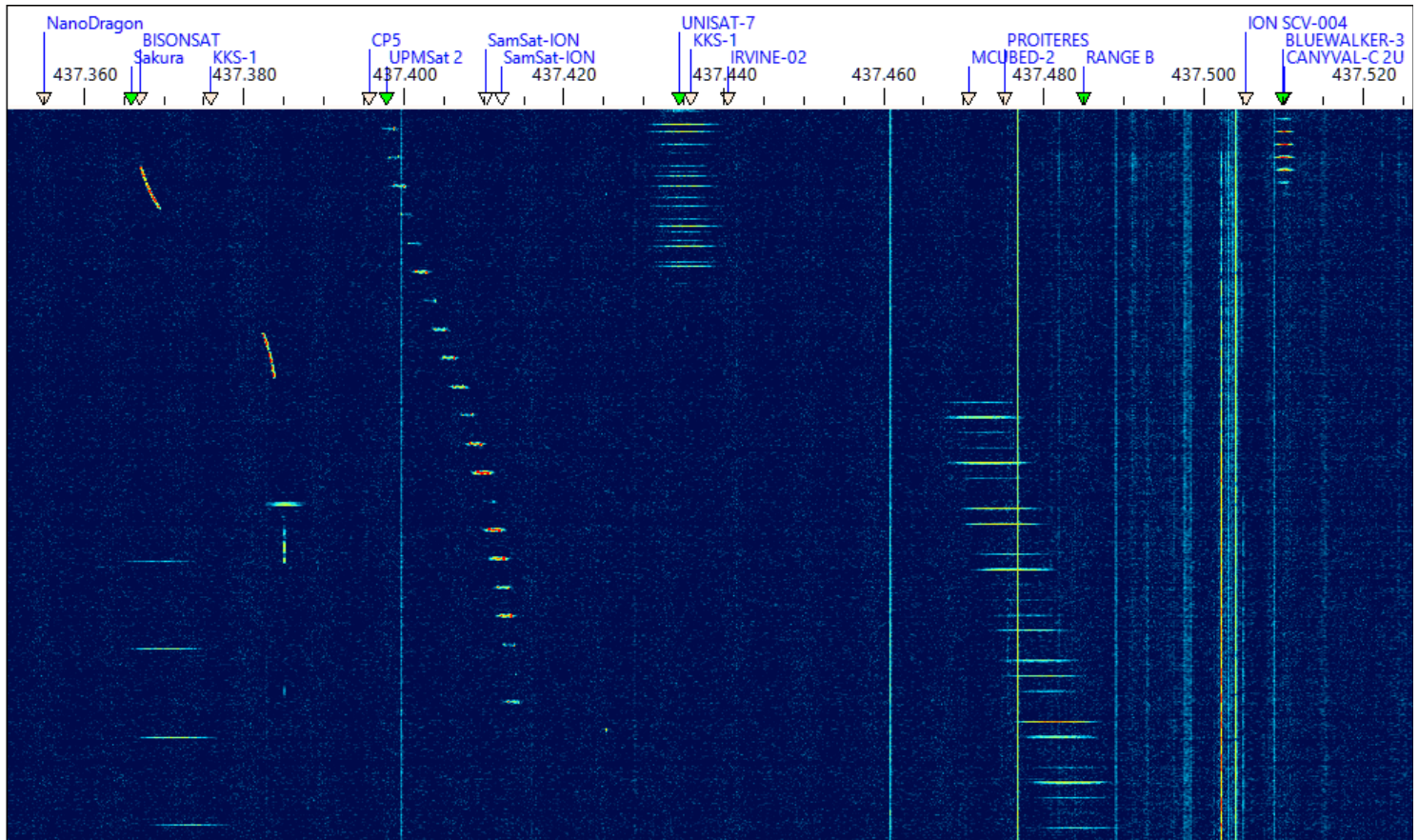
Adjust the height of the frequency scale by dragging the splitter between the scale and the waterfall:



Waterfall Display

Using Waterfall Display

The waterfall display and associated [Frequency Scale](#) is the central piece of SkyRoof that integrates most of the functions available in the application:



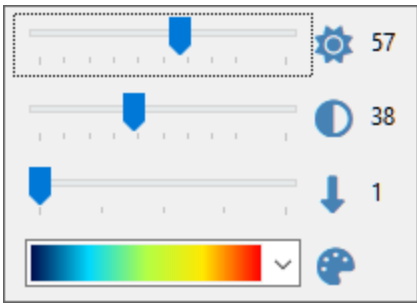
The waterfall spans over 3 MHz of spectrum (depending on the SDR model) and covers the whole satellite segment, 435-438 MHz, on the 70 cm band. On the 2 m band the satellite segment is only 200 kHz wide, 145.8-146 MHz, so it also fits completely in the waterfall.

- Zoom in and out using the mouse wheel
- Pan by dragging the waterfall horizontally with your mouse

A mouse-click on the waterfall display:

- tunes the SDR and external radio to a terrestrial signal
- or, if the frequency is within the transponder segment of a passing satellite, selects that satellite and sets the transponder offset to the clicked signal.

A click on the **Sliders** button in the top left corner of the panel opens the sliders that adjust brightness, contrast and scrolling speed of the waterfall, and select a color palette:



See Also

- [Frequency Scale](#)
- [Doppler Tracking](#)

Waterfall Display Characteristics

Finding and tracking satellite signals on the VHF and UHF bands is a difficult task. These signals are weak because the output power of most satellite transmitters is in the milliwatt range. The fact that an omnidirectional antenna needs to be used to receive all in-range satellites at the same time makes this task even more difficult. On top of that, we need to see the whole 3-MHz frequency segment where the satellite signals may appear, and at the same time we want a very high resolution to examine the structure of the signals, and to tune precisely to CW, SSB and digital transmissions.

The waterfall display in SkyRoof solves these problems by computing power spectra oversampled by a factor of about 100. This has three important consequences:

1. You can zoom in the waterfall display without changing its resolution in Settings, just by spinning the mouse wheel. When zoomed in to the maximum, you can see the signals with a **100x** magnification (20 Hz resolution).
2. When zoomed out to the maximum to see the whole 3-MHz segment, the sensitivity of the waterfall display to narrowband signals improves by about **15 dB** due to oversampling.
3. Spectrum oversampling, however, increases the requirements to the hardware:
 - more CPU power is needed to compute very large spectra;
 - more texture memory is needed in the video card to store the spectra;
 - support of OpenGL 3.3 or higher is required for the zoom function.

Please see the [System Requirements](#) section for the exact requirements.

If your computer does not meet these requirements, you can still use the pass prediction, Doppler correction, radio control and rotator control functions in SkyRoof, just disable the SDR function and close the Waterfall Display panel.

Waterfall Quality

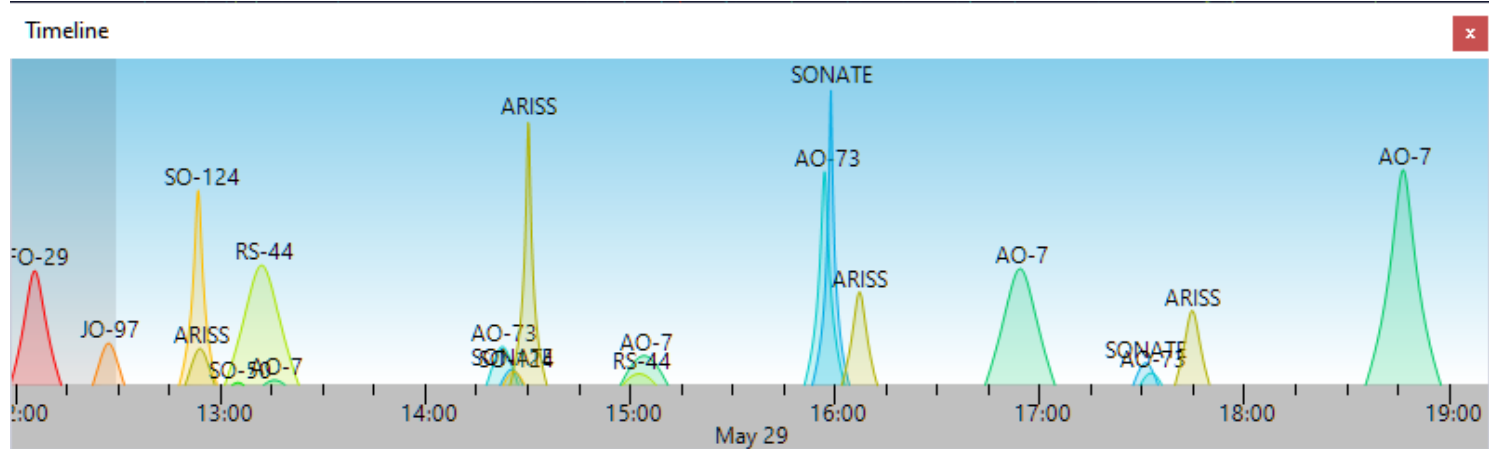
If the waterfall scrolls sluggishly or your video card struggles with the large oversampled spectra, you can lower the **Quality** setting. Open **Tools / Settings**, find the **Quality** field in the Waterfall section, and choose one of the following values:

- **Auto (maximum detail)** - the default. SkyRoof uses the largest spectrum and texture size your hardware can handle, giving the highest resolution and sensitivity.
- **High, Medium, Low (best for older GPUs)** - progressively reduce the internal spectrum and texture size. A lower quality requires less CPU power and less video card texture memory, which improves performance on older or low-end hardware.

The Quality setting only affects the internal resolution of the waterfall image. It does not change the SDR sampling rate, so all signals in the satellite segment remain visible at every quality level.

Time Line

The TimeLine panel shows the satellite elevation chart as a function of time for all satellites in the selected group:



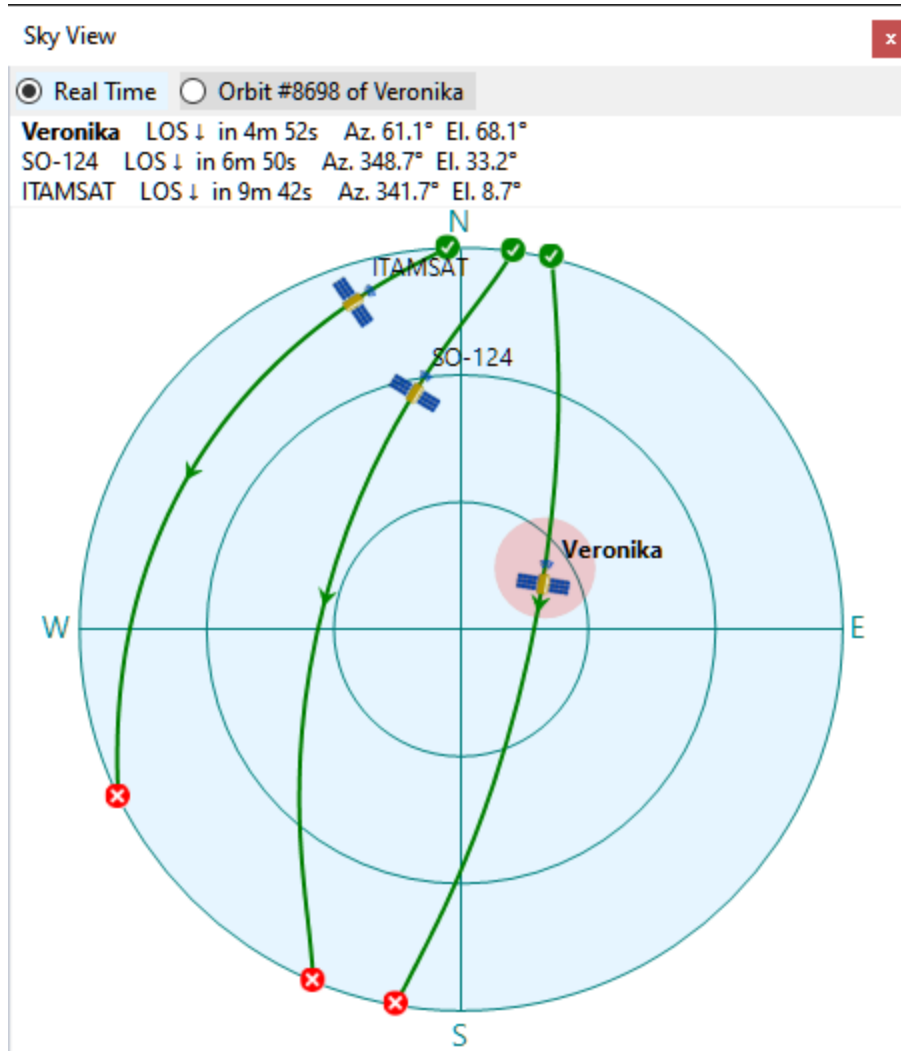
- Zoom in and out using the mouse wheel
- Pan by dragging the chart with your mouse
- Click on the satellite name to make it current, and to view the pass on the [Sky View panel](#).

The dark part of the chart represents the past time.

Sky View

The Sky View panel shows the trajectories of the satellites in the sky, as visible at your location:

The radio buttons at the top switch the chart between the real-time display showing all satellites in the selected group that are currently above the horizon, and a specific pass of a specific satellite.



The pink spot indicates the current antenna bearing if [Rotator Control](#) is enabled.

To select the pass to be displayed, click on it in the [Current Group panel](#), [Satellite Passes panel](#) or [Time Line panel](#).

Click on the satellite name next to the satellite icon to make it selected.

Earth View

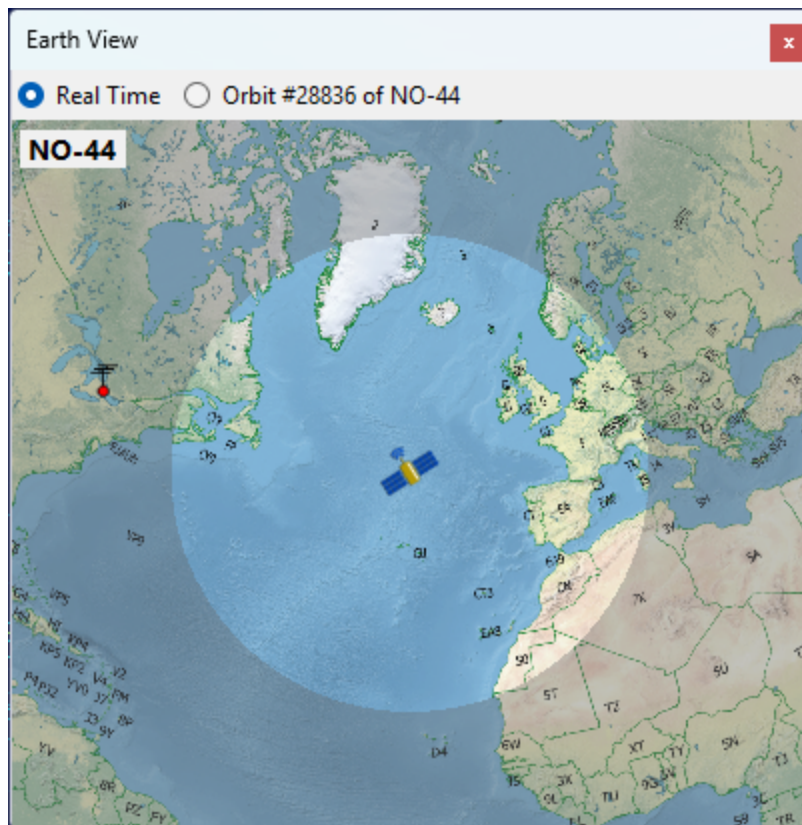
The Earth View panel shows the view of the Earth from the satellite. Use the radio buttons at the top of the panel to switch between two modes:

- **Real Time** centers the view on the satellite and highlights the area it can see at this moment.
- **Selected Pass** centers the view on your station and draws the coverage contour of the currently selected pass.

Use the mouse wheel to zoom the view in and out.

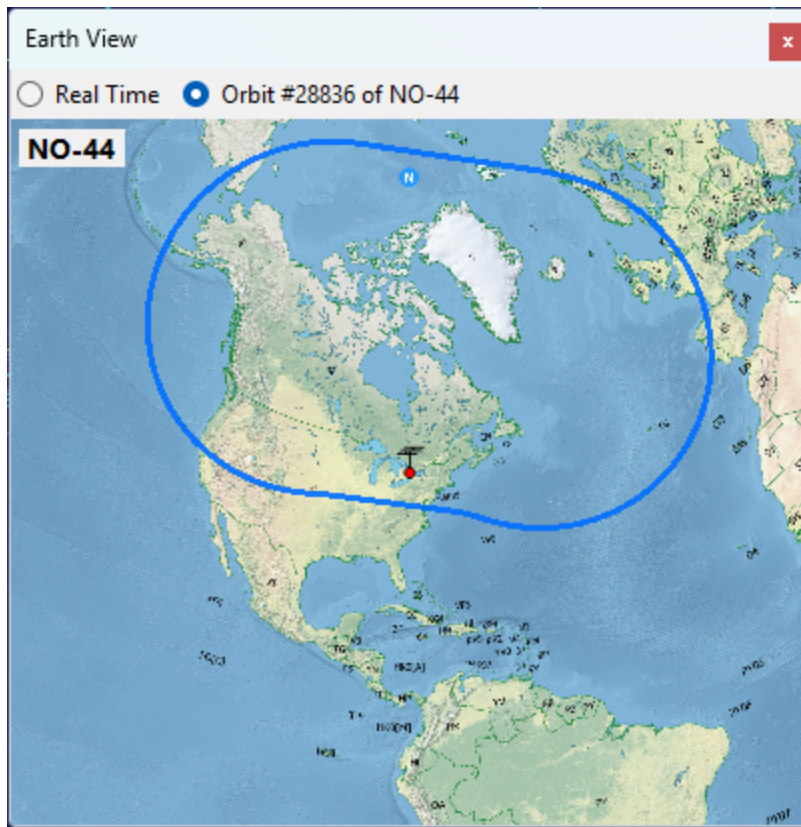
Real Time

The highlighted area is what the satellite can see from its current position. The satellite is above the horizon for the observers located in this area.

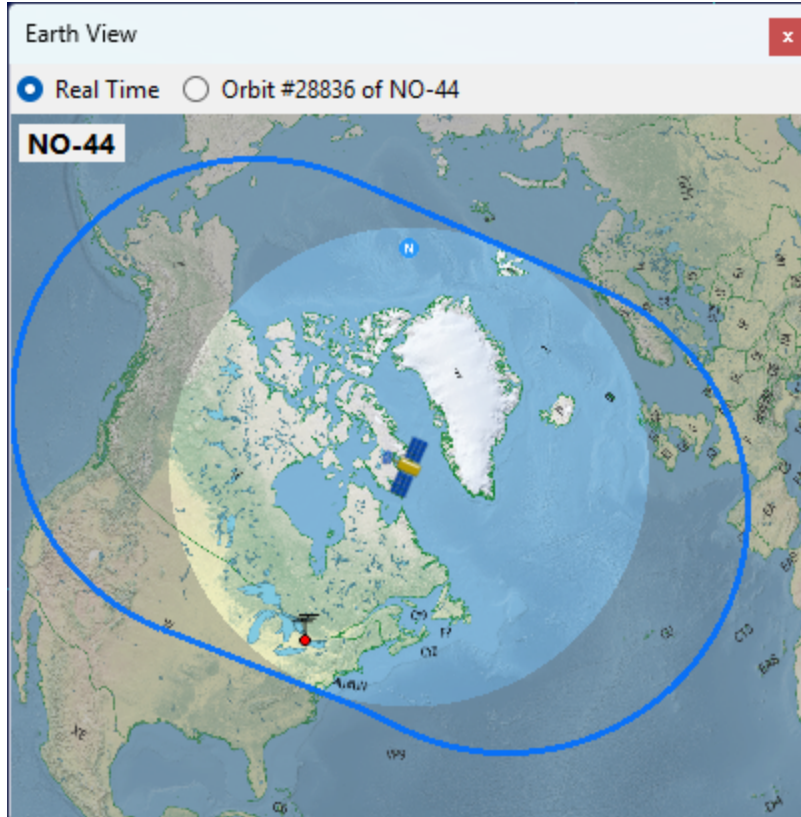


Selected Pass

The blue contour outlines the entire area swept by the satellite footprint during the pass. Any observer inside this contour will have the satellite above the horizon at some point during the pass.



While the pass is in progress, the panel switches back to Real Time mode so you can watch the satellite move across its coverage area:

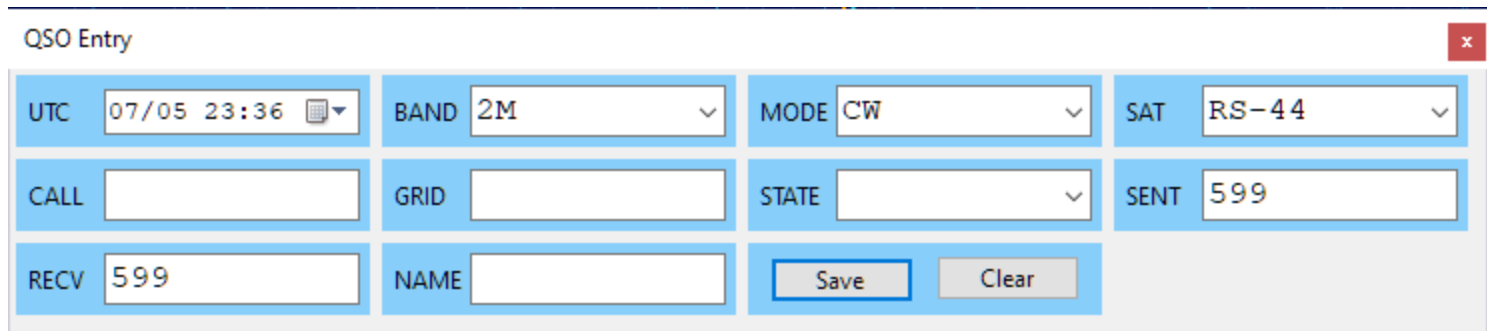


QSO Entry Panel

The QSO Entry panel allows the operator to enter the QSO details with minimum effort and save them to an ADIF file. This file may be later imported to the logging software for award tracking, QSL management, etc.

Configuring The Panel

Click on **View / QSO Entry** to open the panel:



The screenshot shows the 'QSO Entry' panel with a title bar and a close button. The panel contains several input fields and buttons:

UTC	07/05 23:36	BAND	2M	MODE	CW	SAT	RS-44
CALL		GRID		STATE		SENT	599
RECV	599	NAME		Save Clear			

You can either dock the panel somewhere in the main window as described in [Configuring Window Layout](#), or keep it floating. By resizing the panel you can arrange the input fields in a row, a column, or a matrix:

The image shows a software interface with two main panels. On the left is the 'QSO Entry' dialog box, which contains several input fields: 'UTC' (07/05 23:38), 'BAND' (2M), 'MODE' (CW), 'SAT' (RS-44), 'CALL' (empty), 'GRID' (empty), 'STATE' (empty), 'SENT' (599), 'RECV' (599), and 'NAME' (empty). At the bottom of this dialog are 'Save' and 'Clear' buttons. On the right is the 'Earth View' panel, which displays a map of the Americas. A label 'RS-44' is positioned at the top of the map, indicating the satellite's current position.

If you do not need some input fields, disable them in the Settings dialog:

The image shows the 'Settings' dialog box. The 'QSO Entry' section is expanded, showing two settings: 'Visible Fields' set to 'UTC, BAND, MODE, SAT, CALL, GRID, STATE, SE' and 'New file every' set to 'Month'. Below these settings is a preview area labeled 'QSO ENtry'. At the bottom of the dialog are 'OK', 'Apply', and 'Cancel' buttons.

Specify if a new ADIF file should be created every day, month or year, using the **New file every...** setting.

Entering The QSO Data

- **UTC** - this field shows the current UTC time. Click on it to freeze the clock or to make it run again, or enter the date and time manually if saving an old QSO;
- **Band, Mode, Sat** - These fields are populated automatically, based on the currently selected satellite transmitter. You can select different values from the drop-down lists,

or type them in, if needed;

- **Call** - as you enter the callsign, the program tries to guess its grid square, US state and operator's name (see below).
- **Grid, State** - if the program fails to guess these values, or guesses them incorrectly, enter them manually;
- **Sent, Recv** - sent and received reports. Auto-populated with default values when the mode is set, change them manually as needed;
- **Name** - auto-populated if known to the program, otherwise enter manually.

The fields with manually entered values have a blue frame around them, these fields are not auto-populated.

The minimum QSO record consists of **UTC, Band, Mode** and **Call**, all other fields are optional.

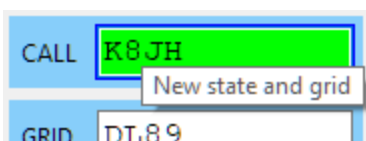
Information Lookup

When the QSO Entry panel opens, it loads all *.adi files in the **Adif** sub-folder of the **Data Folder** and builds the lookup lists of the grid squares, states and operator names associated with the callsigns. These lists are used to auto-populate the corresponding input fields. You can put an adif file with all your previous QSO in that folder to help the program build good lookup lists:

> Afreet > Products > SkyRoof > Adif			
Name ^	Date modified	Type	Size
 2025-07.adi	2025-07-05 19:17	ADI File	1 KB
 old_qso.adi	2025-07-04 19:58	ADI File	5,005 KB

Callsign Status

When the program loads the adif files, it also builds the lists of worked callsigns, grid squares and states. Only the QSO with the **PROP_MODE** field set to **SAT** are counted. As you enter the callsign, the color of the input box changes to indicate the status:



Interfacing with Logging Software

The QSO Entry panel can interface with logging software, using a simple plugin system, to enter the QSO directly to the log and to get the callsign status and lookup values. The plugin DLL has to export 4 functions:

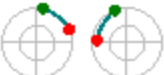
- Init();
- SaveQso(qso_data);
- GetStatus(qso_data);
- Augment(qso_data).

If you are the author of a logger, please contact me directly for details.

QSO Scheduler

The QSO Scheduler panel helps you plan a satellite QSO with a DX station by finding the time windows when the satellite is simultaneously visible from your location and from the DX location.

Click on **View / QSO Scheduler** to open the panel:

QSO Scheduler				
Satellite	RS-44	DX Grid Square	JO20EU	
RS-44 #29969			in 9h 56m 45s	
2026-05-27 06:52:30 to 07:02:13	9 min	Me: 16°, DX: 10°		
RS-44 #29970			in 11h 59m 08s	
2026-05-27 08:54:54 to 08:56:45	1 min	Me: 6°, DX: 2°		
RS-44 #29974			in 19h 38m 11s	
2026-05-27 16:33:56 to 16:43:34	9 min	Me: 3°, DX: 13°		
RS-44 #29982			in 34h 23m 00s	
2026-05-28 07:18:45 to 07:26:28	7 min	Me: 23°, DX: 7°		
RS-44 #29987			in 44h 2m 22s	
2026-05-28 16:58:07 to 17:10:29	12 min	Me: 7°, DX: 7°		
RS-44 #29994			in 56h 47m 43s	
2026-05-29 05:43:28 to 05:54:20	10 min	Me: 4°, DX: 18°		
RS-44 #29995			in 58h 49m 24s	
2026-05-29 07:45:09 to 07:50:32	5 min	Me: 20°, DX: 4°		
RS-44 #30000			in 68h 27m 27s	
2026-05-29 17:22:12 to 17:31:01	7 min	Me: 12°, DX: 2°		

Selecting the Satellite and the DX Station

- **Satellite** - select the satellite for the QSO from the drop-down list. The list contains the satellites in the currently selected group, see [Creating Satellite Groups](#);
- **DX Grid Square** - enter the 4- or 6-character Maidenhead grid square of the DX station. The input box turns pink while the value is incomplete or invalid.

Your own grid square and altitude are taken from the user details in the [Settings Window](#). The DX station is assumed to be at sea level.

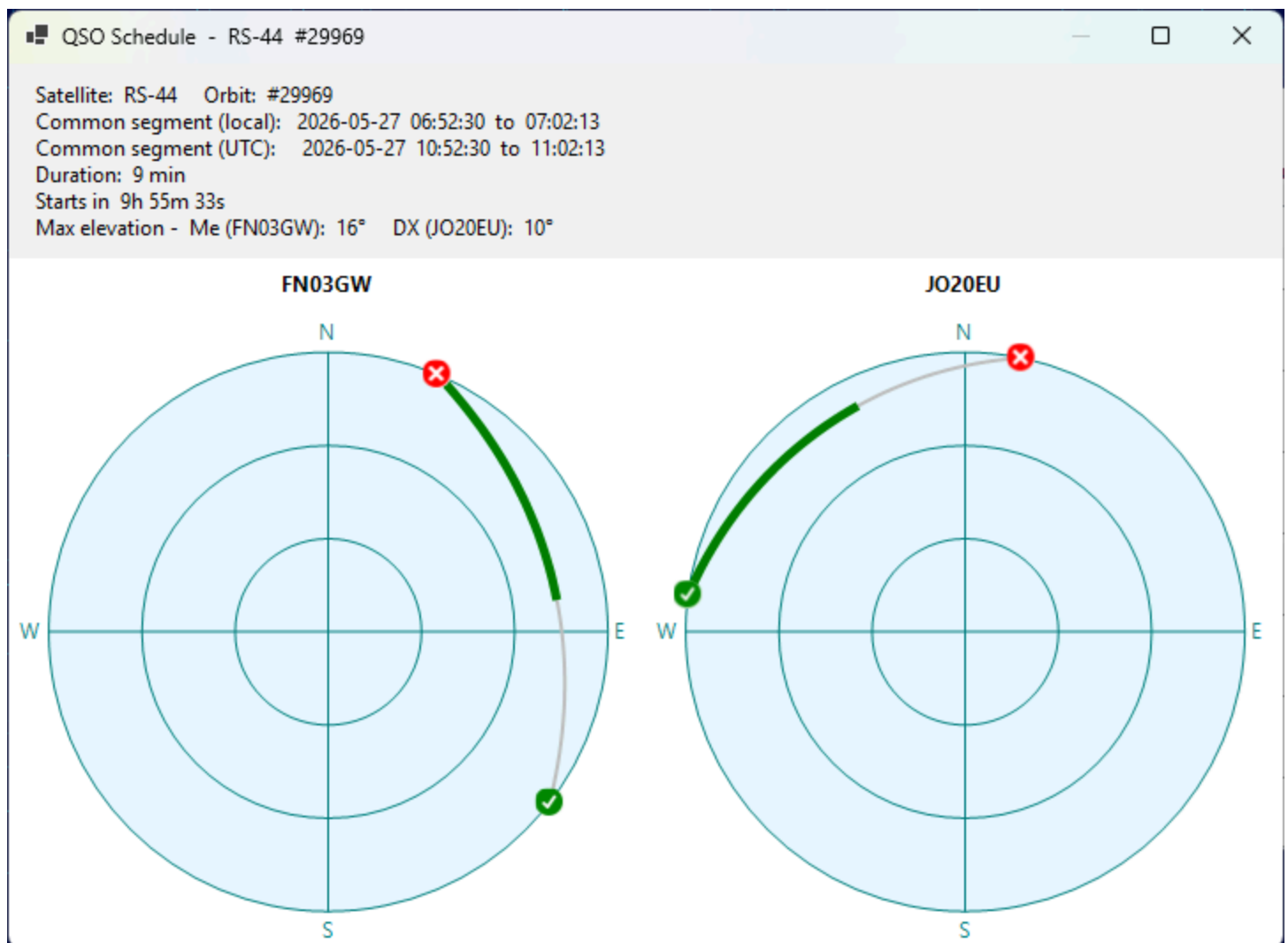
Reading the Prediction List

Each item in the list represents one common visibility window for the next 14 days, sorted by start time:

- **Satellite name** is highlighted in light yellow for VHF downlinks and light cyan for UHF downlinks;
- **Orbit number** follows the satellite name;
- **Wait time** on the upper right shows how long until the window starts, or **Now** in green if the window is already in progress;
- **Start and end times** of the common segment are displayed in local time (the label shows **Geostationary** instead for geostationary satellites);
- **Duration** of the common segment and the maximum elevation of the satellite from each station during that segment;
- **Two mini sky-views** on the right show the satellite path during the common segment, as seen from your station (left) and from the DX station (right). The green dot marks the start of the segment, the red dot marks the end.

Pass Details

Click on a list item to open the QSO Schedule window with more information about the pass:



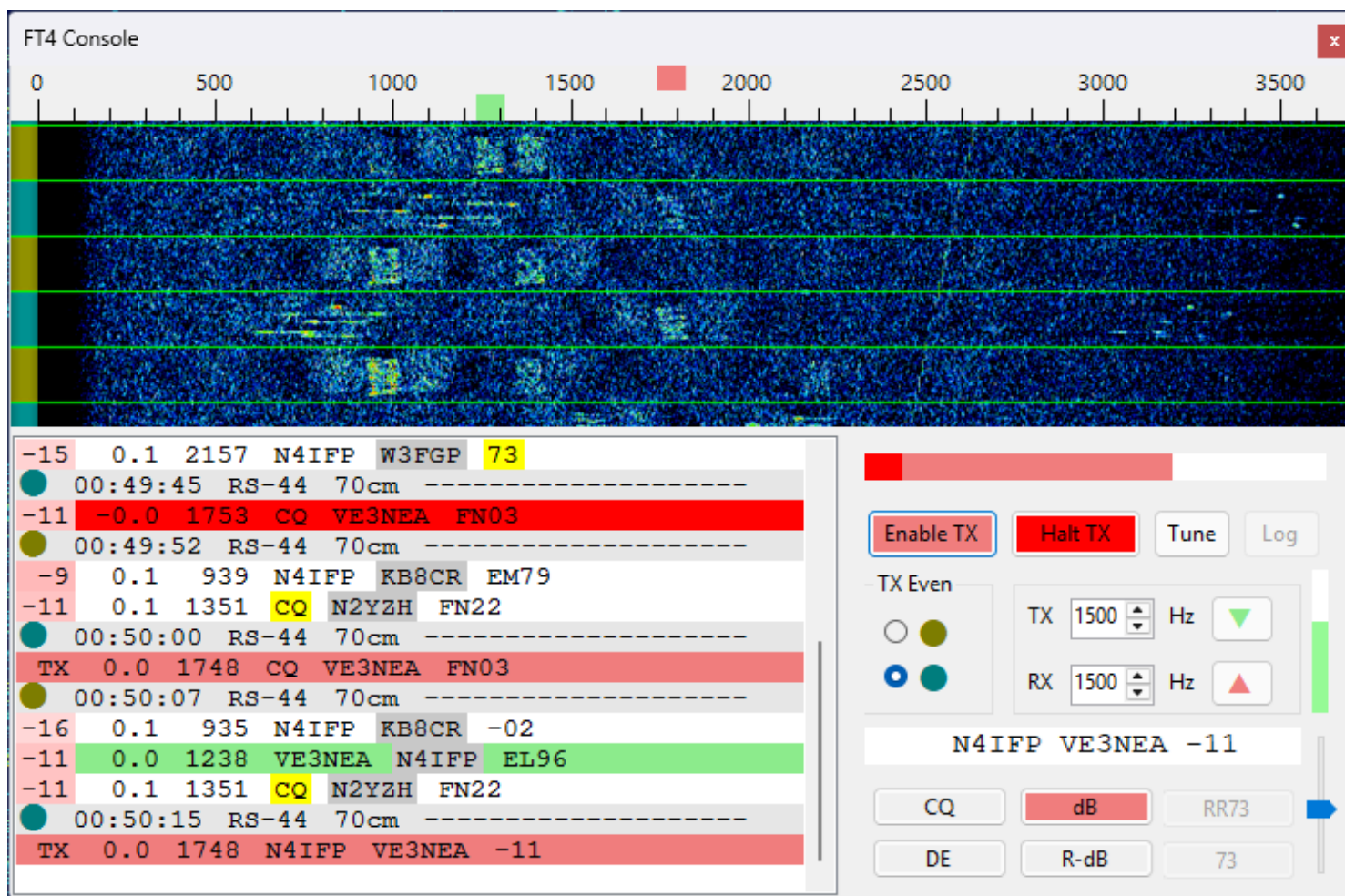
The window displays the full sky view for the pass from each station, with the entire pass arc drawn in silver and the common segment highlighted in green. The green check mark indicates the AOS point, the red X mark indicates the LOS point. The text above the charts lists the satellite, orbit number, common segment in local and UTC time, duration, time until the segment starts, and the maximum elevation of the satellite from each station during the common segment.

FT4 Console Panel

FT4 Console provides a streamlined, satellite-optimized interface for conducting FT4 QSOs via satellites with linear transponders (e.g., RS-44, AO-73, JO-97). While similar results can be obtained by running two instances of WSJT-X side by side with SkyRoof, the Console removes the need for external programs and simplifies the workflow. It offers satellite-specific features not available in general-purpose FT4 software, and an intuitive user interface that improves efficiency and usability.

Be sure to configure the [FT4 Console settings](#) before you start using the FT4 Console panel.

User Interface



Audio Waterfall Display

Click with the left mouse button on the waterfall or on the frequency scale to set RX and TX audio tones:

- **click** - set the RX tone;
- **Shift-click** - set the TX tone;
- **Ctrl-click** - set the RX and TX tones.

Move the mouse cursor over an FT4 signal trace on the waterfall to see sender's callsign.

Message List

The message list shows transmitted and received messages, including your own messages coming back from the satellite.

The list auto scrolls as the new messages are added. To disable auto-scrolling, for example, when looking at the old messages, just scroll the list up manually. To resume auto-scrolling, scroll it down to the last entry.

When the mouse cursor is moved over the list, auto-scrolling is temporarily disabled to prevent clicks on a wrong entry. The frozen status is indicated by a blue frame around the list panel. The list un-freezes when you click on an entry or move the cursor out of the list, and when the cursor has not been moving for 2 seconds.

When the mouse cursor is over a received message, a label on the waterfall display indicates the frequency of the decoded signal and sender's callsign.

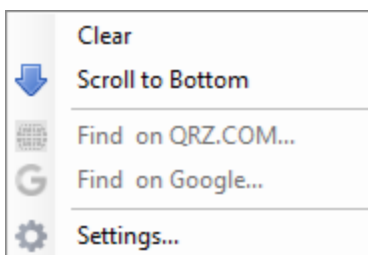
Right-click on the message list to open the popup menu. This menu has commands to clear the list and to scroll down to the bottom.

The first column in the received messages is the SNR of the received signal. It is color-coded, with weak signals represented with pale colors and strong ones with bright colors.

Sender's callsigns are color-highlighted based on their status from the logging interface. See [QSO Entry](#) for information about callsign colors.

The popup menu has commands to

- clear the message list;
- scroll to the bottom and enable auto-scrolling;
- find the callsign on QRZ.COM or on Google;
- open the Settings dialog and expand the FT4 settings.



Control Area

- **Enable TX button** - enable transmissions;

- **Halt TX button** - abort current transmission;
- **Tune button** - transmit a tone for tuning. This button works only when Enable TX is turned off. Similarly, the Enable TX works only when Tune is off;
- **Log button** - log the current QSO. Disabled until the first QSO is made;
- **TX Odd/Even** - select odd or even time slots for transmission;
- **TX #### Hz** - transmitted audio tone;
- **RX #### Hz** - primary received audio tone;
- **up and down arrow buttons** - set RX tone equal to TX tone or vice versa. The buttons are color-coded, e.g., if you click on a green arrow, the green maker on the frequency scale moves;
- **amplitude bar** - shows the peak amplitude of the input audio, see the **FT4 Console Settings** section for details;
- **TX Gain slider** - allows you to adjust the amplitude of FT4 audio signals sent to the radio;
- **message text** - the current message to be transmitted;
- **message buttons** - allow you to select the message to transmit. A single click on a button changes the message immediately, even during the transmission. Only the messages applicable in the current stage of the QSO are enabled. The currently selected message is shwon in pink.

Here are the examples of the messages:

Run Sequence

Button	Example
CQ	CQ VE3NEA FN03
dB	AA0AA VE3NEA -10
RR73	AA0AA VE3NEA RR73

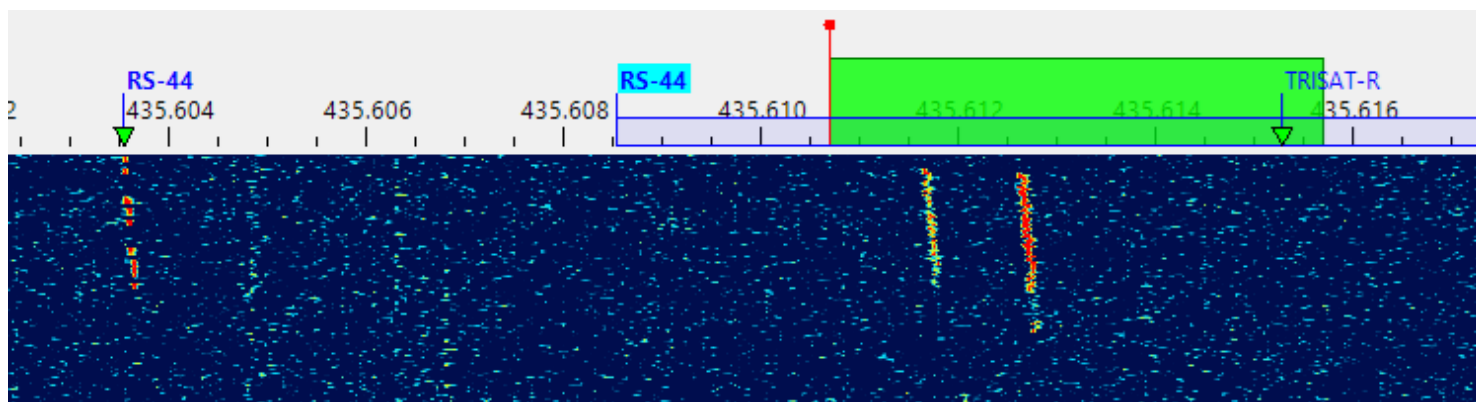
S&P Sequence

Button	Example
DE	AA0AA VE3NEA FN03
R-dB	AA0AA VE3NEA R-10
73	AA0AA VE3NEA 73

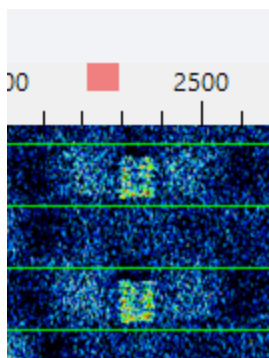
Making Contacts

At the time of this writing RS-44 was the best choice for FT4 contacts, so we will use this satellite in the examples below. When the satellite rises above the horizon and you start receiving its signals, do a quick check:

- verify your Downlink Manual Correction. This is easy to do with an SDR receiver: the label of the RS-44 CW beacon, shown on the left hand side in the screenshot below, should be aligned with its signal trace on the wideband waterfall display. If there is a mismatch, you can correct it as described in the [Frequency Scale](#) and [Frequency Control](#) sections;
- check that the receiver (SDR or external) is tuned to the frequencies where the FT4 signals are present. Usually FT4 stations operate a few kHz above the lower end of the transponder passband. You can just drag your receiver passband (green rectangle) to the required frequency.

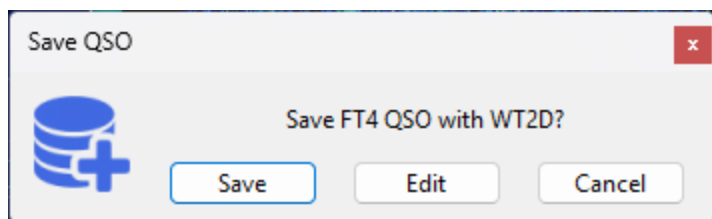


Now you can click on Tune or Enable TX to start transmitting. If all is good, you will receive your own signals back from the satellite:



The received signals are often slightly offset from the intended frequency (the pink mark on the scale), as shown above. This should be corrected by adjusting Uplink Manual Correction. You can do this manually in the [Frequency Control](#) widget, or automatically, by **Ctrl-click**'ing on your own received message (with red background by default) in the message list. Automatic correction works only if the error is less than 1 kHz. If it is greater than that, you have to find out the cause. Large frequency errors could be caused by bad SDR calibration, inaccurate system clock, old TLE data, wrong downlink offset correction, etc.

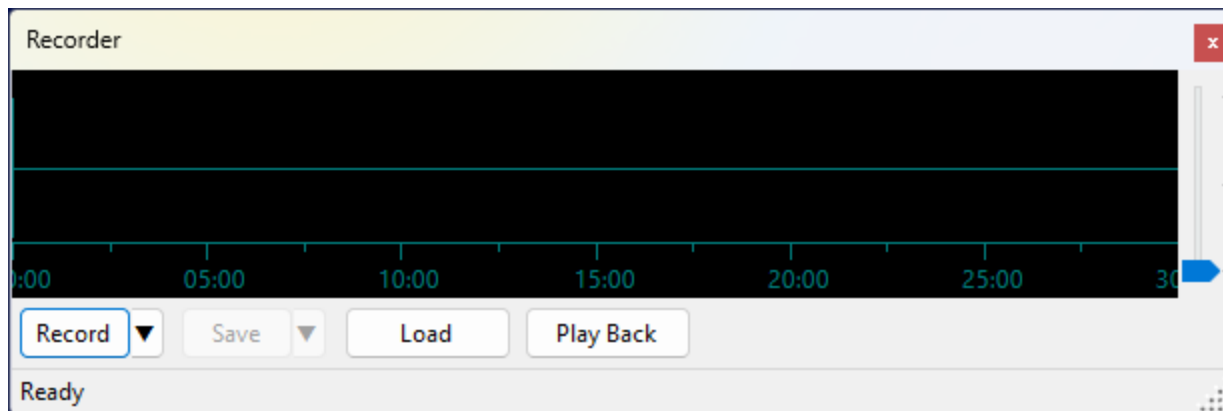
Wait until someone answers your CQ, or call another station by clicking on their message. A single click is enough, no need to double-click. The QSO sequence is followed automatically, and the Save QSO dialog pops up when the QSO is completed:



Click on **Save** to save the QSO , or click on **Edit** to load the QSO details in the QSO Entry form. See [QSO Entry](#) for information about editing and saving QSO.

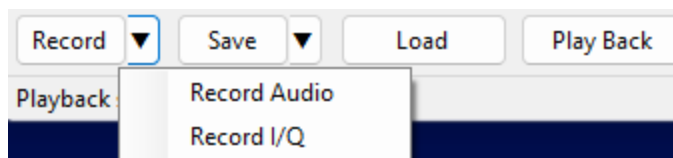
Recorder Panel

The Recorder panel records the received audio or I/Q signal and plays it back later. Up to 30 minutes of data can be stored in the recording buffer.



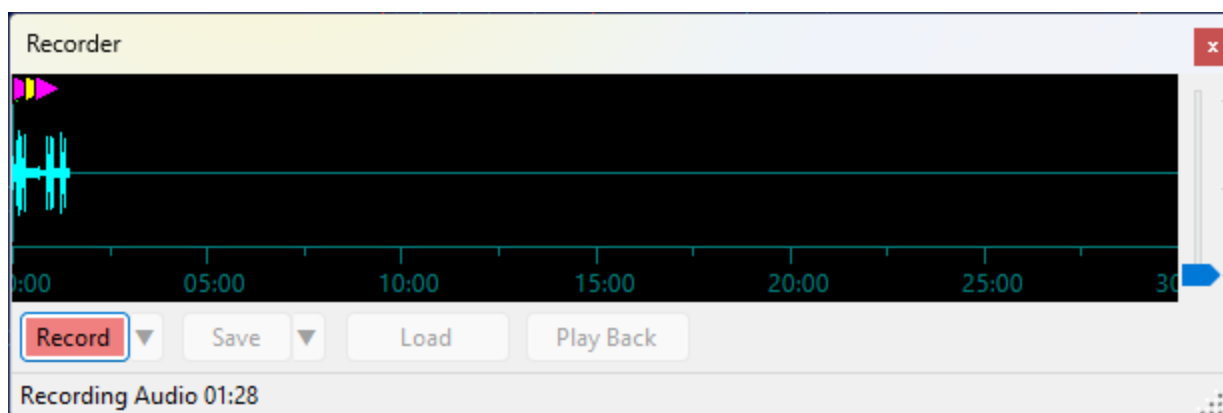
Recording

Click **Record** to start recording audio. To choose what to record, click the ▼ button next to Record:



- **Record Audio** — records the demodulated audio signal (mono);
- **Record I/Q** — records the complex baseband I/Q signal (stereo).

The button turns red and the status bar shows the recording type and elapsed time while recording is active. Click **Record** again to stop.



The waveform display shows the recorded signal growing from the left. The full 30-minute buffer width is always shown during recording, so the waveform expands to fill it as the recording progresses.

Event Markers

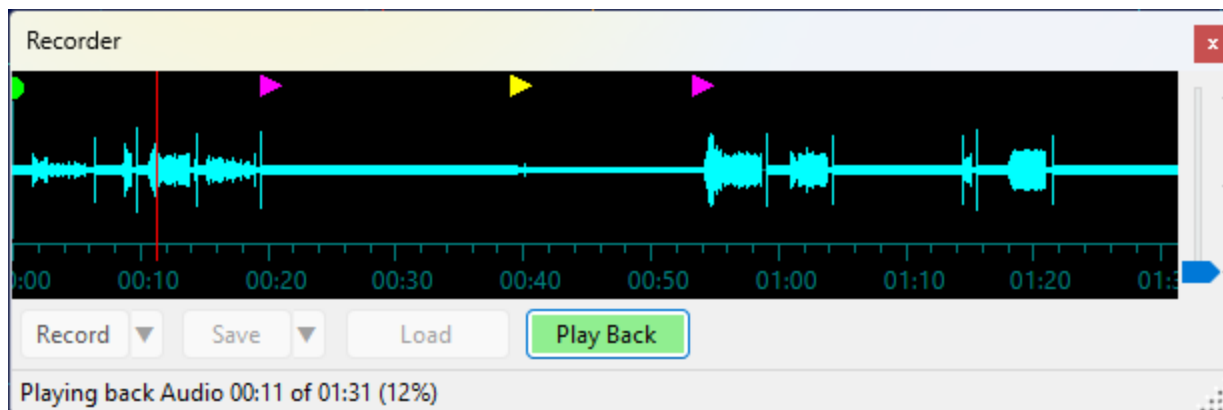
While recording is active, the panel automatically tracks changes in the selected satellite, transmitter, and mode, as well as QSOs saved in the [QSO Entry](#) panel. These events appear as colored markers along the top edge of the waveform:

- **green circle** — satellite selection changed;
- **magenta triangle** — transmitter selection changed;
- **yellow triangle** — demodulation mode changed;
- **cyan square** — QSO saved.

Move the mouse cursor over a marker to see a tooltip with the details of that event.

Playback

Click **Play Back** to play back the recording. The button turns green and a red vertical line moves across the waveform to show the current position. The status bar shows the elapsed time, total duration, and progress percentage.

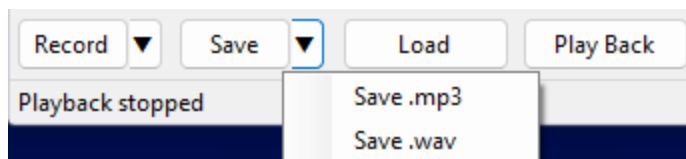


Click anywhere on the waveform during playback to seek to that position.

Click **Play Back** again to stop playback.

Saving and Loading

Click **Save** to save the recording. To choose the file format, click the ▼ button next to Save:



- **Save .mp3** — available for audio recordings only; saves a compressed MP3 file resampled to 16 kHz;

- **Save .wav** — saves a 48 kHz 32-bit float WAV file. Audio recordings are saved as mono; I/Q recordings are saved as stereo (left channel = I, right channel = Q).

Click **Load** to load a previously saved .wav or .mp3 file into the buffer for playback.

The Save and Load dialogs open in the **Recordings** subfolder of the [data folder](#) by default.

Playback Routing

During playback, audio is sent to the same soundcard that is selected in [Settings](#) for SDR audio output. Both audio and I/Q are also routed through the same output channel (VAC or UDP) that is configured for the [Output Stream](#) function in the [Settings](#), so external applications can receive the played-back signal just as they would a live signal.

Waveform Display

The vertical gain slider on the right edge of the waveform adjusts the display amplitude in dB, making quiet signals easier to see. It does not affect the recording or playback level.

How to Decode Telemetry from PEARL-1C

I/Q or Audio data, streamed via VAC or UDP, may be used to decode telemetry transmitted by the satellites. There is a number of telemetry decoders to choose from. One such decoder is [gr_satellites.exe](#) command line tool, its installation instructions are [here](#).

This command runs gr_satellites.exe to decode telemetry of the PEARL-1C satellite using the I/Q UDP stream from SkyRoof:

```
(base) C:\Ham>gr_satellites 58342 --udp --udp_port 7355 --udp_raw --iq --samp_rate 48e3 --hexdump
```

This command decodes the same satellite via I/Q output to VAC :

```
(base) C:\Ham>gr_satellites 58342 --audio "CABLE Output (VB-Audio Virtual Cable)" --samp_rate 48000 --iq
```

To use the second command, you will need a virtual audio cable, such as [VB-Audio](#).

Run one of these commands, then in SkyRoof [settings](#):

- select either I/Q to VAC or I/Q to UDP, depending on the command you use;
- set Gain, dB to 0;
- select the VAC in the list of audio devices;
- click on the Output Stream label on the status bar to enable the output.

Example output from gr_satellites.exe v.5.7.0:

```
pagesize :debug: Setting pagesize to 4096 B
top_block_impl :debug: Using default scheduler "TPB"
udp_source :info: Listening for data on UDP port 7355.
***** VERBOSE PDU DEBUG PRINT *****
((transmitter . 9k6 FSK downlink))
pdu length =          64 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: f9 11 01 83 43 33 a9 e7 a4 10 00 00 11 00 00 00
0020: 00 00 00 00 7a 0f 01 00 00 00 00 00 00 00 00 00
0030: 05 00 00 00 77 fb 01 00 67 aa 00 00 00 00 00 00
*****
***** VERBOSE PDU DEBUG PRINT *****
((transmitter . 9k6 FSK downlink))
```

```
pdu length =          88 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: fb 11 01 81 43 33 00 00 00 00 00 00 00 00 00 00
0020: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0030: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0040: 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00 00
0050: 00 00 00 00 00 00 00 00 00
*****
***** VERBOSE PDU DEBUG PRINT *****
((transmitter . 9k6 FSK downlink))
pdu length =          56 bytes
pdu vector contents =
0000: 9c 86 aa 8e a6 62 e0 a0 8a 82 a4 98 86 e1 03 f0
0010: 06 11 01 82 43 33 83 8a 01 00 9e 00 00 00 04 00
0020: 00 00 88 ff f8 ae 00 00 80 01 00 00 0f 02 00 00
0030: 00 00 00 00 00 00 00 00
```

How to Receive Images from NOAA-15 and NOAA-19

An external program is needed to decode and display the weather images transmitted by the NOAA-15 and NOAA-19 satellites. In this tutorial we will use [WXtoIMG](#).

Installing VAC

A virtual audio cable, VAC, is required to pass the satellite signals demodulated in SkyRoof to the image decoding program. Download and install [VB-Audio](#) if you do not have it yet, and reboot your system. Be sure to get the latest version (2024) of VB-Audio, the old version may not work correctly.

Setting Up WXtoIMG

- download [wxinst21102-beta.exe](#) and run the installer, then start WXtoIMG;
- enter your latitude and longitude when prompted;
- the program tries to download the TLE data from a URL that no longer works. Download the [weather.txt](#) file manually and save it to the WXtoIMG data folder. Type this in File Explorer to find the data folder:

```
%appdata%\WXtoImg
```

- tick **Satellite / Autodetect APT** in the menu;
- click on **Options / Recording Options** and select the **Record only when active...** radio button. Select the VAC cable in the **soundcard** drop-down list;
- click on **File / Record** in the menu, then
 - select **Record and auto process**;
 - tick **Create images**;
 - click on the **Auto Record** button.

Configuring SkyRoof

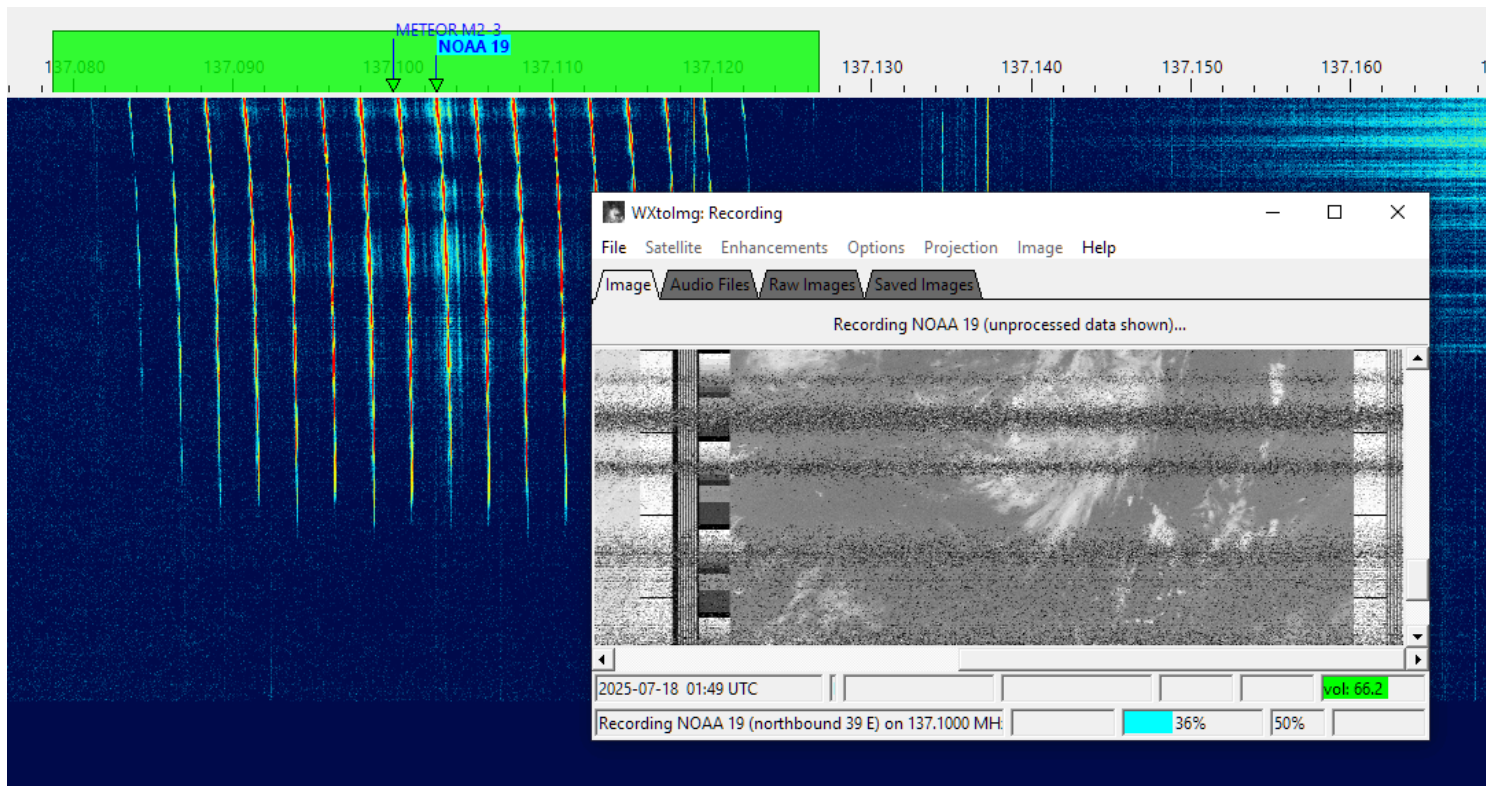
In SkyRoof:

- select the NOAA-15 or NOAA-19 satellite. If it is not in the current group, add it using the [Satellites and Groups](#) dialog;
- Select **APT** transmitter on the 137 MHz band from the list of transmitters;

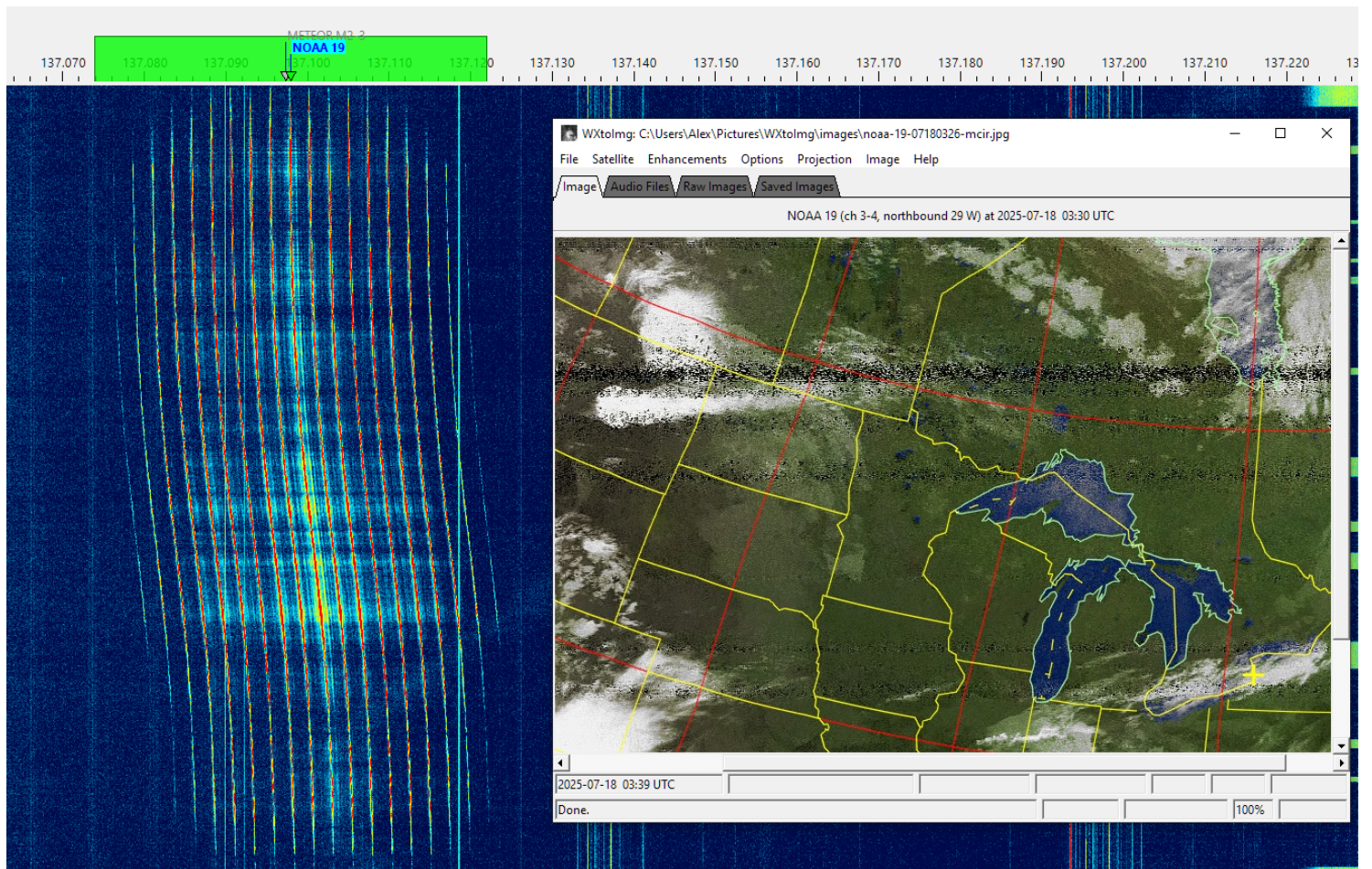
- Select **FM_D** downlink mode in the drop-down list on the toolbar;
- in the **Output Stream** section of the [Settings](#) window:
 - select **Audio to VAC**;
 - set Gain, dB to 0;
 - select the VAC in the list of audio devices;
 - click on the Output Stream label on the status bar to enable the output.

Receiving Images

When the selected satellite raises above the horizon, WXtoIMG will start decoding and produce an image like this:



At the end of the pass WXtoIMG will post-process received data and display an enhanced image:



How to Receive SSTV from ISS

There are two popular programs that demodulate SSTV: **MMSSTV** and **RX-SSTV**, either one may be used with SkyRoof to receive SSTV transmissions from ISS.

Installing VAC

A virtual audio cable, VAC, is required to pass the satellite signals demodulated in SkyRoof to the SSTV decoding program. Download and install [VB-Audio](#) if you do not have it yet, and reboot your system. Be sure to get the latest version (2024) of VB-Audio, the old version may not work correctly.

Setting Up MMSSTV

- download [MMSSTV](#), install it and run the program;
- click on **Option / Setup MMSSTV** in the menu to open the settings dialog, and:
 - click on the **Misc** tab;
 - select your VAC in the **Sound Card / In** drop-down list;
 - enter "48000" in the **Clock / Hz** box.
- in the main window:
 - click on the **RX** tab;
 - in the **RX Mode** panel right-click on the second and third button, and select "PD120" and "PD180" respectively;
 - click on **RX Mode / Auto**.

Setting Up RX-SSTV

- download [RX-SSTV](#), install it and run the program;
- in the **Sound Card Selection** window that pops up on start-up select your VAC as **Sound Input**;
- click on **Setup / SSTV Engine** in the menu, click on the **Misc** tab, and set **Clock / Hz** to "12000";
- click on **Setup / RX-SSTV** in the menu, under **User-defined buttons** set the first two buttons to "PD120" and "PD-180";
- in the main window:
 - click on the **RX** tab;
 - click on the **PD120** or **PD180** button, depending on the mode used at ISS (see below).

Setting Up SkyRoof

In SkyRoof:

- select the **ARISS** satellite. If it is not in the current group, add it using the [Satellites and Groups](#) dialog;
- Select ISS transmitter that is used to send SSTV (see below);
- Select **FM** downlink mode in the drop-down list on the toolbar;
- in the **Output Stream** section of the [Settings](#) window:
 - select **Audio to VAC**;
 - set Gain, dB to 0;
 - select the VAC in the list of audio devices;
 - click on the Output Stream label on the status bar to enable the output stream.

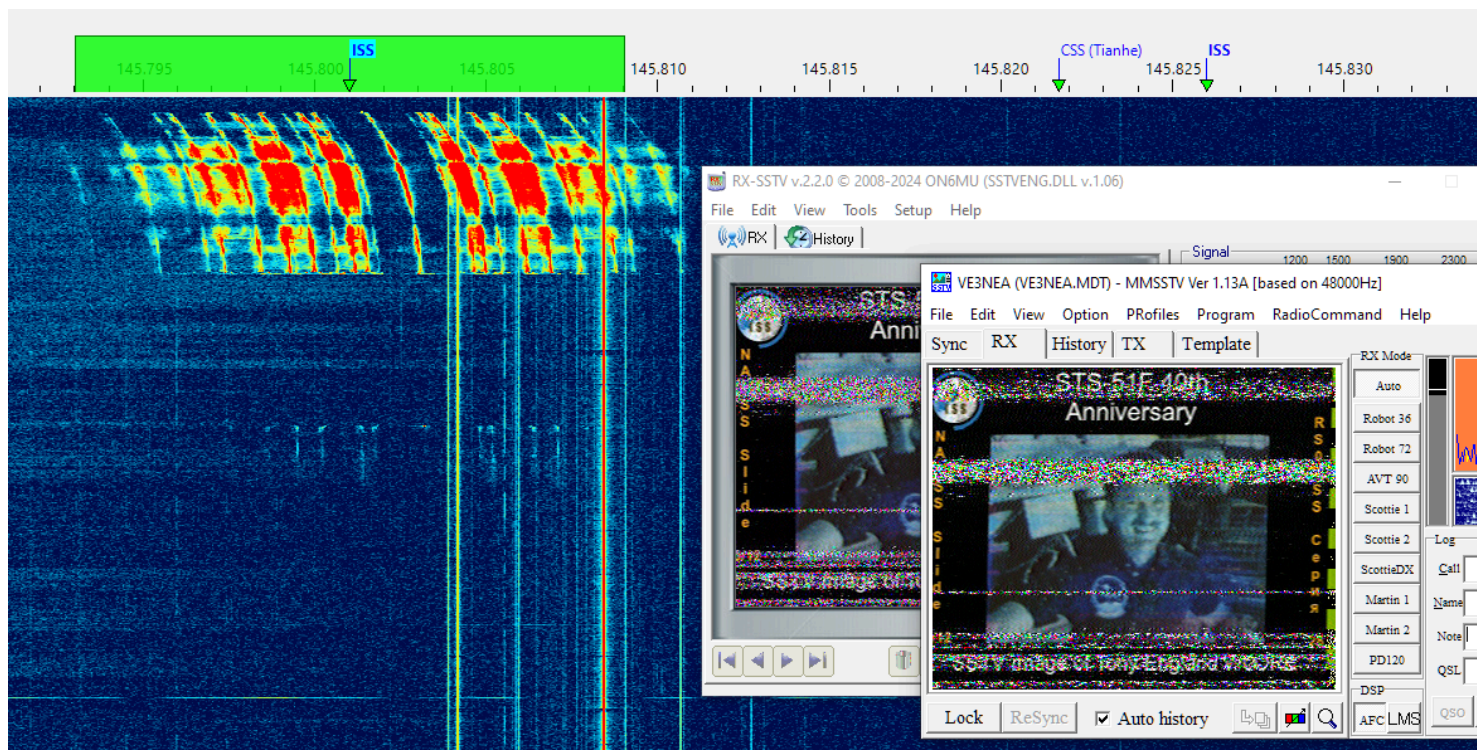
Receiving SSTV

Find out when the next SSTV activity on ISS will take place, usually it happens every few months and lasts for up to a week. One good source of information about such activities is the [ARISS web site](#)[↗]. You will need to know:

- when SSTV signals will be transmitted;
- which transmitter on ISS will be used, usually it is one of these:
 - **Mode U - SSTV** on 437.800 MHz;
 - **Mode V Imaging** on 145.800 MHz;
- which SSTV mode will be used, usually it is either **PD120** or **PD180**. In these modes the image transmission takes 120 s and 180 s respectively.

In SkyRoof, select ARISS in the list of satellites, and the transmitter that is used in the current SSTV activity. When ISS rises above the horizon and its signals start to appear on the waterfall, the RX-SSTV or MMSSTV program automatically starts decoding the image. If for some reason it doesn't, click on the SSTV mode button (PD120 or PD180) to start it manually.

Here is an example of an image received from ISS in July 2025. Both SSTV decoders seem to have the same image quality, which is mainly determined by the signal strength variations:



How to Receive FSK & AFSK Telemetry and other info using UZ7HO packet radio TNCs

by Marcus PY2PLL

An external program is needed to decode and display common information usually disseminated using FSK or AFSK. In this tutorial the UZ7HO sound modems will be used. More information about these modems can be found at the [UZ7HO Personal page](#) under **Packet-Radio** - English version.

Installing VAC

A virtual audio cable, VAC, is required to pass the satellite signals demodulated in SkyRoof to the decoding program. Download and install [VB-Audio](#) if you do not have it yet, and reboot your system. Be sure to get the latest version (2024) of VB-Audio, the old version may not work correctly.

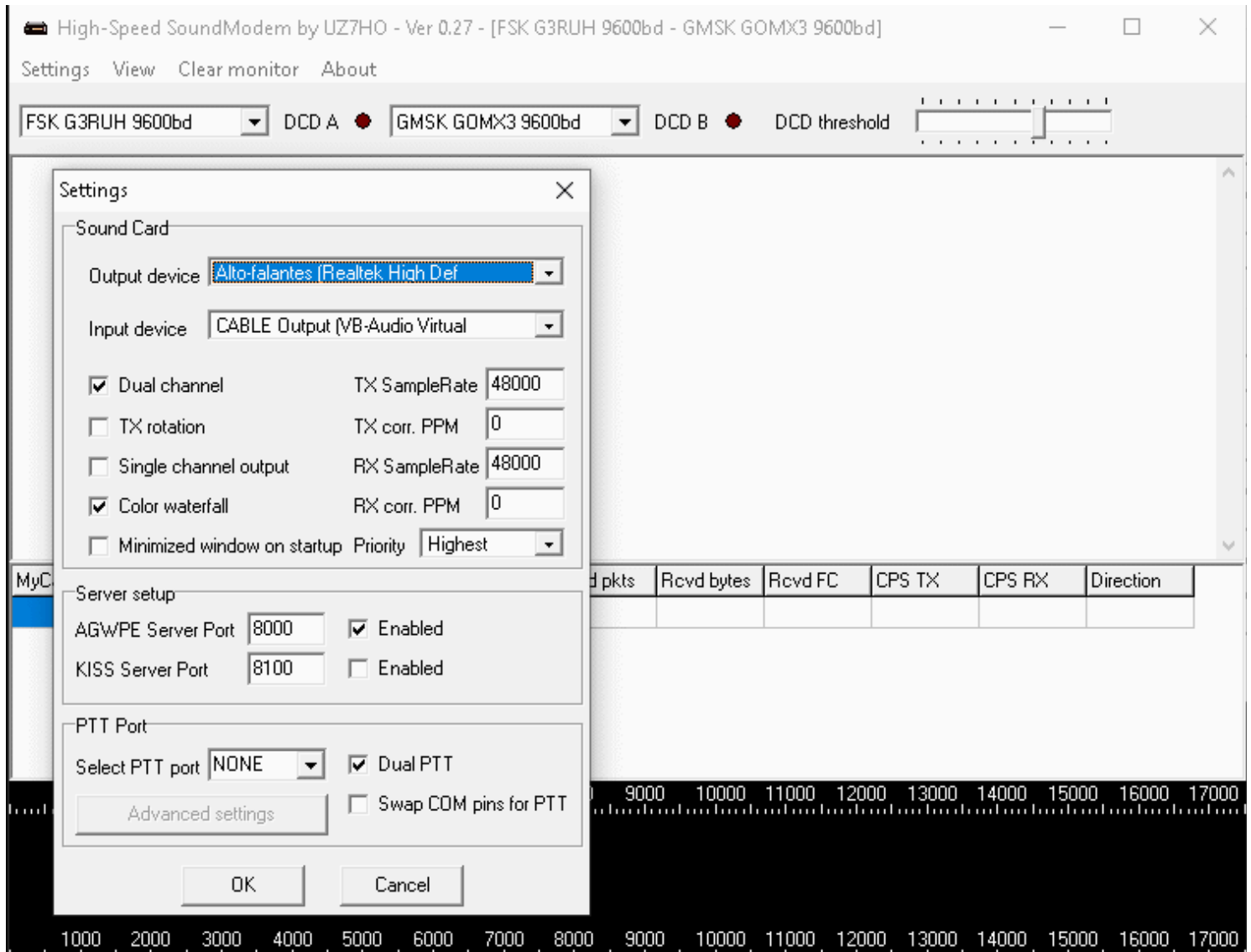
Setting Up The Sound Modems

- Download and extract zipped files content to new folders -- one folder for each executable -- somewhere on your PC storage. For AFSK, e.g. the common modes for APRS, AX-25, 1200bps packet and other similar modes, use this [low speed modem](#); for the higher speeds and formats such as 9600/19200 G3RUH FSK, etc., use this [high speed modem](#). Sometimes there is false virus detection or unsafe download alert from some web browsers. As already recommended in the SkyRoof [FAQ](#), scan both links using one of the several online virus scanning services that you can use to check the link.

P.S.: there are a few extra modems that can be tested for other modulation formats such as BPSK, GMSK. etc. in [this archive](#). But this How-To only covers AFSK and FSK associated to **FM_D** SkyRoof mode. For the UZ7HO itself, there is a more detailed user guide at [Soundmodem User Guide v114](#)

- Once downloaded and extracted, create corresponding desktop shortcuts to easy programs launch;
- The following examples will cover **Veronika** or **Tigrisat** satellites FSK data;
- Configure the sound modems audio setup using the already installed VACs. This setup is under Settings => Devices tab. Despite the fact that those modems can be used to TX and RX, the sample picture below assumes that the use is RX only;

- Select **FSK G3RUH 9600bd** in one of the drop down menus and adjust **DCD slider** up to the point that the "led" blinks a little bit.



Configuring SkyRoof

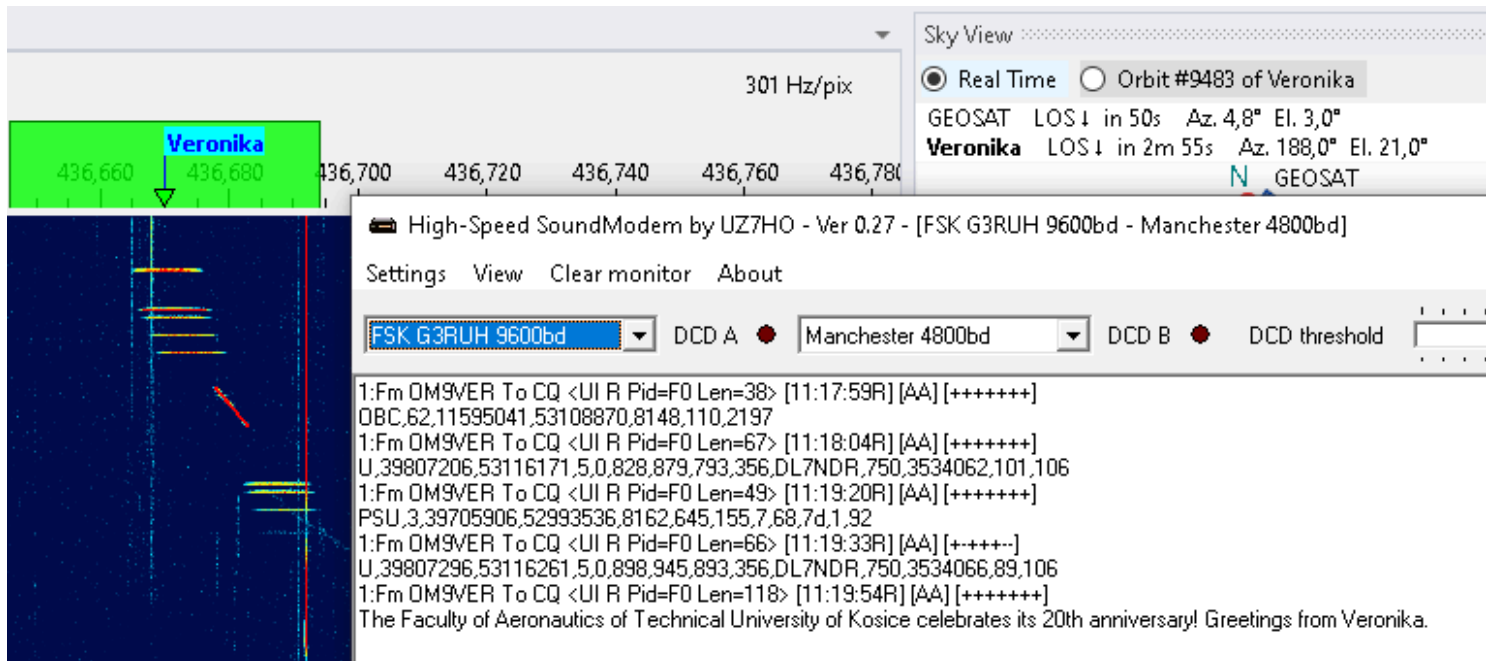
In SkyRoof:

- select the Veronika satellite. If it is not in the current group, add it using the [Satellites and Groups](#) dialog;
- Select **GFSK9K6 AX25** transmitter on the UHF band from the list of transmitters;
- Select **FM_D** downlink mode in the drop-down list on the toolbar;
- in the **Output Stream** section of the [Settings](#) window:
 - select **Audio to VAC**;
 - set Gain, dB to 0;

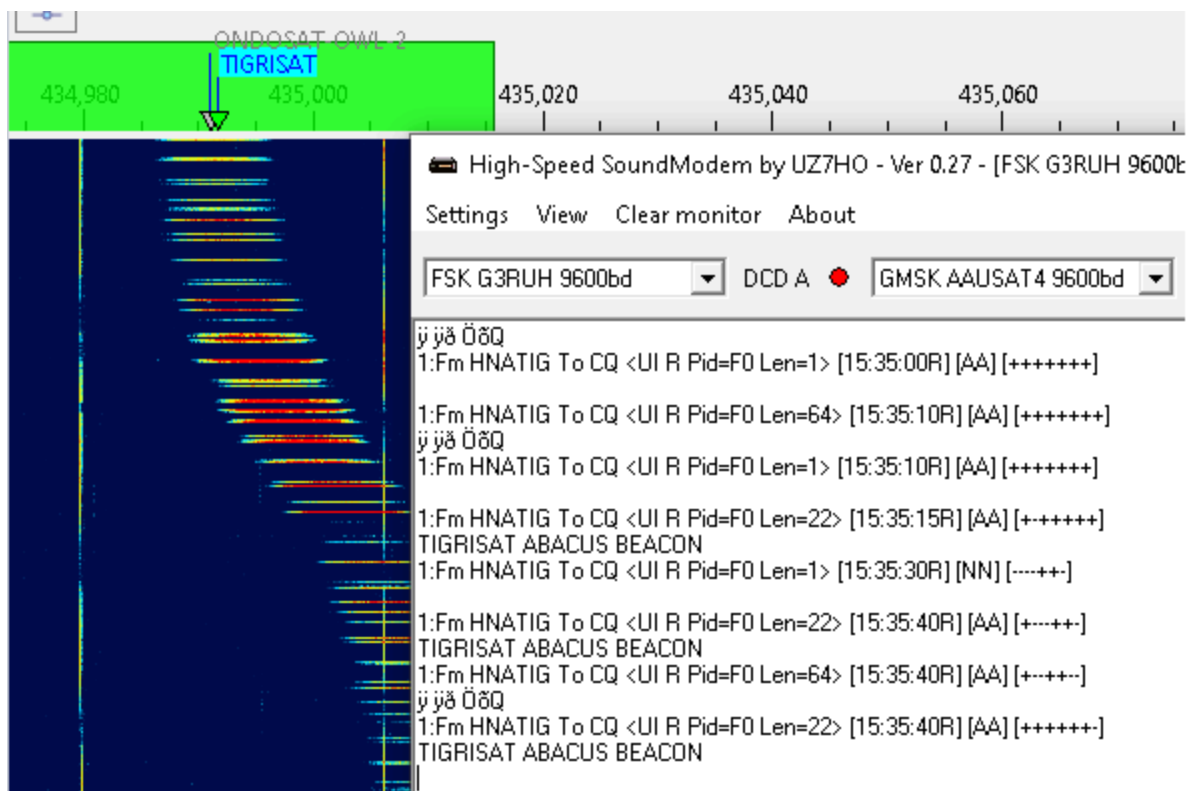
- select the VAC in the list of audio devices;
- click on the Output Stream label on the status bar to enable the output.

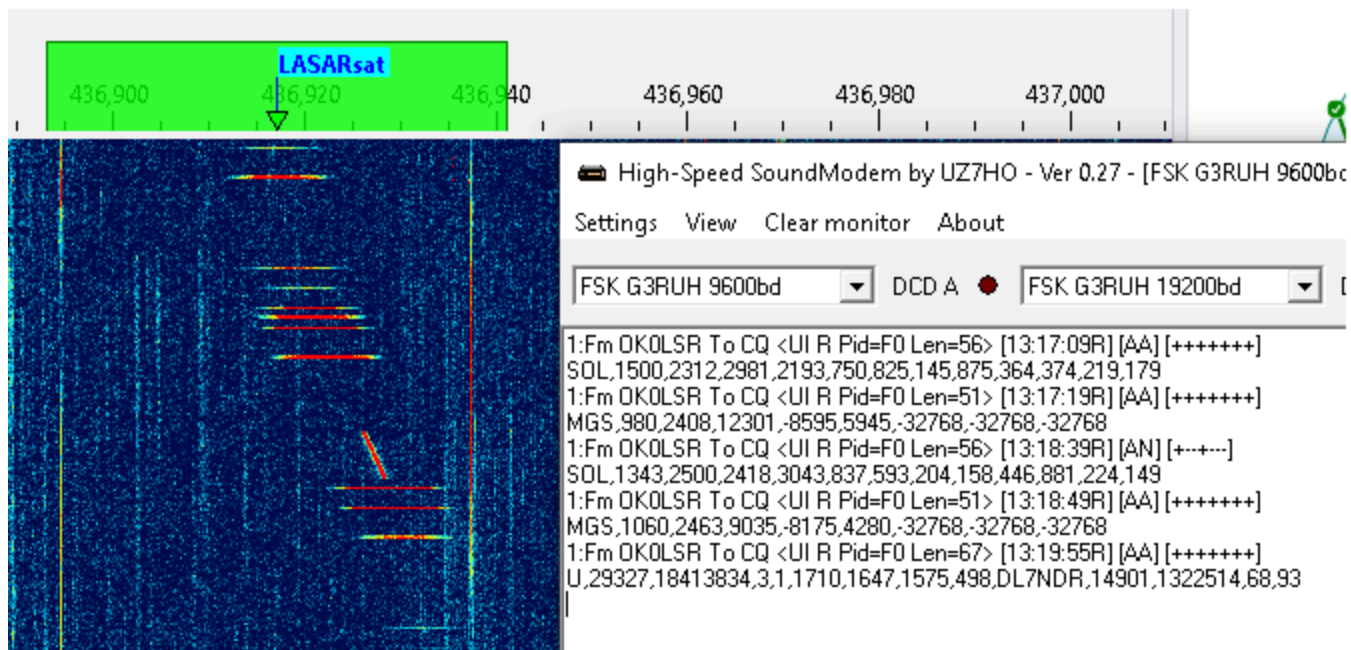
Receiving Data

When the selected satellite raises above the horizon, the UZ7HO FSK Modem will start decoding and produce output like this:



For Tigrisat, the decoded telemetry should be like the picture below:





Satellite Data

Data Sources

SkyRoof obtains satellite data from several sources:

- [SatNOGS DB](#) is the main source of satellite data. It is a frequently updated, crowd-sourced dataset that contains detailed information about all satellites transmitting in the Ham bands;
- [JE9PEL Satellite List](#) is another dataset with information about the satellites, maintained by Mineo Wakita JE9PEL, that, in particular, includes the callsigns of the satellites. This dataset presents the frequencies in an undocumented format, so its data are included in the SkyRoof database only FYI.
- [LoTW](#) - The ARRL LoTW service accepts satellite QSO only if the satellite abbreviation is one of those published on their [web site](#). These abbreviations are stored in a file in the [Data folder](#), you can view them in the [Satellite Details window](#).
- [AMSAT Live OSCAR Satellite Status Page](#) accepts satellite observations with their own satellite abbreviations, these abbreviations are stored in a file in the [Data folder](#).

TLE

The satellite orbit elements ([TLE](#) data) are downloaded from **SatNOGS DB**.

SatNOGS obtains these data from different sources and makes the latest and most reliable data available on their web site. The source of TLE and its creation time are shown in the [Satellite Details window](#) or [panel](#):

Satellite Details	
SO-124	
a.k.a. HADES-R	
AMSAT	SO-124
▼ Orbit	
TLE	<u>2025-05-29 12:37 (Space-Track.org)</u>
Period, min	94

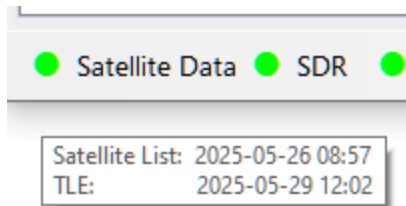
and in the mouse tooltip of the satellite:

AO-91	43017	12m 55s
FO-29	SO-124, HADES-R	
AO-123	NORAD: 62690	
RS-44	status: alive	
SO-50	countries: ES	
ARISS	TLE: 2025-05-29 12:37 (Space-Track.org)	
	period: 94 min.	
	inclination: 97°	

Automatic Updates

SkyRoof automatically downloads the satellite list every 7 days, and TLE data every 24 hours.

The mouse tooltip of the Satellite Data label on the status bar shows the last download time:



The light next to the label turns yellow if the satellite data are not up to date.

Manual Updates

In addition to automatic downloads, the data may be manually downloaded at any time using the **Tools / Download All Satellite Data** and **Tools / Download Only TLE** menu commands.

Loading TLE from File

If your system is not connected to the Internet, you can load TLE data from a local file using the **Tools / Load TLE from File** menu command. Two TLE formats are supported:

- **.json** - TLE data from the SatNOGS web site, recommended ([download](#));
- **.txt** - 3-line TLE data in a text file, available from many sources, e.g. CelesTrak ([download](#)).

Note that TLE import cannot add new satellites, it only loads orbital elements for the satellites already in the database.

AMSAT Satellite Status

[AMSAT Live OSCAR Satellite Status Page](#) is a crowd-sourced, real-time Ham satellite status page.

Posting Status Data

You can post your satellite status observations the the AMSAT web site either by filling the submission form on their site, or using the right-click menu of the satellite labels on the [Frequency Scale](#). A valid Ham callsign must be entered in the [Settings window](#) for this function to work.

Downloading Status Data

Set the **Amsat Satellite Status / Enable** option in the Settings window to **true** to enable automatic downloads of the satellite status information from the AMSAT web site. The statuses are shown on the [Current Group](#) panel, the green and red icons represent the active and inactive status respectively.

Satellite status data are downloaded once an hour. You can manually download it at any time using the **Tools / Download AMSAT Statuses** menu command.

Doppler Tracking

The [SGP4](#) algorithm used in SkyRoof to compute the Doppler offset produces very accurate results for the LEO satellites, typically within tens of Hertz, if it receives accurate input data. For best results, ensure that the following conditions are met.

Home Location

Make sure that your grid square is accurate. Correct it in the [Settings window](#) if necessary.

System Time

Your system clock should be accurate to a second. Get one of those little programs that run in the system tray and periodically synchronize your clock with the time servers on the Internet. [NetTime](#) is one such program.

PPM correction

Find the PPM correction factor of your SDR radio as described in the [Calibrating PPM Correction](#) section and enter it in [SDR settings](#).

TLE Data

SkyRoof downloads the TLE data automatically every 24 hours. Some sources claim that TLE may be updated once a week, but that would not be enough for accurate Doppler tracking, especially for the satellites that perform frequent orbit corrections. When in doubt, download TLE manually as described in the [Satellite Data](#) section.

Transmitter Frequency Correction

Most satellite transmitters transmit on the frequencies that differ from the nominal values by up to a few kHz. A one-time correction described in the [Frequency Control](#) and [Frequency Scale](#) sections eliminates this error.

Data Folder

SkyRoof keeps all of its data in the **data folder**.

- Click on **Help / Data Folder** in the main menu to open this folder in File Explorer.
- To open the folder when the program is not running, type this in File Explorer:

```
%appdata%\Afreed\Products\SkyRoof`
```

Data Files

- **Settings.json** - this is the file where all user-defined settings are stored;
- **amsat_sat_names.json** - satellite names used on [AMSAT Live OSCAR Satellite Status Page](#). The [Frequency Scale](#) section explains how to post your observations to this page;
- **lotw_sat_names.json** - the list of satellite abbreviations accepted by [LoTW](#);
- **Satellites.json** - the satellite database compiled from the downloaded data;
- **cat_info.json** - lists the CAT capabilities of a generic simplex radio;
- **wsjtx_wisdom.dat** - optimal FFT transform settings found by automatic testing.

Folders

- **Logs** - contains the log files with error messages and other information;
- **Adif** - QSO records stored in the ADIF format;
- **Downloads** - a copy of the satellite data downloaded from various sources, kept for troubleshooting;
- **Palettes** - definition of the color palettes used by the waterfall display. Add your own palette as a text file with "html" color codes. Pick the color codes at [htmlcolorcodes.com](#);
- **sat_images** - cached satellite thumbnails shown by the [Satellite Photo](#) widget. Delete this folder to force re-download.

Smart Antenna Rotation

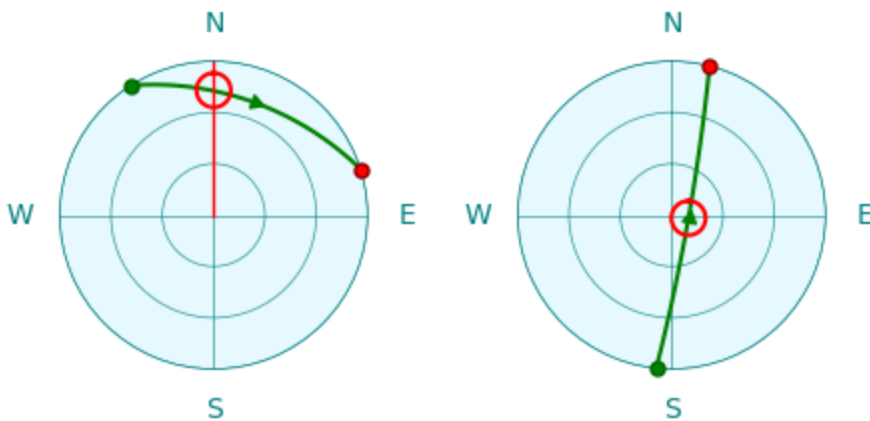
SkyRoof uses a new antenna rotation algorithm, based on dynamic programming, that globally optimizes the tracking of the satellite pass and ensures the minimum possible communication downtime. The algorithm makes full use of the extended range of azimuth and elevation available in the modern rotators.

Limitations of the Simple Approach

The satellite pass prediction algorithm computes the satellite azimuth in the range of **0° to 360°** and elevation in the range of **0° to 90°**. A simple antenna rotator just sends these azimuth and elevation values to the controller. This works most of the time, but there are cases when this requires the antenna to rotate by 180° or 360°. While the antenna is making this full circle, or half-circle, it does not point at the satellite, causing a significant downtime in the communication. The charts below show two such cases.

In the first chart the path crosses the 360° azimuth line, and the antenna needs to make a full circle counter clockwise to get to 0°.

In the second chart, when the satellite is at its highest elevation point, its azimuth quickly changes from South to North, by about 180°. This rotation may take up to a minute, causing a significant downtime.



Rotator Capabilities

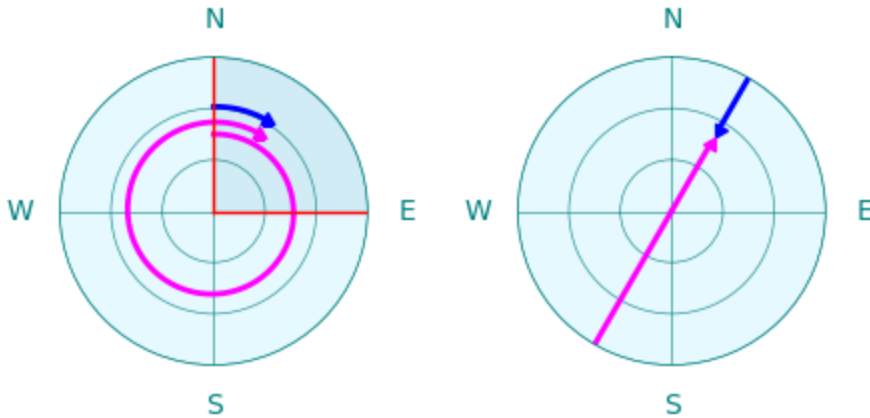
Many antenna rotators have an extended range of azimuth and elevation that goes beyond the **0° to 360°** for the azimuth and **0° to 90°** for the elevation. This allows the controller to point the antenna in the same direction in more than one way. We will use [Yaesu G-5500DC](#) , one of the most popular satellite antenna rotators, as an example. This rotator has the following rotation range:

- **0° to 450°** for the azimuth;

- **0° to 180°** for the elevation.

With this azimuth range, there is a 90° segment where every point may be reached in two ways. The first chart below shows that Azimuth=30° and Azimuth=390° point in the same direction.

With this elevation range, every point may be reached in two ways, with elevation < 90° and > 90°, as shown in the second chart.



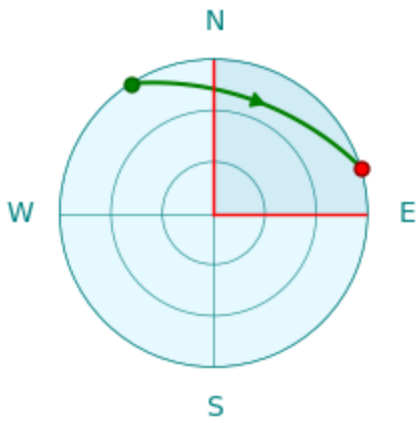
Note also that the rotation speed of G-5500DC is different in the horizontal and vertical planes. Here is what appears in its specifications:

Rotation time (approx.):
 Elevation (180°): 65 sec.
 Azimuth+(360°): 60 sec.

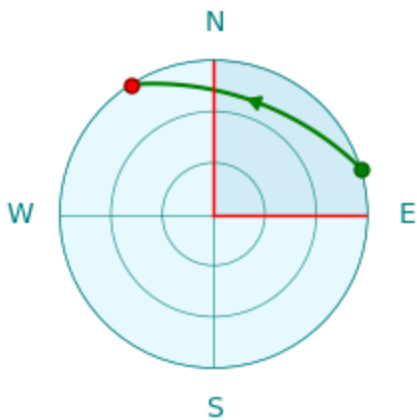
The SkyRoof algorithm assumes that vertical rotation is two times slower than horizontal.

Algorithm

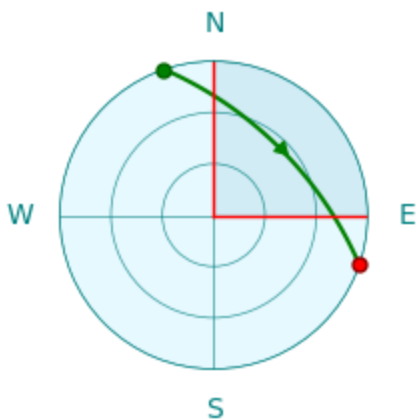
The SkyRoof algorithm that finds the optimum tracking strategy for the given satellite path is not rule-based, it uses the global optimization approach to find the optimal solution. The examples below show how it handles certain special cases.



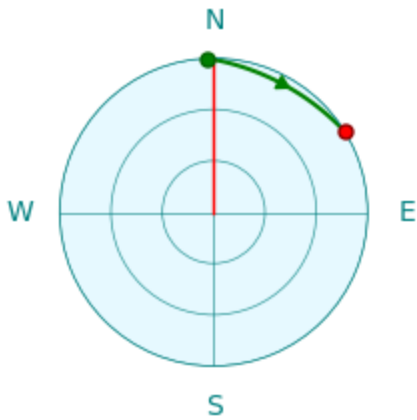
This is an easy case. The azimuth changes from 330° to 80° , so the rotation starts at 330° and, when the azimuth crosses 360° , continues in the same direction, up to 440° , which is the same direction as 80° .



The same path, but in the opposite direction. The simple approach described above, that is used in many simple antenna rotation programs, no longer works. If we start at Azimuth= 80° , we hit the 0° limit and cannot continue. The algorithm in SkyRoof looks at the whole pass and chooses to start at 440° instead of 80° so that the pass is tracked without interruption.

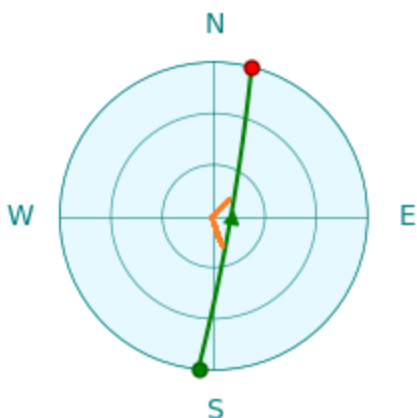


Now the path crosses both 0° and 450° lines. In this case the 90° overlap is not enough to ensure uninterrupted tracking. The dynamic algorithm still finds a solution: it uses the elevations $> 90^\circ$. Tracking starts at (Az= 170° , El= 180°) and ends at (Az= 280° , El= 180°). Again, all azimuths are in the $0^\circ - 360^\circ$ range, ensuring uninterrupted tracking.



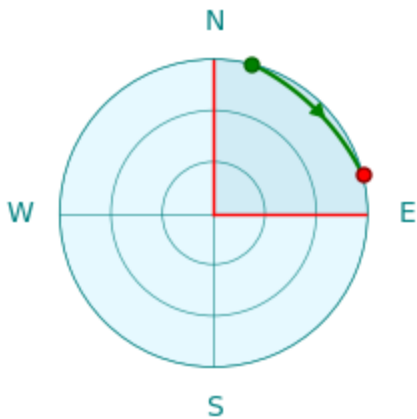
SkyRoof has a parameter in the rotator control settings that determines the steps in which the antenna is rotated, to prevent frequent starts and stops of the motors. Indeed, if the antenna beam width is, say, 60° , there is no point in rotating it in 1° steps. The default step size is 5° .

The path in the chart above crosses 360° , but it extends beyond the red line only by a couple of degrees. If the overshoot does not exceed the rotation step, the algorithm is smart enough to start the path at 0° so that 360° rotation is not required. This works even if the rotator does not have an extended azimuth range.



This is a high elevation pass. As we have seen before, it requires the antenna to make a fast rotation by about 180° in azimuth. The rotators that support elevations of 0° to 180° can use elevation $> 90^\circ$ for the second half of the path, thus eliminating the 180° azimuth change. However, to switch from the elevation $< 90^\circ$ to $> 90^\circ$, the antenna must deviate

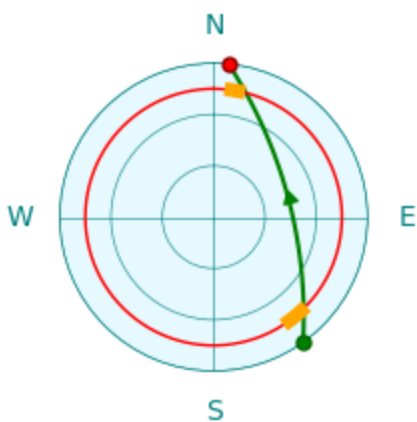
from the path a little bit and pass through the zenith (90°), as shown with the orange lines in the chart above. Dynamic programming plans this deviation in an optimal way.



In many cases there is more than one way of tracking the same path. For example, the path above may be tracked in three different ways:

- ($Az=20^\circ$, $El=0^\circ$) $\rightarrow$ ($Az=70^\circ$, $El=0^\circ$)
- ($Az=380^\circ$, $El=0^\circ$) $\rightarrow$ ($Az=430^\circ$, $El=0^\circ$)
- ($Az=200^\circ$, $El=180^\circ$) $\rightarrow$ ($Az=250^\circ$, $El=180^\circ$)

When the algorithm finds multiple optimum paths, it chooses the one which has the starting point closest to the current antenna direction.



Some antenna rotators have a turning range that does not cover the whole upper hemisphere. The chart above shows the case where the rotator has an elevation range of 15° to 90° . This could be a limitation of the hardware or just an obstructed horizon that makes lower elevation angles useless. The algorithm obeys these restrictions, it does not go beyond the red line. It follows the path shown in the chart with orange lines.

Frequently Asked Questions

Q: I downloaded SkyRoof, and my virus scanner shows an infection. Is it real?

A: Most likely, it is a false detection. However, it is always a good idea to test the download links **before** downloading **any** software. There are several online virus scanning services that you can use to check the link. [VirusTotal](#) and [Hybrid Analysis](#) are just two examples.

Copy the download link and paste it in the virus scanner page. The scanner will download the file, test it with multiple antivirus programs, and show you the results.

In the screenshot below the download link of SkyRoof 1.5 beta was tested with VirtusTotal, and all virus scanners, except one, agreed that the file was clean. When you see something like this, you know that it was a false alarm.

If the file is clean, you can add it to the exception list of your virus scanner and safely install it. For Windows Defender follow [these instructions](#), for other anti-virus products follow the instructions in their documentation.

To prevent such problems in the future, please submit the file to Microsoft for analysis using this link:

<https://www.microsoft.com/en-us/wdsi/filesubmission>

The screenshot shows the VirusTotal web interface. At the top, the VirusTotal logo is visible. The main content area displays the analysis results for the URL `https://github.com/VE3NEA/SkyRoof/releases/download/v.1.5-beta/SkyRoof.exe`. A warning icon indicates that 1/97 security vendors flagged this URL as malicious. The interface includes tabs for DETECTION, DETAILS, and COMMUNITY. A table titled 'Security vendors' analysis' lists various vendors and their results.

Security vendors' analysis	Do you want to automate checks?
URLhaus	Malicious
Abusix	Clean
Acronis	Clean
ADMINUSLabs	Clean
AILabs (MONITORAPP)	Clean
AlienVault	Clean
AlphaSOC	Clean
Antiy-AVL	Clean
Artix Against 419	Clean

Q: The right part of the SkyRoof toolbar does not fit in the screen, even though the screen resolution is 1980x1280.

A: This happens because your **text size** setting in Windows is too high. For example, if it is set to 200%, the effective screen width is only $1900 / 2 = 950$ pixels. To fix this, right-click on the Desktop, click on **Display Settings** and set the text size to a lower value.

Q: How can I run two instances of SkyRoof?

A: By default, only one instance of SkyRoof can run at any time, but there is a work around. Make a copy of the SkyRoof.exe file in the same folder, but with a different name, e.g., SkyRoof_2.exe. Each exe will have its own settings, its own data folder, and will run independently of the other instance.

Q: I found an error in the satellite database: one of the satellite transmitters is marked as active, while it has been inactive for years. How can I correct the error?

A: SkyRoof uses [satellite data](#) from the [SatNOGS database](#)[↗]. This database is crowd-sourced, so anyone can suggest changes. If you find an error in a satellite record (shown in the [Satellite Details](#) and [Satellite Transmitters](#) panels), click on the "SatNOGS" link in the Satellite Details panel to open the SatNOGS web site, then click on Transmitters, and select Edit from the drop-down menu. You may need to create an account with them (free) in order to submit changes.

Q: How can I add a satellite that is currently not listed in SkyRoof?

A: The only way to add a satellite to SkyRoof is to add it to the SatNOGS database. Here is how you can do this:

- navigate to the [SatNOGS DB web site](#)[↗];
- click on **Sign Up / Log In** in the top right corner of the page;
- create a free account, or log in to an existing one;
- click on **All Satellites**;
- in the **Actions** drop-down list select **Suggest New Satellite*.

See this [Wiki page](#)[↗] for details.

Q: My video card does not have an OpenGL 3.3 driver. Is there a way to use a software-only implementation of OpenGL?

A: Some users have had success with the OpenGL 3.3 functionality implemented in the software, though this approach is very CPU-intensive. You can try [this](#)[↗] or [this](#)[↗] solution. Download the 64-bit version of the library and extract all DLL's from the archive to the SkyRoof folder.

Q: How do I prevent CTCSS from turning off when using SkyRoof with FM satellites and an Icom radio?

A: If you're experiencing an issue where the CTCSS tone turns off when transmitting or changing frequencies with SkyRoof and an Icom radio, the solution is to disable the "Auto Repeater" setting in the radio. This setting causes the radio to automatically disable repeater-related features like CTCSS when the frequency is changed, assuming you're no longer on a repeater. Disabling "Auto Repeater" prevents this behavior and keeps CTCSS

enabled during frequency changes. This workaround was highlighted in a helpful [YouTube video](#).

Q: What are all those offset settings for?

A:

- **Downlink Manual Correction:** the frequencies of most satellite transmitters slightly differ from their published values. The purpose of this setting is to compensate for the difference, you set it once for each satellite and never change it again;
- **RIT:** just as in any HF radio, it is used to quickly switch the receiver between two frequencies close to each other. A typical use case is when someone answers your CQ slightly off the frequency and you want to tune to it without changing your main operating frequency;
- **Uplink Manual Correction:** this setting allows you to precisely align your own signal received back from the satellite with your receiver frequency. It needs to be set only once per satellite, and the offset is usually very small;
- **Transponder Offset:** the position of your operating frequency within the transponder passband. This offset is included in the Corrected Downlink Frequency shown on the toolbar, the actual value of the offset is shown only on the mouse tooltip of the frequency display. There are many ways to tune within the transponder passband, such as click on the desired frequency, drag the green passband rectangle, spin the mouse wheel over the rectangle, etc. These methods work only when the transponder is selected in the Transmitter drop-down box.
- **Doppler correction:** this correction is computed and applied automatically, it keeps your RX and TX signals in sync when you tune through the transponder segment.

Q: How to troubleshoot the rotator control?

A:

1. Set up and enable rotator control as described in [Setting Up Rotator Control](#).
 - if the Rotator light on the status bar is red, the problem is with the SkyRoof to rotctld.exe connection;
 - if the light is green, then the SkyRoof to rotctld.exe connection is fine, the problem is with the rotctld.exe to rotor connection.
2. Add the `-vvvvv` option to the command line that starts rotctld.exe, and restart it. This option tells rotctld.exe to print detailed information to the console window (the black

window that opens when rotctld.exe is started).

3. Read the [rotctld.exe documentation](#) that will help you to understand information in the console window.
4. Possible causes of SkyRoof to rotctld.exe connection problems:
 - rotctld.exe failed to start;
 - SkyRoof is configured to use a port number different from the rotctld.exe listening port.
5. Possible causes of the rotctld.exe to rotator connection error problems:
 - the rotator is not connected to the computer or not powered on;
 - the COM port number or Baud rate specified on the command line is wrong - then the COM port fails to open;
 - the rotator model specified on the command line is wrong - then the commands sent to the rotor are rejected or have no effect.
6. Make rotator control work without SkyRoof first: use **rotctl.exe**, an interactive version of rotctld.exe, and send commands to the rotator manually, by typing them in the console window. Rotator control uses three commands, "P", "p" and "S" (set position, read position, stop rotation). Once you make rotctl.exe work with the rotator, run rotctld.exe with the same command line parameters.